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MIL-STD-361-8(TM)
15 JULY 1992

SUPERSEDING
(See 6.5.)

MILITARY STANDARD

MANUALS, TECHNICAL:

MAINTENANCE INSTRUCTIONS

AMSC A6757

AREA TMSS

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FOREWORD

1. This military standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DOD).

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAMC Materiel Readiness Support Activity, ATTN: AMXMD-EP, Lexington, Kentucky 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

3. This document is one part of an 11-part bookform standard to be used in the preparation and acquisition of technical manuals that are essential to support (operate and maintain) the various types of equipment and weapon systems within the Department of the Army.

4. This document covers the maintenance instructions requirements for the preparation of technical manuals.

5. Other parts of this multipart standard are:

MIL-STD-361-1	-	Writing Style and Format Requirements.
MIL-STD-361-2	-	Comprehensibility Requirements.
MIL-STD-361-3	-	Requirements for Graphics.
MIL-STD-361-4	-	Quality Assurance Requirements.
MIL-STD-361-5	-	Requirements for Packaging.
MIL-STD-361-6	-	Introductory Information with Theory of Operations Requirements.
MIL-STD-361-7	-	Operator and Preventive Maintenance Instructions.
MIL-STD-361-9	-	Troubleshooting Procedures.
MIL-STD-361-10	-	Lubrication Instructions.
MIL-STD-361-11	-	Repair Parts and Special Tools List (RPSTL).

6. By incorporating standardized and service-unique technical manual requirements, this standard and its dash-numbered companion parts, along with the general acquisition specification MIL-M-63008, Acquisition/Preparation Requirements, support the acquisition of technical manuals.

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1. SCOPE

1.1 Scope. This document establishes the content and procedural requirements for preparing maintenance instructions for technical manuals (TMs), revisions, supplements, and changes.

1.2 Purpose. This document supplements the various requirements in the other parts to this 11-part standard for specific types of technical manuals and related publications and does not alone deliver any technical data. (NOTE: The figures in this document are examples only. The text of this document takes precedence over the figures.)

1.3 Applicability. The applicability of the requirements contained in this document is governed by the equipment, systems, or weapon systems for which the technical manuals are being developed. This document is applicable for use by the Department of the Army.

1.4 Selective application and tailoring. This document contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in this document are the responsibility of the contracting activity and shall be accomplished through the use of MIL-M-63008. The applicability of some requirements is also designated by one of the following statements:

- unless requirement is specifically excluded by the contracting activity; or
- as/when specified by the contracting activity.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2.)

SPECIFICATIONS

MILITARY

MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition.
MIL-M-63005	-	Manuals, Technical: For Preparation for Shipment of Army Aircraft.
MIL-M-63008	-	Manuals, Technical: Acquisition/ Preparation Requirements.
MIL-M-63043	-	Manuals, Technical: Missile System Check Procedures.
MIL-M-63044	-	Manuals, Technical: Missile System Equipment Unit-Under-Test (UUT) Procedures.
MIL-W-63150	-	Weapons and Support Materiel Quality Assurance, Provisions for.

STANDARDS

MILITARY

MIL-STD-12	-	Abbreviations for Use on Drawings, and in Specifications, Standards, and Technical Documents.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-120	-	Gage Inspection.
MIL-STD-209	-	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-361-1	-	Manuals, Technical: Writing Style and Format Requirements.

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MIL-STD-361-2	-	Manuals, Technical: Comprehensibility Requirements.
MIL-STD-361-3	-	Manuals, Technical: Requirements for Graphics.
MIL-STD-361-4	-	Manuals, Technical: Quality Assurance Requirements.
MIL-STD-361-5	-	Manuals, Technical: Requirements for Packaging.
MIL-STD-361-6	-	Manuals, Technical: Introductory Information with Theory of Operations Requirements.
MIL-STD-361-7	-	Manuals, Technical: Operator and Preventive Maintenance Instructions.
MIL-STD-361-9	-	Manuals, Technical: Troubleshooting Procedures.
MIL-STD-361-10	-	Manuals, Technical: Lubrication Instructions.
MIL-STD-361-11	-	Manuals, Technical: Repair Parts and Special Tools List (RPSTL).
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1388-2	-	DOD Requirements for a Logistic Support Analysis Record.
DOD-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment, Excluding Electrically Initiated Explosive Devices (Metric).
MIL-STD-45662	-	Calibration Systems Requirements.

HANDBOOKS

MILITARY

MIL-HDBK-263	-	Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment, Excluding Electrically Initiated Explosive Devices (Metric).

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MIL-HDBK-275	-	Guide for Selection of Lubricants, Fluids, and Compounds for Use in Flight Vehicles and Components.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PL 89-272	-	Resource Conservation and Recovery Act (as amended by PL 91-512, PL 93-611, and PL 94-580).

(Copies of Resource Conservation and Recovery Act are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

AMC-R 702-32	-	Critical Safety Item Program.
AR 95-16	-	Weight and Balance: Army Aircraft.
AR 190-11	-	Physical Security of Arms, Ammunition, and Explosives.
AR 190-13	-	The Army Physical Security Program.
AR 700-22	-	Worldwide Ammunition Reporting System (WARS).
AR 746-1	-	Packaging of Army Materiel for Shipment and Storage.
AR 750-1	-	Army Materiel Maintenance Concepts and Policies.
AR 750-2	-	Army Materiel Maintenance Wholesale Operations.
DA Form 2404	-	Equipment Inspection and Maintenance Worksheet.
DA Form 2408-17	-	Aircraft Inventory Record.
DESCOM-R 702-1	-	Depot Quality Systems.
SB 11-573	-	Painting and Preservation Supplies Available for Field Use for Electronics Command Equipment.
SB 742-1	-	Ammunition Surveillance Procedures.

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TB 43-0118	-	Field Instructions for Painting and Preserving Communications-Electronics Equipment.
TB 43-0209	-	Color, Marking and Camouflage Painting of Military Vehicles, Construction Equipment, and Material Handling Equipment.
TM 43-0139	-	Painting Instructions for Army Materiel.
TB 750-93-1	-	Functional Grouping Codes: Combat, Tactical, and Support Vehicles and Special Purpose Equipment.
TM 55-1500-204	-	General Aircraft Maintenance Manual.
TM 55-1500-335	-	Nondestructive Inspection Methods.
TM 55-1500-342	-	Army Aviation Maintenance Engineering Manual, Weight and Balance.
TM 55-1500-345	-	Painting and Marking of Army Aircraft.

(Copies of these regulations and pamphlets are available from the U.S. Army Publications Distribution Center, 2800 Eastern Blvd, Baltimore, MD 21220-2896.)

2.2 Order of precedence. In the event of a conflict between the text of this document ("This document" includes all volumes of MIL-STD-361 series.) and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

a. AVIM - Aviation Intermediate Maintenance.

b. AVUM - Aviation Unit Maintenance.

c. BIT - Built-In Test.

d. CAGEC - Commercial and Government Entity Code.

e. CPC - Corrosion Prevention Control.

f. DMWR - Depot Maintenance Work Requirement.

g. DOD - Department of Defense.

h. DODISS - Department of Defense Index of Specifications and Standards.

i. ECP - Engineering Change Proposal.

j. ESD - Electrostatic Discharge.

k. FGC - Functional Group Code.

l. FMC - Fully Mission Capable.

m. LOAP - List of Applicable Publications.

n. LRU - Line Replacement Unit.

o. LSA - Logistic Support Analysis.

p. LSAR - Logistic Support Analysis Record.

q. MAC - Maintenance Allocation Chart.

r. MTOE - Modified Table of Organization and Equipment.

s. NDTI - Nondestructive Testing Information.

t. NHA - Next Higher Assembly.

u. NSN - National Stock Number.

v. PMCS - Preventive Maintenance Checks and Services.

w. QA - Quality Assurance.

x. RAM - Reliability, Availability, Maintainability.

y. RCM - Reliability Centered Maintenance.

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z. RPSTL - Repair Parts and Special Tools List.

aa. SC - Supply Catalog.

ab. SMR - Source, Maintenance, Recoverability.

ac. TBO - Time Between Overhaul.

ad. TM - Technical Manual.

ae. TMDE - Test, Measurement, and Diagnostic Equipment.

3.2 Adjust. To maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.

3.3 Align. To adjust specified variable elements of an item to bring about optimum or desired performance.

3.4 Assembly. Two or more parts or subassemblies joined together to perform a specific function and capable of disassembly (e.g., power shovel-front, fan assembly, audio frequency amplifier).

NOTE

The distinction between an assembly and subassembly is determined by the individual application. An assembly in one instance may be a subassembly in another where it forms a portion of an assembly.

3.5 Auxiliary equipment. Equipment, accessories, or devices which, when used with basic equipment, extend or increase its capability (e.g., Modified Table of Organization and Equipment (MTOE) items, etc.).

3.6 Bulk material. Material issued in bulk for manufacture or fabrication of support items (e.g., sheet metal, pipe tubing, bar stock, or gasket material); excludes expendable items. (See 3.17.)

3.7 Calibrate. To determine and cause corrections to be made or to be adjusted on instruments or test, measuring, and diagnostic equipment used in precision measurement. Consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.

3.8 Commercial and Government entity code (CAGEC). A five-position code assigned to manufacturers or nonmanufacturers organizational entities or contractors.

3.9 Component. A constituent part not normally considered to be capable of independent operation; a piece part.

3.10 Contracting activity. The Department of Defense (DOD) component, activity, or organization of a using military service,

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or that organization delegated by a using service, that is responsible for the selection and determination of requirements for TMs.

3.11 Depot-level maintenance. Maintenance that is beyond the capability of the unit, direct support, and general support activities. Depot-level maintenance normally consists of overhaul, recondition, manufacture, repair, or modification and requires technical assistance beyond lower maintenance level capability.

3.12 Direct support maintenance. Maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit, or other portion either on the system or after it is removed. The replace function for this level of maintenance is indicated by the letter "F" appearing in the third position of the source, maintenance, recoverability (SMR) code. An "F" appearing in the fourth and fifth positions of the SMR code indicates complete repair is possible at the direct support maintenance level.

3.13 Disassemble. The step-by-step taking apart (or breakdown) of a spare or functional group-coded item to the level of its least componency identified as maintenance-significant (i.e., assigned a source, maintenance, and recoverability (SMR) code for the category of maintenance under consideration).

3.14 End item. A final combination of end products, component parts, or materials that is ready for its intended use (e.g., tank, mobile machine shop, aircraft, receiver, rifle, recorder).

3.15 Equipment. One or more units capable of performing specified functions.

3.16 Equipment, complex. Equipment with numerous sophisticated systems installed, including integrated systems and built-in test (BIT) capability.

3.17 Expendable items. Items, other than repair parts, that are consumed in use (e.g., paint, lubricants, wiping rags, tape, cleaning compounds, sandpaper).

3.18 Fully mission capable (FMC). Equipment that can perform all of its combat missions without further endangering the lives of crew or operators in a combat environment.

3.19 Functional group code (FGC). A basic (usually two-position) group code assigned to identify major components, assemblies, and subassemblies to a functional system. Subordinate subfunctional groups/subassemblies are coded to relate back to the basic (top position) FGC in a sequential, next higher assembly (NHA) relationship (i.e., top-down breakdown structure).

3.20 General support maintenance. Maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit, or

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other portion either on the system or after it is removed. The replace function for this level of maintenance is indicated by the letter "H" appearing in the third position of the SMR code. An "H" appearing in the fourth and fifth positions of the SMR code indicates complete repair is possible at the general support maintenance level.

3.21 Hardtime scheduled maintenance. Hardtime maintenance is scheduled maintenance conducted at predetermined fixed intervals because of age, calendar, or usage such as operating time, flying hours, miles driven, or rounds fired.

3.22 Inspect. To determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel).

3.23 Line replacement unit (LRU). An item normally removed and replaced as a single unit to correct a deficiency or malfunction on a weapon system or end item of equipment.

3.24 Maintenance function. A group of maintenance tasks performed on a system or on a component of a system.

3.25 Maintenance level. The separation of maintenance activities or functions in the U.S. Army according to the required skills and available facilities.

3.26 Maintenance step. A single maintenance action, such as setting a switch to the OFF position. Usually, a step has one action, but in certain cases, there may be a series of identical actions, such as removing seven bolts.

3.27 Maintenance task. A group of related maintenance steps with a definite beginning and end.

3.28 Modification. A change made in a system, assembly, subassembly, or component.

3.29 Module. A subassembly that, in the area of electronic systems, may be removed and replaced without use of soldering equipment or special tools; a module may be encapsulated.

3.30 On-condition scheduled maintenance. On-condition scheduled maintenance is maintenance or item replacement action performed based upon condition of the item as determined by an evaluation of each item on a scheduled basis.

3.31 Overhaul. That maintenance effort (service/action) prescribed to restore an item to a completely serviceable/ operational condition as required by maintenance standards in appropriate technical publications. Overhaul does not normally return an item to like new condition.

3.32 Part number (P/N). A primary number used to identify an item used by the manufacturer (individual, company, firm,

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corporation, or Government activity) that controls the design, characteristics, and production of the item by means of its engineering drawings, specifications, and inspection requirements.

3.33 Preshop analysis. To determine, prior to beginning maintenance activities, the extent of maintenance required to return the end item, assembly, subassembly, or component to a serviceable condition as specified by the depot level maintenance instructions.

3.34 Rebuild. Consists of those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing tolerances.

3.35 Remove/install. To remove, then install the same item removed. (Compare with "3.38 Replace.") Such action is prescribed by the logistic support analysis (LSA)/maintenance allocation chart (MAC), but is not covered by source, maintenance, and recoverability (SMR) code authorization. (For example: the operator/crew is authorized to remove/install the cannon tube, but the SMR code for the cannon tube is PAOOO, authorizing replacement at unit level.) Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module in a manner to allow the proper functioning of an equipment or system. Performed during service actions, disassembly/assembly procedures, or other maintenance activities.

3.36 Repair. The application of maintenance services (inspect, test, service, adjust, align, calibrate, and/or replace) including fault location/troubleshooting, removal/installation, and disassembly/assembly procedures, and maintenance actions to identify troubles and restore serviceability to an item by correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system.

3.37 Repair part. Any nonreparable component (Supply Class IX item) required for maintenance/repair of an end item/equipment (will be coded Z or B in fourth position of the source, maintenance, recoverability (SMR) code).

3.38 Replace. To remove an unserviceable spare or repair part and install a serviceable counterpart in its place. Replace is authorized by the LSA/MAC and the assigned maintenance level is shown as the third position code of the SMR code.

3.39 Service. Operations required periodically to keep an item operating, i.e., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases.

3.40 Set. A unit and necessary assemblies, subassemblies, and parts connected together or used in association to perform an operational function (e.g., radio receiving set, set measuring

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set, radar, homing set, which includes parts, assemblies, and units such as cables, microphones, and measuring instruments).

3.41 Source, maintenance, recoverability (SMR) code. The five-position code containing supply/requisitioning information, maintenance level authorization criteria, and disposition instruction. The first two positions of the SMR code determine how to get an item. The third position represents who can install, replace, or use the item. The fourth position dictates who can do complete repair on the item. The fifth position represents who determines disposition action on unserviceable items. For further information, refer to MIL-STD-361-11.

3.42 Spare part. Any reparable and recoverable component required for the maintenance or repair of an end item or equipment (will have a recoverability code other than "Z" and will be assigned an FGC or subfunctional group code in the MAC).

3.43 Special tools. Those support items that have single or peculiar application to a specific end item/system. Items include special tools; special test, measurement, and diagnostic equipment (STMDE); and special support equipment.

3.44 Subassembly. Two or more parts that form a portion of an assembly or a component replaceable as a whole, but having a part or parts that are individually replaceable (e.g., gun mount stand, window recoil mechanism, floating piston, intermediate frequency strip, mounting board with mounted parts, power shovel dipper stick).

3.45 System. A regularly interacting or interdependent group of items forming a unified whole to perform a specific operational function.

3.46 Technical manuals (TMs). Manuals that contain instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual displays, electronic imbedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures.

3.47 Test. To verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards.

3.48 Time between overhaul (TBO) items. Those items having a definite retirement schedule within a defined overhaul interval, e.g., those items which must be replaced within a system, assembly, subassembly, or component between scheduled overhauls.

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3.49 Top-down breakdown. The pyramidal breakdown of an end item, with the top item being the complete end item. The process of breakdown is established from the engineering drawing structure in an NHA progression until the lowest reparable in each family tree group is identified. All nonreparables (spare parts) can be identified in like manner to establish their NHA relationships.

3.50 Unit. An assembly or any combination of spare and repair parts mounted together, normally capable of independent operation in a variety of situations.

3.51 Unit maintenance. The responsibility of, and performed by, a using organization on its assigned equipment. It normally consists of inspecting, servicing, lubricating, adjusting, and replacing parts, minor assemblies, and subassemblies. The replace function for this level of maintenance is indicated by the letter "O" in the third position of the SMR code. An "O" appearing in the fourth and fifth positions of the SMR code indicates complete repair is possible at the unit maintenance level.

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4. GENERAL REQUIREMENTS

4.1 General style requirements.

4.1.1 Conformance to standard. The general style and format for maintenance instructions shall be in accordance with MIL-STD-361-1.

4.1.2 Technical content preparation. The following baseline requirements shall be used, as specified by the contracting activity:

a. Logistic support analysis record (LSAR). The technical data and instructions developed by the requirements of MIL-STD-1388-2, DOD Requirements for a Logistic Support Analysis Record (LSAR), shall be used as the baseline to prepare TMs.

b. Maintenance allocation chart (MAC). The MAC shall be used as the baseline to prepare TMs. The MAC shall be prepared from the equipment top-down breakdown and be in functional group code sequence to consolidate and identify those groups on the list which involve identified maintenance functions. Logistic support Analysis (LSA) Output Report 04, Maintenance Allocation Summary will be used when available as source data for the final approved MAC.

When specified by the contracting activity and if a waiver of standard requirements is granted, TB 750-93-1 may be used as the baseline for preparing TMs.

4.1.3 Security classification markings. All classified TMs and material shall be safeguarded and identified in accordance with MIL-STD-361-1.

4.1.4 Use of color. Use of color shall be in accordance with MIL-STD-361-1 and MIL-STD-361-3.

4.2 Contents. The instructions shall provide all essential information required by user personnel for performing all required unit, direct support, general support, and overhaul/depot level maintenance on equipment, systems, and weapon systems. Maintenance instructions shall be in sufficient detail to enable the user of the TM to receive, process, clean, service, operate, test, troubleshoot, repair, inspect, and return to an acceptable performance standard all components of the equipment in a minimum of time with the skills, tools, test equipment, and spare parts authorized by the LSA/MAC. Instructions shall cover all items comprising the equipment, such as accessories, components, wiring, and junction boxes. Task procedures shall be presented in the order in which they are to be performed.

4.2.1 Development of text and graphics. The requirements of MIL-STD-361-1 and MIL-STD-361-3 shall be applied. Repetitious

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information shall be avoided except for clarity or where considerable maintenance time would be saved by repeating information to avoid excessive cross-referencing between TMs or within a TM. Refer to MIL-STD-361-1 for specific instructions on referencing. Emphasis shall be placed on the specific steps of the maintenance procedures to be followed, the results that may be expected or desired, and the corrective measures required when the expected results are not obtained. Unless requirement is specifically excluded by the contracting activity, the LSAR output reports shall be the data used to prepare the text, MAC, and repair parts and special tools list (RPSTL) for the TM.

4.2.2 Environmental conditions. As applicable, the maintenance instructions shall include the required environmental control data and information. The instructions shall include information on any special maintenance required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment.

4.2.2.1 Environmental regulation compliance. All maintenance procedures shall be presented in such a manner that current environmental regulations implemented by federal, state, and local governments are complied with. All regulations contained in the Resource Conservation and Recovery Act (PL 89-272, as amended by PL 91-512, PL 93-611, and PL 94-580) shall be complied with.

4.2.3 Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-STD-361-1.

4.2.4 Abbreviations. Abbreviations shall be in accordance with MIL-STD-361-1 and MIL-STD-12.

4.2.5 Preparation of illustrations. The general manner of preparation of charts, tables, schematics, and functional diagrams shall be in accordance with MIL-STD-361-1 and MIL-STD-361-3.

4.2.6 Cable fabrication. When cable assemblies are not supplied but are required for bench test setup, instructions shall be provided for fabricating interconnecting cable assemblies from spares and bulk supplies. The part number, drawing number, and manufacturer or designer for each part of the cable assembly shall be shown, as well as identification of wires, connectors, pin connections, and letters or other designators.

4.2.7 Measurements and tolerances. All numerical data shall be presented in a consistent, clear, easily understood manner. Refer to MIL-STD-361-1.

4.2.8 Safety devices and interlocks. The purpose and location of all safety devices and interlocks shall be given in conjunction with the pertinent procedures.

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5. DETAILED REQUIREMENTS

5.1 General. Maintenance instructions for TMs shall be prepared in accordance with the detailed requirements specified below.

5.1.1 Arrangement of combined TMs. Refer to MIL-M-63008.

5.1.2 TM maintenance levels. TMs shall be prepared in accordance with this document at the maintenance level(s) listed below, as provided by the contracting activity:

Unit maintenance manual; or aviation unit maintenance (AVUM) manual (-20)

Unit and direct support maintenance manual; or aviation unit (AVUM) and aviation intermediate maintenance (AVIM) manual (-23)

Unit, direct support, and general support maintenance manual (-24)

Direct support maintenance manual; or aviation intermediate maintenance manual (AVIM) (-30)

Direct support and general support maintenance manual (-34)

General support maintenance manual (-40)

Overhaul/depot level maintenance manual (-50)
OR
Depot maintenance work requirement (DMWR)

5.2 Preparation of TMs.

5.2.1 Text content. The TMs shall contain all essential information required to maintain and support the equipment for which these TMs are prepared. Sound engineering principles and techniques, LSA, engineering analysis, service experience, performance data on similar items, and all other reliability, availability, maintainability (RAM) data available shall be used in the preparation of specific instructions.

5.2.2 Use of maintenance allocation chart (MAC). Unless requirement is specifically excluded by the contracting activity, TMs containing unit or unit aviation level maintenance shall have a MAC prepared in accordance with 5.18.2.

5.2.3 Combined operator's and maintenance instructions TM. When approved by the contracting activity, a combined operator's and maintenance instructions TM may be prepared. Refer to 5.13 and MIL-M-63008 for arrangement of combined TMs.

5.2.4 Maintenance instructions TM. A separate maintenance instructions TM shall be prepared if a separate operation

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instructions TM is required. (Refer to MIL-STD-361-7.) The maintenance instructions shall provide information necessary for operational checkout, inspection, troubleshooting/fault isolation, repair, replacement, remove/install, and test, in accordance with LSA/MAC and SMR coding, required to keep the equipment in operating condition. The maintenance level(s) of the instructions shall be the level(s) provided by the contracting activity.

5.2.5 Maintenance instructions with repair parts and special tools list (RPSTL). For simple equipment, not requiring a separate RPSTL, a maintenance instructions TM with a RPSTL shall be prepared. For preparation of a RPSTL, refer to MIL-STD-361-11.

5.2.6 Combined TM identification. The inclusion of RPSTLs in a combined maintenance level TM shall be indicated by adding "Including Repair Parts and Special Tools List" to the title.

5.2.7 Combined unit, direct support, and general support level maintenance TM (-24). When specified by the contracting activity, a combined unit, direct support, and general support level maintenance TM may be prepared. The title pages shall identify the combined coverage. Levels of maintenance shall be in separate paragraphs/chapters. Refer to 5.11.

5.2.8 Maintenance and inspection for ammunition. As applicable, the TM shall include inspection and maintenance of ammunition. See 5.16 for instructions.

5.2.8.1 Maintenance and inspection for ammunition segment. The maintenance and inspection for ammunition information shall be prepared as a separate section, chapter, or appendix of the maintenance TM or as a separate TM, as specified by the contracting activity.

5.3 Arrangement of maintenance TMs. The arrangement of the TM shall depend on the equipment and the level of maintenance that the TM shall cover. Refer to 5.13 and MIL-M-63008 for arrangement of combined operator and maintenance TMs.

5.3.1 Numbering of chapters, sections, and appendixes.

a. Chapters. Unless requirement is specifically excluded by the contracting activity, if a chapter does not apply to a particular item of equipment, the remaining chapters shall be renumbered so that no gaps appear in the numbering sequence.

b. Sections. Unless requirement is specifically excluded by the contracting activity, if a section does not apply to a particular item of equipment, the remaining sections shall be renumbered so that no gaps appear in the numbering sequence.

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c. Appendixes. Unless requirement is specifically excluded by the contracting activity, if an appendix does not apply to a particular item of equipment, the remaining appendixes shall be renumbered so that no gaps appear in the numbering sequence. Appendixes shall be numbered in the order in which they are first referenced in the body of the TM and as explained in the How To Use This Manual portion of the TM. (Refer to MIL-STD-361-6.)

5.3.2 Front matter. Front matter shall be prepared in accordance with MIL-STD-361-6 (content) and MIL-STD-361-1 (format).

5.4 Introduction chapter. The introduction chapter shall be prepared in accordance with MIL-STD-361-6. The information and instructions of this chapter shall depend on the complexity of the equipment involved.

5.5 Unit or aviation unit maintenance TM (-20). Unit or aviation unit maintenance as prescribed by the LSA/MAC and SMR-coded items shall be described in detail. The maintenance instructions chapter(s) shall present all required instructions to maintain equipment in an FMC status.

5.5.1 Types of maintenance instructions chapters.

a. Depending on the type and complexity of the equipment, the TM shall contain the following maintenance instructions chapters that apply:

(1) End item maintenance chapter(s) (e.g., M16A1 rifle, M109/M109A1 truck, aircraft).

(2) Component maintenance chapter(s) (e.g., engine, gun carriage, landing gear).

(3) Assembly maintenance chapter(s) (e.g., feeder assembly, control assembly).

(4) Subassembly maintenance chapter(s) (e.g., elevation control assembly, weapon controller grip assembly).

(5) Auxiliary equipment maintenance chapter(s) (i.e., equipment (e.g., MTOE items) not part of, but authorized for use with, the end item).

(6) Software maintenance chapter(s) (e.g., drawings, machine cards).

(7) Ammunition maintenance chapters(s) (e.g., inspection, care, handling).

b. All applicable maintenance chapters shall meet the following requirements:

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(1) One chapter shall contain overall end item maintenance. For simple equipment, such as a basic rifle, only one maintenance instructions chapter shall be included.

(2) Complex end items, where each subsystem is maintained independently of the others, shall have additional maintenance instructions chapters for each subsystem. These chapters would be in addition to the overall end item maintenance instructions chapter.

(3) For equipment composed of two or more integrated end items where each end item has maintenance-significant components, each of these components shall have individual sections within the end item chapter to which they apply. For complex end items, the sections may be paragraphs within a component section. For TMs covering end items without multiple subsystems, each component may have an individual chapter.

(4) Troubleshooting procedures shall be referenced by using either a troubleshooting symptom index or a corrective procedure column in an operational checks table. Instructions shall be in accordance with MIL-STD-361-9.

(5) Maintenance instructions chapters shall contain the sections listed below, as applicable:

Section I (Normally) — Repair Parts; Tools; Special Tools; Test, Measurement and Diagnostic Equipment (TMDE); and Support Equipment.

NOTE

As a rule, this section will appear only in the overall maintenance chapter.

Where Section I is essentially a reference to the applicable RPSTL, the above data shall be included in Chapter 1 and this section shall be retitled as appropriate.

Section II — Service Upon Receipt.

NOTE

If extensive, service upon receipt instructions may be included in a separate chapter; otherwise this section shall appear (only) in the overall end item maintenance chapter.

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Section III	Equipment Check Procedures (for Missile Equipment).
OR	
Section III	Equipment/User Fitting Instructions (for Personal Use Equipment).
Section IV	Preventive Maintenance Checks and Services (PMCS), Lubrication Instructions, and Mandatory Replacement Parts.

NOTE

All PMCS procedures shall normally be located in the overall or introductory maintenance chapter. PMCS procedures data may be provided in a chapter or a separate TM.

Section V	Troubleshooting.
Section VI	Maintenance Procedures.
Section VII	Preparation for Storage or Shipment.

(a) For complex equipment, the sections may be paragraphs within a component section.

(b) Where multiple maintenance chapters are required, Sections I, II, and IV may be included in a single chapter; Sections V and VI may each be a single chapter with its own title; and Section III data may be included in the chapter titled Maintenance Instructions.

5.5.2 Section I - repair parts, special tools, TMDE, and support equipment. This section shall contain the following paragraphs, as applicable.

5.5.2.1 Common tools and equipment.

a. This paragraph shall contain a statement similar to the following: "For authorized common tools and equipment, refer to the Modified Table of Organization and Equipment (MTOE), CTA 50-970, or CTA 8-100, as applicable to your unit."

b. The tool kit (box) assigned to the mechanic (on a 1-per-mechanic-by-MOS basis) shall be identified in this paragraph by nomenclature and supply catalog (SC) number. No tool in the kit shall be further identified. Other tools required for performance of all tasks for the

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maintenance levels covered in the TM shall be identified in the initial setup and shall be referenced to the Tool Identification List appendix. (See figure 1 for example.) "Other tools" includes tools which are part of/components of shop sets authorized to sections/teams; tools authorized by RPSTL and CTA 50-970; special and fabricated tools; and items of TMDE.

c. When specified by the contracting activity, the above information may be included for on-site repair tasks at the general support level of maintenance.

5.5.2.2 Special tools, TMDE, and support equipment. This paragraph shall make reference to the RPSTL and the MAC. When no special tools or equipment are required, it shall be so stated. If tools are to be fabricated, instructions and illustrations shall be provided in accordance with 5.18.6.

5.5.2.3 Repair parts. This paragraph shall contain a statement similar to one of the following:

Repair parts are listed and illustrated in the repair parts and special tools list (TM number of RPSTL).

or

Repair parts are listed and illustrated in Appendix ___ (insert appropriate appendix letter) of this manual.

5.5.3 Section II - service upon receipt.

a. This section shall provide the information required for the user to ensure that the equipment will be adequately inspected, serviced, and operationally tested before it is subjected to its normal everyday use. These procedures shall cover unpacking, processing, and similar actions. Unpacking instructions shall be provided when special design reusable containers are designated for either the end item or for components that are authorized for replacement or when equipment damage could result. Instructions shall be included to report the empty container through supply channels or, if a component is being replaced, to package the unserviceable component in the empty container in the same manner that the new component was packaged.

b. As applicable, this section shall contain procedures for performing visual inspection of ammunition received from the ammunition supply facility. This inspection shall include verification that ammunition received was that requisitioned. Instructions for a condition check of the shipment (pallets, containers, boxes, and legibility of markings) shall be included. Instructions to note the quantity of each lot for recording purposes shall be included. (See figure 2.)

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c. When inspection procedures are in other TMs, this section shall not duplicate that information, but shall reference the applicable TM.

d. For complex equipment (e.g., missiles, electronic communication systems) that requires extensive service upon receipt, this section shall be further subdivided. The following divisions shall be used, as applicable:

(1) Site and shelter requirements.

(2) Service upon receipt of materiel.

(3) Installation instructions.

(4) Preliminary servicing and adjustment of equipment.

(5) Circuit alignment.

(6) Electrostatic discharge (ESD) control standards.

(7) Nuclear hardness survivability requirements.

5.5.3.1 Site and shelter requirements. This paragraph shall contain the information described in the following paragraphs.

5.5.3.1.1 Siting. Siting instructions peculiar to the equipment shall be detailed and illustrated. Applicable operational and maintenance features shall be considered, such as:

a. Location.

b. Proximity to power sources.

c. Effective ranges.

d. Terrain requirements to avoid screening, reflections, ground clutter, and other poor operational conditions due to terrain.

e. Technical requirements.

f. Shelter locations.

g. As applicable, instructions on compensating for adverse siting conditions shall be included.

h. When the equipment being covered contains large components such as towers and antennas that require orientation to a base line during siting, the orientation procedure shall be covered.

i. If the equipment is mobile and the type that is oriented during installation, orientation shall be covered.

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5.5.3.1.2 Shelter requirements. Whenever the equipment will normally be housed in a permanent or semipermanent shelter (other than a military truck, van, or transportable shelter) during use, the TM shall state:

a. Amount of floor, wall, and height space required.

b. A plan for a typical layout.

c. Required weight capacity of the building floor.

d. Dimensions required for installed equipment.

e. Total weights that the floor must support and the area in square feet over which the total weight will be distributed, if several different types of equipment are installed.

f. Environmental conditions (e.g., venting).

g. Power requirements.

h. Unusual requirements specific to equipment, such as air-conditioning.

Architectural and engineering data on beam sizes, lengths, bending moments, and required supports shall not be included.

5.5.3.2 Service upon receipt of materiel. Service upon receipt procedures shall be in a format as shown in figure 2 and shall cover topics in 5.5.3.2.1 through 5.5.3.2.3.

5.5.3.2.1 Unpacking. As a minimum, the following information shall be provided:

a. Any special sequence of action necessary to protect the equipment.

b. If a special design reusable container is involved for either the end item or for components which are authorized for replacement, instructions shall be included to report or reenter the empty container through supply channels or if a component is being replaced, to package the unserviceable component in the empty container in the same manner that the new component was packaged.

c. Step-by-step instructions for unpacking. Illustrations shall be provided as necessary.

d. Listing of all tools, test equipment, supplies, and parts necessary to unpack the equipment.

e. Manhour requirements and total manhours required for unpacking the equipment.

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5.5.3.2.2 Checking unpacked equipment. Any inspection procedures peculiar to the equipment shall be provided. Instructions for a condition check of the shipment (including that of pallets, containers, boxes, and legibility of markings) shall be included. In addition, the following shall be inserted exactly as stated here:

"a. Inspect the equipment for damage incurred during shipment. If the equipment has been damaged, report the damage on SF 361, Transportation Discrepancy Report.

b. Check the equipment against the packing slip to see if the shipment is complete. Report all discrepancies in accordance with applicable service instructions (e.g., for Army instructions, see DA PAM 738-750).

c. Check to see whether the equipment has been modified."

5.5.3.2.3 Processing unpacked equipment. Complete instructions for processing the unpacked equipment (e.g., removing excess lubricant from a new rifle) shall be included, provided they do not conflict with any warranty provisions. As applicable, the following topics shall be discussed:

a. Listing of all tools, test equipment, support equipment, supplies, and parts required to process the equipment.

b. Any special skills required by processing personnel.

c. All caustic, corrosive, and/or toxic material used during processing shall be identified and applicable cautions and warnings given.

d. Instructions on safe disposal of waste products generated during processing actions.

e. Manhour requirements and total manhours required for processing the equipment.

5.5.3.3 Installation instructions. These instructions shall include all information necessary to install the equipment properly, to make the necessary interconnections, and to lubricate, calibrate, and adjust the equipment as required.

5.5.3.3.1 Tools, test equipment, and materials required for installation. The tools, test equipment, and materials required for installation shall be listed. Items listed shall be those available to installers either furnished as part of the equipment, prescribed by the LSA/MAC, or available in the service supply system. When essential, the instructions for the use of these tools shall be illustrated.

5.5.3.3.2 Assembly of equipment. This paragraph shall contain step-by-step instructions for assembling equipment that has been shipped unassembled. If necessary for clarity, these procedures shall be illustrated. When the equipment is to be shelf or rack

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mounted, these instructions shall also describe assembly of the rack, if necessary, and installation of the equipment in the rack.

5.5.3.3.3 Installation instructions.

a. Step-by-step instructions shall be provided for all actions (including placing, mounting, and attaching) for:

(1) Cable and wiring interconnections.

(2) Proper use of special tools.

(3) Other actions required for installation of the equipment covered.

b. Steps in the instructions shall be in the sequence that ensures the most efficient, rapid, and accurate installation possible.

c. Installation instructions shall be illustrated as necessary, with all dimensions that must be maintained in placing, mounting, or attaching indicated on the illustration(s).

d. When initial adjustments can be made efficiently during installation, such adjustments shall be included in the installation procedure.

e. These instructions shall not include connection to outside lines.

f. When the equipment is designed and intended for use in more than one type of installation (e.g., field, fixed station, and mobile), instructions shall be included for each type of installation involved.

g. If performance of any step in the installation instructions requires the assistance of personnel from a higher level of maintenance, this shall be stated in a note similar to that below:

NOTE

The following installation procedure must be made with the assistance of (insert level) maintenance personnel (include MOS if applicable).

h. Installation instructions shall be considered complete only when they include instructions for:

(1) All required installation options (e.g., ESD control requirements).

(2) Accessory items.

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(3) Auxiliary items (those that extend or increase equipment capability).

(4) Grounding of the equipment for both safety and proper operation.

(5) Torque requirements.

(6) Any additional tasks required.

5.5.3.3.4 Cable diagrams. Cable diagrams shall be provided as specified by the contracting activity. If equipment covered is complex, cable diagrams may be included as foldouts or a separate volume/appendix. If a separate volume or appendix is required, a reference to this coverage shall be included.

a. Special instructions and cautions shall be provided for any mating connectors that call for a special procedure either to make the proper connection or to prevent damage to the connector.

b. Except for cables, a wiring diagram fully identifying each wire to be connected, by color code or wire number if applicable, shall be provided. This diagram shall show the location of each pertinent terminal, which shall be identified by number or other marking, if available, or by position if neither is available.

c. All alternate connection patterns required for various modes of operation shall be shown and explained.

d. Only one diagram shall be used to illustrate interconnection patterns which appear more than once within the same equipment.

5.5.3.3.5 Installation of plug-in items. Diagrams showing the location of items that are not installed in the equipment when received shall be included or referenced. Step-by-step instructions shall be provided whenever special techniques or connections are required.

5.5.3.3.6 Special applications. Installation procedures common to all special applications shall be included. Details of installation that are peculiar to the item in which the basic equipment is being installed shall be omitted (e.g., items requiring special treatment when in vehicle or aircraft).

5.5.3.3.7 Van and shelter installations. The following information shall be provided only to the extent required for unit maintenance.

a. Instructions for the removal and replacement of each nonpermanent unit shall be included.

b. When the equipment is permanently installed in vans or shelters, installation instructions shall not be included.

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c. Diagrams and instructions pertaining to electrical and interconnection wiring, exclusive of wiring peculiar to the equipment on which the installation is being made (e.g., headlight, ignition wiring), shall be included.

d. Cable run locations, equipment locations, circuit breaker panels, and other similar details shall be covered.

5.5.3.4 Preliminary servicing, adjustment, and calibration of equipment. These instructions shall be step-by-step for all lubrication, checks, and adjustments to be made on newly installed equipment. Steps or actions that may result in injury to personnel or damage to equipment shall be preceded by appropriate warnings or cautions. Illustrations showing the location of items such as controls, parts, and check points shall be provided or referenced.

a. Instructions shall be provided for making checks and adjustments that must be made before equipment is put into operation. As applicable, these instructions shall include:

(1) Checks for interconnections.

(2) Checks for adequate clearance for rotating or moving devices.

(3) Checks of initial settings of all controls that must be preset before power can be applied.

(4) All other checks needed to determine that power can be applied without damaging the equipment or injuring personnel.

b. After the above checks are made, all other checks required to ensure proper operation of the equipment shall be made. These checks and instructions shall include all necessary items, but they shall be limited to those that are realistic and practical. The following shall be covered, as applicable:

(1) Grounding, including earth ground connections, earth conditioning for conduction, and a check of the grounding circuit for negligible resistance.

(2) Firm seating and connection of all plug-in parts, mating connectors, jacks, and plugs.

(3) Cable and wire harness routing, dressing, and fastening.

(4) Cautions against damaging transistors, diodes, and other electrically sensitive items.

(5) Replacement of all covers, inspection and access doors, and plates.

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(6) Operation of safety interlocks and switches.

(7) Operation of ventilating louvers and intake and exhaust ports.

(8) Operation and content of liquid cooling systems.

(9) Lubricants and corrosion and prevention control (CPC) procedures.

(10) Switch and control settings that are preset at installation (installer's adjustments).

(11) Presetting and adjustment of automatic controls.

(12) Terminal connections.

(13) Required terminal or capacitor strapping.

(14) Preliminary test measurements.

(15) Presetting operator's controls. (Reference the operating instructions, as applicable.)

(16) Normal operating checks. (Reference operating instructions, as applicable.)

(17) After-installation orientation.

(18) Burn-in of parts.

(19) ESD control standards.

(20) After operations, shutdown, checks, and inspections.

(21) Calibration requirements.

5.5.3.5 Circuit alignment. These instructions shall include circuit alignment procedures to be performed.

5.5.3.5.1 External connections. Instructions shall be provided covering all connections to external lines required for each installation option. Connection instructions shall conform to the requirements for installing wiring and cabling interconnections.

5.5.3.5.2 Switch settings, patch panel connections, and internal control settings. Instructions shall be provided for all switch settings, patch panel connections, and internal control settings required for each installation option and mode of operation.

5.5.3.5.3 Alignment procedures. Instructions shall be provided for all alignment procedures, including any variations required for different installation options and modes of operation.

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5.5.4 Section III - equipment check procedures or equipment/user fitting instructions.

a. As applicable, checks for missile system equipment shall be included as follows:

(1) The check procedures shall be prepared in accordance with MIL-M-63043.

(2) Schematic diagrams and supporting functional explanations shall be limited to the data absolutely required to perform the check procedures. Normally, this is limited to block and functional diagrams supported by functional explanations of inputs, outputs, and specific electrical values required to be measured when performing the check procedures.

(3) Reference to the maintenance instructions or the actual instructions shall be included for repairing or replacing all items identified in the check procedures tables. Only those illustrations which are essential for performing the repair procedures shall be included.

b. As applicable, equipment/user fitting instructions for personal use equipment shall be included.

5.5.5 Section IV - preventive maintenance checks and services (PMCS), lubrication instructions, and mandatory replacement parts. The content of PMCS shall be based upon the principles of reliability centered maintenance (RCM) logic and shall include applicable scheduled corrosion inspections. The checks and services shall be in a tabular form using the format shown in figure 3. The PMCS procedures shall be arranged in a logical sequence requiring minimum time and motion on the part of the person(s) performing them and shall be so arranged that there will be a minimum interference between persons performing the checks simultaneously on the same end item. An introduction paragraph shall explain how the procedures are to be used and shall contain an explanation of all columns and codes. The explanation for the Item No. column shall detail how the item numbers are used when recording results of PMCS on DA Form 2404, Equipment Inspection and Maintenance Worksheet. When specified by the contracting activity, the first page following the introduction shall consist of an illustration of the equipment which shall include a routing diagram by which the PMCS will be performed. At the beginning of the PMCS there shall be an initial setup paragraph prepared in accordance with 5.5.7.1.

5.5.5.1 PMCS table.

a. Item no. column. Item numbers shall be assigned to procedures in numerical consecutive order.

b. Interval column. Information in this column shall show the designated interval when each check is to be

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performed. (See figure 3 for example of completed interval column.)

c. Item to be checked or serviced column. The items listed in this column shall be identified in as few words as will clearly identify the item, usually the common name (e.g., bumper, gas can and mounting bracket, front axle, etc.).

d. Procedure column. This column shall contain a brief description of the procedure by which each check is to be performed, as well as any information required to accomplish each check or service, including lubrication, appropriate tolerances, adjustment limits, and instrument gage readings. Whenever practical, essential illustrations shall be integrated with the text/table. If information does not exceed 120 words or 10 lines, it shall be duplicated in the PMCS table, not referenced. Any required reference to more extensive information must be approved by the contracting activity. Whenever replacement or repair is recommended, this column shall reference the applicable maintenance instruction. Lubrication instructions shall be included as required to present all applications and procedures, lubricants and lubrication points. Refer to MIL-STD-361-10 and figures 3 and 4.

e. Not fully mission capable if: column. This column shall contain a brief statement of the condition (e.g., malfunction, shortage) that would cause the covered equipment to be less than fully ready to perform its assigned mission.

5.5.5.2 Mandatory replacement parts. This paragraph shall contain a table listing all items that must be replaced during PMCS, whether they have failed or not. The table shall reflect the interval at which these items must be replaced, whether hardtime maintenance or on-condition maintenance is the determining factor. Refer to figure 5.

5.5.6 Section V - troubleshooting. This section shall contain those checks and corrective actions required to isolate defects which can be corrected by performance of maintenance allocated to the TM user by the MAC. Troubleshooting procedures shall begin with fault/symptom and should lead to single fault isolation; however, if the corrective action indicated for a fault must be performed at a higher maintenance level, the user shall be told to notify the next higher level of maintenance. Refer also to MIL-STD-361-9.

5.5.7 Section VI - maintenance procedures.

5.5.7.1 Initial setup. The initial setup is part of each maintenance paragraph and lists the information required by the user before starting the detailed maintenance procedure. This shows how to prepare the work area, what tools are needed, and other essential information. The format should be arranged to

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present the information clearly, make optimum use of page space, and be consistent throughout the TM. Refer also to figure 1 for an example of an initial setup layout and how it keys to a referenced appendix by item number. Detailed requirements are listed below:

a. Applicable configurations. When the procedure does not apply to all configurations of the equipment, different configurations covered by the same procedure shall be brought to the user's attention. The method for identifying the information that applies to each configuration shall be explained. Omit this portion if the same instructions apply to all configurations.

b. Test equipment. All test equipment required to perform the procedure shall be listed by name and part/model number designation, if this information is not contained in an overall list elsewhere in the TM. If such a list exists, refer to it [e.g., Multimeter (Item 4, App C)] instead of repeating the information throughout the TM.

c. Tools and special tools. The tool kit to be used in maintenance of a particular equipment shall be listed by tool kit number, supply catalog (SC) or TM number if this information is not contained in an overall list elsewhere in the TM. If such a list exists, refer to it [e.g., Fixed open end wrench set (Item 47, App G)] instead of repeating the information throughout the TM.

d. Materials/parts. All expendable items and support materials shall be listed. The item number and appendix which lists these items shall be given. Mandatory replacement parts (i.e., those parts which must be replaced during repair procedures) shall be listed by name (and part number, if any). The number, quantity, or size necessary to complete the task shall be listed, when applicable. When a mandatory replacement parts appendix exists, it shall be referred to.

e. Personnel required. Personnel required to perform a task shall be identified if the task requires more than one. The MOS designation is not necessary, but it may be included in those cases where more than one MOS is required.

f. References. All detailed maintenance procedures or other TMs which will be needed shall be listed here. References may be to a paragraph, page, figure, or TM number, as applicable. Only list references not listed in equipment conditions. They will also be referenced when needed for each task, but this summary allows a technician to select everything needed at the start of the job without checking through the task listing. It also makes selections of frames easy if microfilm storage and retrieval are used. Refer also to MIL-STD-361-1.

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g. Equipment conditions. Any special equipment conditions required before the procedure can be started shall be listed here and cross-referenced to the appropriate paragraph in the text for setting up the condition.

h. Special environmental conditions. Any special environmental conditions (such as ventilation, lighting, or temperature) that are required shall be listed here. The reason that such conditions are needed shall be explained.

5.5.7.2 Maintenance procedures required.

a. This section shall contain maintenance procedures which are the responsibility of the unit maintenance or aviation unit maintenance TM user as authorized by the LSA/MAC and SMR coded items. Headings for procedures/tasks shall be identified by the same functional group titles and codes as those used in the applicable MAC and RPSTL. Procedure format shall be in accordance with MIL-STD-361-1.

b. As applicable, maintenance instructions shall reference or contain all procedures required for care and handling of all hazardous material, e.g., ammunition, radioactive components, or ESD sensitive items, including prevention of deterioration due to rough handling, exposure to adverse weather conditions, or other hazards. Visual inspection and safety criteria shall be provided to determine item serviceability. Instructions shall also contain procedures for disposition of defective ammunition. These procedures shall include use of cleaning materials and paint authorized for use in the specified maintenance operations. Inspection, repair, and packaging shall be described and illustrated as required.

c. Instructions for CPC procedures shall reference applicable CPC publications or, when peculiar to the equipment, shall contain applicable procedures.

d. National stock numbers (NSNs) shall not be used in procedural steps, illustrations, or illustration legends of maintenance TMs. Part numbers shall not be used in procedural steps, illustrations, or illustration legends, except when essential for identification.

e. Aviation maintenance TMs shall reference procedures in TM 55-1500-204-25/1, as applicable.

5.5.7.3 Arrangement of procedures.

a. All applicable maintenance procedures shall be arranged in the order in which they appear in the MAC. Each major item of equipment shall be fully covered before proceeding to the next.

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b. Maintenance procedures (such as inspect and repair) may be combined when appropriate for the item involved.

5.5.7.4 Maintenance. Complete instructions for maintenance shall provide the proper techniques and detailed procedures required. Each maintenance procedure shall provide step-by-step instructions in the order in which the work is most logically accomplished. Any unusual or critical steps shall be described in detail (e.g., handling radioactive materials, specifying ESD control requirements, specifying quality assurance (QA) checks). Illustrations shall be used to improve understanding and clarity. (See figure 6.) Tools shall be illustrated or described only when the tool or procedure is peculiar or abnormal. Maintenance procedures shall be presented in the general order listed below, as applicable:

a. Servicing.

b. Ground handling.

c. Operational check.

d. Inspection - installed items.

e. Removal.

f. Disassembly.

g. Cleaning.

h. Inspection-acceptance/rejection criteria.

i. Nondestructive testing inspection.

j. Repair or replacement.

k. Alignment.

l. Painting.

m. Lubrication.

n. Assembly.

o. Test/inspection.

p. Installation.

q. Adjustment.

r. Radio interference suppression.

s. Placing in service.

t. Testing.

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5.5.7.4.1 Nuclear hardness. If equipment to be operated, maintained, or overhauled has nuclear survivability requirements (i.e., overpressure and burst, thermal radiation, electromagnetic pulses, or transient radiation effects on electronics), applicable cautions shall be incorporated to ensure that hardness of equipment is not degraded during operation and maintenance. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.5.7.4.2 Electrostatic discharge (ESD). This paragraph shall describe the ESD control standards for the protection of electrical and electronic parts, assemblies, and equipment, where applicable. During such maintenance functions/tasks as removal, disassembly, inspection, repair/replace, assembly, testing, and packaging, there shall be identification of all ESD sensitive items by class along with the protective/control measures to be taken. Of primary importance in the identification of these class 1 items requiring stringent protection are class 1 chips on a printed circuit board and the point assembly which requires less stringent protection or none at all, and when the installation of a class 1 item in its next higher assembly changes the ESD requirement. Refer also to MIL-STD-361-1.

5.5.7.5 Servicing.

a. Instructions shall cover replenishment of fuel, oil, hydraulic or other fluids, oxygen/nitrogen/other gases, and tire pressure, plus any other such items/materials (except for lubricants) required for complete servicing of the equipment.

b. Servicing instructions shall be supplemented with a diagram showing locations of regular and emergency servicing points.

c. Reference shall be made to the servicing diagram and to the list of expendable and durable items for identification of fuel, oil, and other material used. All expendable items used in the servicing instructions shall be identified by military/federal standard nomenclature and designation number.

d. The precautions to observe in servicing a particular tank or reservoir (e.g., grounding and prevention of fire hazards) shall be stated clearly.

e. Illustrated instructions shall be included regarding access to any out-of-the-way or unusual places requiring service.

f. Reference shall be made to data in other parts of the text pertinent to servicing, such as tire pressure and capacities of tanks.

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5.5.7.6 Ground handling.

a. Ground handling shall be presented in concise form, giving the following:

(1) Description, instructions, and necessary precautions for ground handling of the equipment, including any information needed in extreme cold, heat, humidity, or dust.

(2) Equipment dimensions and clearances and weights of components.

(3) Instructions for folding and unfolding appropriate parts, such as rotor blades or wings, rudders, and fins.

(4) Description, instructions, and pertinent illustrations as necessary to clarify text.

(5) Description and instructions for operating any ground equipment involved.

b. Arrangement of information shall be consistent with the following order of topics:

(1) Towing. Method of towing, both forward and backward, supplemented with diagrams or illustrations, as necessary, indicating location of towing lugs or rings.

(2) Jacking. Method of lifting the equipment by means of jacks, supplemented with a diagram showing location of jacking points and pads; methods to be used for jacking; and instructions for any precautions (blocking, supporting, storing, or other) that may be necessary during this operation.

(3) Parking. Parking methods utilizing parking brakes, control locks, and chocks, if applicable.

(4) Mooring. Instructions and reference diagram(s) showing location of securing points for mooring the equipment under adverse environmental conditions.

(5) Covering. Instructions for the installation of equipment covers and tiedown devices shall be included, as applicable. Illustrations showing installation and stowage locations of the covers may be included, or a separate illustration may be referenced.

(6) Hoisting. Methods of hoisting the equipment (or equipment sections) by means of hoisting slings, shall be included. Instructions for hoisting shall

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include a diagram showing hoisting points and attachment of slings in accordance with MIL-STD-209.

(7) Sling loading. Methods of sling loading equipment, supplemented with diagrams showing sling points, equipment requirements, and attaching points, will be described in concise detail in accordance with MIL-STD-209. Special precautions to be observed to prevent damage to the equipment or its components shall be stated.

(8) External power. Instructions and necessary precautions for use of external power, such as attaching external power unit to the equipment and applying electrical power, shall be included. Recommended type auxiliary power unit and rating required for starting the equipment shall be stated.

5.5.7.7 Operational check. Instructions for operational checks shall:

a. Include the techniques and methods required to ensure the satisfactory performance of the item.

b. Reference the operator's instructions for preoperational checks for starting, runup, and shutdown procedures as required.

c. When so specified in 5.5.1b(4), include a corrective procedures column to provide a means of referencing troubleshooting procedure.

5.5.7.8 Inspection of installed items. Instructions shall cover inspection of components, assemblies, or parts installed on the equipment. Inspection will be performed with the item in its normally installed position/condition, considering accessibility and visibility of the item being inspected. The purpose of the inspection is to determine if the item is damaged, deteriorated, or incomplete to the extent that it should be replaced/repaired.

5.5.7.9 Removal. Instructions shall be provided in the logical removal sequence prescribed by the FGC, with illustrations as required. (Refer to figure 7.) When applicable, instructions shall be included for checking and recording gear wear patterns, backlash, ESD protective control measures, measurements/tolerances for determining thickness of shims and purpose for shims, and separating and indexing parts for the assembly.

5.5.7.10 Disassembly. Instructions shall cover disassembly of components, assemblies, or subassemblies, as required, to the extent specified by the MAC and SMR coded items.

a. Precision matched or mated components, assemblies, subassemblies, or parts (other than common hardware), including ESD sensitive items, shall be marked, handled,

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and stored to preclude damage and ensure reassembly and installation in their matched positions.

b. These instructions shall not include separation of bonded, press-fitted, soldered, welded, or riveted parts, or the removal of electronic circuitry parts, unless such removal is necessary to clean, inspect, or test separately.

c. Illustrations shall be included for the disassembly. (Refer to figure 8.)

d. Cleaning after disassembly in accordance with 5.5.7.11 shall be included as required.

e. Information shall be furnished pertaining to components which require relubrication after the equipment has been cleaned.

5.5.7.11 Cleaning. Cleaning procedures, methods, special equipment, and materials that are required shall be specified. Instructions shall be provided for corrosion prevention treatment of metal parts after cleaning, as applicable:

a. All materials used in the cleaning and corrosion prevention of equipment, components, or parts shall be referenced and contained in the expendable and durable items list. (See MIL-STD-361-7.)

b. Procedures shall include precautions to avoid damage of components and to prevent the entrance of water or other solvents into electrical components, ducts, or like openings.

c. Warnings and cautions shall be provided whenever incompatible chemicals or cleaning compounds are used or mixed. Identify whether the danger is to personnel or equipment and state the effect (e.g., gases, fumes, caustic, fire).

5.5.7.12 Inspection-acceptance/rejection criteria. Detailed inspection requirements shall be provided to include acceptance/rejection information sufficient to determine that new, repaired, and used components, assemblies, and subassemblies conform to the wear limits, fits, and tolerances established. ESD control protective measures and CPC procedures shall be provided.

5.5.7.13 Nondestructive testing inspection (NDTI).

a. For aircraft, TM 55-1500-335-23 shall be the only NDTI document referenced in the NDTI procedures, and technical provisions of this TM shall be adhered to unless requirement is specifically excluded by the contracting activity.

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b. Individual NDTI procedures shall be specified for each part requiring NDTI. In order to satisfy this requirement, the following shall be included in aircraft TMs:

(1) If penetrant is required, identification of the particular TM 55-1500-335-23 process that is applicable.

(2) If magnetic particle inspection is required, the specific TM 55-1500-335-23 method, the type of magnetization, and amount of current or ampere turns.

c. The reject criteria shall be specified in all cases. This shall be done by means of a blanket statement, individual criteria for a part, or a combination of both.

d. Unless requirement is specifically excluded by the contracting activity, instructions for use of visible dye penetrants shall not be included as part of NDTI instructions. See TM 55-1500-335-23.

e. When several NDTI methods are permitted, the relative order of preference shall be specified.

f. Instructions for removing primer and/or paint shall be included in TMs that require the removal process as part of NDTI procedures. If a part requires a special process, this procedure must be contained within the NDTI procedure for that part.

g. Cleaning requirements prior to, during, and after NDTI shall be specified. If a part has a built-in bearing, then a procedure to protect the bearing must be included in the NDTI procedure.

5.5.7.14 Repair or replacement. Essential instructions shall be provided for repair or replacement to restore an item to a completely serviceable or FMC status.

a. The instructions shall provide for the repair of a part or the substitution of a new or serviceable part in a manner to allow the proper functioning of the equipment.

b. Items authorized to be manufactured or fabricated will be identified by part number, and reference shall be made to the illustrated list of manufactured items appendix for criteria and bulk materials/parts required to make the item. (See 5.18.6.)

c. ESD standards, ESD sensitive items and protective/control measures to be taken, and CPC procedures shall be identified.

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5.5.7.15 Alignment. Detailed instructions shall be provided for alignment procedures to adjust specified variable elements of an item to bring about optimum or desired performance.

5.5.7.16 Painting. Instructions shall be referenced or provided for required painting, refinishing, and marking of assembled components, assemblies, subassemblies, or end item as applicable. Reference may be made to TM 55-1500-345-23, TM 55-1500-204-25/1, SB 11-573, AR 746-1, TB 43-0209, TB 43-0118, TM 43-0139, or others as appropriate.

5.5.7.17 Lubrication. When applicable and approved by the contracting activity, lubrication instructions shall be included early in the maintenance actions, i.e., within PMCS or under Servicing. Pertinent mandatory lubrication instructions, CPC procedures, and pertinent general lubrication instructions not contained elsewhere shall be provided here.

a. If lubrication instructions are provided here, and not in a separate publication, they shall include all mandatory statements from MIL-STD-361-10, along with a statement to the effect that the instructions are mandatory.

b. Lubricant types and abbreviations for flight vehicles and components shall be as specified in MIL-HDBK-275; and for ground equipment systems, lubricants, functional fluids, preservatives, and specialty products shall be in accordance with MIL-HDBK-113. Required abbreviations not covered in the appropriate handbook shall be furnished by the contracting activity.

5.5.7.18 Assembly. Step-by-step procedures shall be included for assembling items disassembled or removed that make up the components, assemblies, or subassemblies. (See figure 9.)

a. Precision-matched or mated parts marked during disassembly shall be noted.

b. Instructions shall be included for checking and recording gear wear patterns, backlash, shimming requirements, and the indexing of parts to ensure proper alignment during assembly, as applicable. The purpose of shims shall be given, e.g., adjust backlash, prevent metallurgical reaction, etc.

c. Torque requirements, values, and sequences shall be indicated, as applicable. Only critical torques shall be indicated in task steps. All noncritical torques will be covered by the torque limits appendix and a reference thereto. Torque values shall be given for all structural attaching hardware, fluid couplings (fuel, oil, hydraulic, pneumatic, etc.), and connections, as applicable. Torque values shall include torque correction factors when crowfoot extensions, thread lubricants, and cadmium-plated screws/nuts are used. Torque values identified in the tasks must reflect torque wrenches authorized to personnel

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targeted to perform tasks. Upon completion of torque action, instruction on use of an orientation mark (striping) shall be given.

d. Equipment that requires calibration after assembly shall be indicated, and reference shall be made to the publication containing the applicable calibration procedure.

e. Instructions that are not step-by-step, such as "reverse the disassembly procedure," shall not be used. Curtailed text may be used when supported by illustrations.

f. ESD standards, ESD sensitive items and protective/control measures to be taken, and CPC procedures shall be identified.

5.5.7.19 Test/inspection. Procedures shall be provided for testing and inspection during or after assembly to ensure proper assembly of the item. Correct method of testing; procedures for making tolerance checks; and procedures for inspection of distance measurements (e.g., clearance, end play, backlash) shall be provided. Measurement criteria and tolerances shall reflect the TMDE available to the user.

5.5.7.20 Installation. Necessary steps and procedures for installation of the item shall be provided.

a. Inspection procedures for checking alignment and adjustment of the item during the installation sequence shall be provided. These instructions shall include a statement that adjustment, servicing, testing, and/or an operational check is required, as applicable.

b. Instructions that are not step-by-step, such as "reverse the removal procedure," shall not be used.

c. Peculiar requirements for lockwiring, installing cotter pins, and similar operations shall be included with applicable references to the expendable and durable items list. (See MIL-STD-361-7.)

d. Shelf-life items, mandatory replacement parts, etc., shall be included.

e. Pressurizing and charging with gas shall be described along with all safety requirements.

5.5.7.21 Adjustment. This paragraph shall contain all adjustment instructions that may be required before operating the part, system, or end item.

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5.5.7.22 Radio interference suppression.

a. Primary components in the suppression system and replacement of these primary components shall be described and illustrated.

b. Secondary components shall be referenced to pertinent maintenance procedures containing removal and installation instructions.

c. Testing radio interference suppression components shall be described.

5.5.7.23 Placing in service. Instructions not previously covered that may be required for an assembly, component, or end item shall be provided. Instructions such as removal of an item from storage and preparation for installation on an end item shall be provided. Final servicing checks, calibration, leak checks, charging, pressurizing, and operational checks shall be included, as applicable.

5.5.7.24 Testing. Instructions shall be provided to test the performance of components, assemblies, and subassemblies prior to installation in the end item.

5.5.8 Section VII - preparation for storage or shipment. This section shall include the following special instructions, as required:

a. Security procedures (See AR 190-11 and AR 190-13.) and special transportation requirements for sensitive items (security, terrorists, etc.).

b. Special preservation, packaging, packing, marking, ESD protective/control measures, and shipping instructions, including use of special design reusable containers.

c. Special use of corrosion-preventive compounds, moisture barriers, and desiccant materials.

d. Directions or reference to instructions for applying special identifying, shipping, and precautionary markings to shipping containers; to include security classification, special temperature requirements, and shelf life.

e. Instructions for placing equipment in, and for removing it from, administrative storage will be provided by the contracting activity in accordance with AR 750-1 and MIL-STD-361-6.

f. Proper handling, blocking, and bracing of basic load ammunition when being transported in trucks and other tactical vehicles.

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g. Basic load storage, quantity-distance, class, and storage compatibility groupings, storage temperatures, stacking limits, and other pertinent storage requirements (for conventional and chemical ammunition TMs only).

h. As applicable, pre-embarkation inspection of materiel in units alerted for overseas movement information shall be included as provided by the contracting activity.

i. Equipment systems storage such as aircraft and missiles. (See 5.9.)

j. Special transportation requirements for sensitive items (security, terrorists, etc.).

5.5.8.1 Shipment of aircraft. Data for shipment of aircraft shall be covered in separate -S series manuals prepared in accordance with MIL-M-63005.

5.5.8.2 Aviation ground support equipment TMs. Aviation ground support equipment TMs shall reference TM 55-1500-204-25/1 for general technical information on preparation for storage or shipment.

5.6 Direct support maintenance or AVIM TM (-30) (other than aircraft). This TM shall describe in detail the direct support maintenance or aviation intermediate maintenance prescribed by the LSA/MAC and SMR coded items. The actual number of chapters in a TM depends on the maintenance concept and complexity of the equipment. (Refer to MIL-M-63008.) Required theory of operation shall be included in the applicable maintenance chapter.

5.6.1 Front matter. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.6.2 Chapter 1 - introduction. Refer to MIL-STD-361-6.

5.6.2.1 Section I - general information. In addition to the requirements of MIL-STD-361-6, add the following requirements.

5.6.2.1.1 Maintenance forms, records, and reports. Where applicable, add reference to SB 742-1, Ammunition Surveillance Procedures, and AR 700-22, Worldwide Ammunition Reporting System (WARS).

5.6.2.1.2 Calibration. Equipment requiring calibration shall be indicated and reference shall be made to the publication containing the applicable calibration procedure.

5.6.2.2 Section II - equipment description and data. This section shall include a reference to the equipment data in the operator's TM plus any data required by direct support or aviation intermediate personnel but not included in the operator's, unit, or AVUM TMs. Reference may be made to material in other publications, provided they are available at direct support or aviation intermediate maintenance levels. Information will not be duplicated except for emphasis or clarity.

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5.6.3 Inspection and test of conventional and chemical ammunition or components containing radioactive materials chapter. As specified by the contracting activity, this chapter shall reference separate ammunition TMs or radioactive materials procedures, or shall consist of the following sections.

5.6.3.1 Section I - general. This section shall contain a statement to the effect that inspection criteria are provided to assure that maintenance performed will restore the items to an acceptable quality level. It shall also reference regulations and technical publications relating to policy responsibility and procedures applicable to ammunition stockpile reliability, ammunition surveillance, radioactive materials procedures, and quality evaluation programs. The types of inspections contained in this section shall, at a minimum, include a premaintenance inspection to be conducted during unpacking, in-process inspections, and final acceptance inspection.

5.6.3.2 Section II - classification of materiel defects. This section shall describe the inspection methods or techniques used to detect defective components/end items being processed. It shall also include "Classification of Materiel Defects" tables for ammunition components and packaging and packing material. The tabulated data shall include a listing of components, categories of defects (minor, major, critical) attributable to each component, a listing of defects by category, a reference to the paragraph of the TM containing corrective action, the inspection methods used to determine if corrective action was accomplished, and the acceptable quality level established for each defect.

5.6.3.3 Section III - visual inspection. This section shall contain visual inspection criteria for the packing of the items in conformance with the inspection criteria noted in 5.6.3.1 above.

5.6.3.4 Section IV - function testing. This section shall contain detailed procedures and criteria for function testing. Diagrams and instructions for fabrication of test fixtures shall be included when required. Where ammunition is required for function testing weapons, it shall be identified in this section by Department of Defense Ammunition Code (DODAC), NSN, and nomenclature, to include dummy rounds.

5.6.3.5 Section V - evaluation of defects. This section shall contain a paragraph covering evaluation of item defects and a paragraph covering evaluation of packing defects. Both paragraphs shall contain guidelines to be used in establishing a uniform system of examination for deterioration or damage. Definitions of minor, major, and critical defects shall be included. References to the lower maintenance levels shall be included when appropriate.

5.6.3.6 Section VI - classification of defects. This section shall contain a classification of defects (i.e., minor, major, or

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critical) for both functioning and nonfunctioning categories. The criteria shall conform to the publications noted in 5.6.3.2.

5.6.3.7 Section VII - disposition of lots.

a. For missiles, the contracting activity shall provide directions relating to disposition of lots.

b. For other than missiles, this section shall contain the following paragraphs verbatim:

"Each lot of materiel shall be inspected and screened 100 percent if one critical nonfunctioning defect is observed. If a critical functioning defect occurs, save remaining pieces and components and suspend the lot from local issue and use. Submit malfunction reports as prescribed in AR 75-1. Disposition instructions will be furnished by the US Army Materiel Command."

"A 'lot' of materiel is acceptable for issue and use if the acceptable criteria as indicated in (insert applicable table number) are met."

"For conventional and chemical ammunition, report all lots of materiel rejected under the applicable serviceability table to: Commander, U.S. Army Armament, Munitions, and Chemical Command, ATTN: AMSMC-DSM, Rock Island, IL 61299-6000, for disposition instructions. Include a statement describing the capability and workload situation of your organization as to whether you are capable of reworking/demilitarizing the item."

5.6.4 Maintenance instructions chapter(s). See 5.5 through 5.5.8 for Sections I through VII and adjust for direct support or aviation intermediate maintenance.

a. Provide instructions for missile system equipment units under test in accordance with MIL-M-63044.

b. For ammunition, procedures shall prescribe the responsibilities for performing such operations as care and preservation of items and stocking authorized replacement components; packing material and consumable supplies; and recovering items such as cardboard inner boxes and wooden boxes needed for maintenance of stocks. Reference shall be made to the RPSTL for replacement parts and any special tools required, to the expendable and durable items list appendix for the list of expendable materials, and to the ammunition marking appendix for appropriate marking information, as applicable.

c. In cases where an assembly (such as truck engine) is part of a larger end item, any direct support maintenance of the assembly installed (e.g., engine in truck) will be in the end item TM. Direct support maintenance of the assembly removed from its end item will be covered in the

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assembly's direct support TM. Differences in configuration for end item application shall be included.

5.6.4.1 Section VIII (or as applicable) - pre-embarkation inspection of materiel in units alerted for overseas movement. When applicable, the information for this section shall be included as provided by the contracting activity.

5.6.5 Appendixes. Refer to 5.18.

5.7 General support maintenance TM (-40). This TM shall describe in detail the general support maintenance prescribed by the LSA/MAC and SMR-coded items. The actual number of maintenance chapters in a TM depends on the maintenance concept and complexity of the equipment. (Refer to MIL-M-63008.) Required theory of operation shall be included in the applicable maintenance chapter.

5.7.1 Front matter. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.7.2 Chapter 1 - introduction. Refer to MIL-STD-361-6.

5.7.2.1 Section I - general information. Refer to 5.6.2.1.

5.7.2.2 Section II - equipment description and data. This section shall include a paragraph referencing the equipment data in the operator's TM. Any data required by the general support level but not included in the operator's, unit, or direct support TMs shall be included. References may be made to material in other publications, provided they are available at the general support maintenance level. Information will not be duplicated except for emphasis or clarity.

5.7.3 Inspection and test of conventional and chemical ammunition or components containing radioactive materials chapter. See 5.6.3 and adjust to general support maintenance.

5.7.4 Maintenance instructions chapter(s). See 5.5 through 5.5.8 for Sections I through VII and adjust for general support maintenance.

a. Provide instructions for missile system equipment units under test in accordance with MIL-M-63044.

b. For ammunition, procedures shall prescribe the responsibilities for performing such operations as care and preservation of items and stocking authorized replacement components; packing material and consumable supplies; and recovering items such as cardboard inner boxes and wooden boxes needed for maintenance of stocks. Reference shall be made to the RPSTL for replacement parts and any special tools required, to the expendable and durable items list appendix for the list of expendable materials, and to the ammunition marking appendix for appropriate marking information, as applicable.

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c. In cases where an assembly (such as a truck engine) is part of a larger end item, any general support maintenance of the assembly installed (e.g., engine in truck) will be in the end item TM. General support maintenance of the assembly removed from its end item will be covered in the assembly general support maintenance TM.

5.7.4.1 Section VIII (or as applicable) - pre-embarkation inspection of materiel in units alerted for overseas movement. When applicable, information for this section shall be included as provided by the contracting activity.

5.7.5 Appendixes. Refer to 5.18.

5.8 Unit and direct support maintenance or AVUM and AVIM maintenance TMs (-23) (other than aircraft). (For aircraft, see 5.9.) Describe in detail the unit and direct support maintenance or aviation unit and intermediate maintenance prescribed by the LSA/MAC and SMR coded items. The actual number of chapters in a TM depends on the maintenance concept and complexity of the equipment. (See MIL-M-63008.) Required theory of operation for direct support maintenance or aviation intermediate maintenance only, shall be included in the applicable maintenance chapter. Levels of maintenance shall be separated by chapter.

5.8.1 Front matter. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.8.2 Chapter 1 - introduction. Refer to MIL-STD-361-6.

5.8.2.1 Section I - general information. Refer to 5.6.2.1.

5.8.2.2 Section II - equipment description and data. This section shall include a paragraph referencing the equipment data in the operator's TM. Any data required by unit and direct support maintenance or aviation unit and intermediate maintenance but not included in the operator's TM shall be included. References may be made to material in other publications, provided they are available at the maintenance levels covered. Information will not be duplicated except for emphasis or clarity.

5.8.3 Inspection and test of conventional and chemical ammunition or components containing radioactive materials chapter. See 5.6.3.

5.8.4 Maintenance instructions chapter(s). See 5.5 through 5.5.8 for Sections I through VII and adjust to include direct support or aviation intermediate maintenance.

a. Provide instructions for missile system equipment units under test in accordance with MIL-M-63044.

b. For ammunition, procedures shall prescribe the responsibilities for performing such operations as care and preservation of items and stocking authorized replacement components; packing material and consumable

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supplies; and recovering items such as cardboard inner boxes and wooden boxes needed for maintenance of stocks. Reference shall be made to the RPSTL for replacement parts and any special tools required, to the expendable and durable items list appendix for the list of expendable materials, and to the ammunition marking appendix for appropriate marking information, as applicable.

c. See 5.6.4c and adjust as necessary.

5.8.4.1 Section VIII (or as applicable) - pre-embarkation inspection of materiel in units alerted for overseas movement. When applicable, the information for this section shall be included as provided by the contracting activity.

5.8.5 Appendixes. Refer to 5.18.

5.9 AVUM and AVIM TMs (-23) (aircraft). These TMs shall describe in detail the scope of maintenance prescribed by the LSA/MAC and SMR coded items. The actual number of maintenance chapters in the TM will depend on the maintenance concept and complexity of equipment. (See MIL-M-63008.) Levels of maintenance shall be separated by chapter.

a. For AVUM TMs, include requirements in 5.9.1 through 5.9.4.4.3.

b. For AVUM and AVIM combined TMs, include all the requirements of the AVUM TM, plus all intermediate maintenance. When combined maintenance data pertain only to aviation intermediate maintenance, the paragraph or instruction title shall be suffixed by the acronym AVIM (e.g., Removal of governor (AVIM)).

5.9.1 Front matter. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.9.1.1 Warning summary. A warning summary to include first aid data shall be included and shall be in accordance with MIL-STD-361-1. Definitions of warnings, cautions, and notes shall not appear on these pages.

5.9.1.2 List of effective pages. There shall be no list of effective pages included.

5.9.2 Chapter 1 - introduction. In addition to the requirements of MIL-STD-361-6, add the information contained in the following subparagraphs.

5.9.2.1 Section IV - servicing. This section shall contain instructions for replenishment of fuel; oil; hydraulic and other fluids; oxygen, nitrogen, and other gases; tire pressures, and all other such items involved in completely servicing the aircraft. Information contained herein shall not duplicate lubrication information found in Section V.

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a. Reference shall be made to the servicing diagram and to the expendable and durable items list for identification of fuel, oil, and other materials used. The precautions to observe in servicing a particular tank or reservoir, such as grounding and prevention of fire hazards, shall be stated clearly, and instructions shall be included regarding access to any out-of-the-way or unusual places requiring service. Servicing instructions shall be supplemented with a diagram showing locations of regular and emergency servicing points. Items located on each side of the aircraft which require servicing will be illustrated and identified in the legend as right and left side. NO STEP areas on walkways leading to any tank shall be indicated, and necessary precautions included.

b. Reference shall be made to graphs or data in other parts of the text pertinent to servicing, such as tire pressure versus gross takeoff weight, and capacities of tanks.

5.9.2.1.1 Cleaning. Detailed instructions shall be furnished for cleaning and washing the entire aircraft, together with precautions to avoid damage to microswitches and electrical connections and to prevent the entrance of water or other solvents into cabin ducts and other such openings. Instructions shall be included on the removal of the battery, relief tube, power plant, and armament exhaust deposits, or other items or material as necessary. Information shall also be furnished pertaining to components which require relubrication after the aircraft has been washed or steam cleaned.

5.9.2.2 Section V - lubrication. This section shall contain the following:

5.9.2.2.1 Lubrication instructions. Lubrication schedules shall be prepared as required to present all applications and procedures, lubricants, and lubrication points to completely lubricate aircraft.

5.9.2.2.2 Lubrication charts. Lubrication charts shall consist of a main drawing prepared as a 3-dimensional diagram, and such enlarged or detailed views as are considered necessary to identify items which otherwise should be obscured. See figure 10. They shall show all lubrication requirements for all parts of the aircraft requiring periodic lubrication, other than those lubricated by the main engine oil system. The charts shall also indicate type of lubricant, method of application, and frequency. This information shall be presented through the use of a standard symbol. Use of black silhouette figures representing a likeness of the tool used in the application (oil can, grease gun, brush, or hand) shall be the accepted means of presenting application methods on the lubrication chart. Approved abbreviations, as specified in MIL-HDBK-275, shall be used to present lubricant types. In the event a lubricant does not have an abbreviation listed in MIL-HDBK-275, the abbreviation will be provided by the contracting activity. Assigned application symbols, type abbreviations, and frequency shall be placed within the standard

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lubrication symbols. A key shall also be included on the chart to define each application symbol and lubricant abbreviation used. Consecutively numbered notes may be used to specify requirements other than normal. A statement "SEE NOTE 1" shall be placed adjacent to the symbol, as appropriate.

5.9.2.3 Section VI - handling, jacking, mooring, hoisting, and sling loading. Paragraphs in this section shall include any instructions required for blocking and supporting the aircraft during performance of the operation or procedure involved. This section shall contain the following requirements, as applicable.

5.9.2.3.1 Ground handling. This paragraph(s) shall contain instructions and necessary precautions for ground handling the aircraft. This shall include instructions for handling the aircraft in extreme cold, extreme heat, high humidity, and dusty or other unusual or extreme conditions. Aircraft dimensions and clearances, instructions for folding and unfolding rotor blades, and instructions for operation of any ground handling equipment involved shall be included. Illustrations shall be used to the extent necessary to clarify the text.

5.9.2.3.2 Towing. This paragraph(s) shall describe methods of towing backward and forward. If necessary for clarity, a diagram or illustration indicating location of towing lugs or rings shall be included. Limitations and cautions shall be included.

5.9.2.3.3 Jacking. This paragraph(s) shall describe methods of lifting the aircraft by means of jacks. A diagram showing location of jacking points, pads, and methods used for jacking shall be included as necessary. Instructions for shoring or other pertinent precautions shall also be included where applicable.

5.9.2.3.4 Parking. This paragraph(s) shall describe parking methods, including the use of parking brakes, control locks, and chocks, where applicable.

5.9.2.3.5 Mooring instructions. This paragraph(s) shall provide instructions and reference diagrams showing location of securing points for mooring the aircraft under adverse environmental conditions.

5.9.2.3.6 Aircraft covers. This paragraph(s) shall contain instructions for installation of aircraft covers and rotor tiedown devices. Location of installed covers and their stowage locations shall be illustrated, either on illustrations in this portion or elsewhere in the TM.

5.9.2.3.7 Hoisting. This paragraph(s) shall describe methods used with hoisting slings to lift aircraft or aircraft sections. A diagram showing hoisting points and illustrating how to attach slings shall be included.

5.9.2.3.8 Sling loading. Methods of sling loading aircraft, supplemented with diagrams showing sling points, equipment

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requirements, and attaching points, shall be described in detail. Special precautions to prevent damage to the aircraft or its components shall be included.

5.9.2.3.9 Application of external power. Instructions for application of external power, such as attaching external power unit to the aircraft and applying electrical power, shall be included. Precautions, if any, shall be described. Recommended type of auxiliary power unit and rating required for starting aircraft shall be stated.

5.9.2.4 Section VII - preventive maintenance inspections.

5.9.2.4.1 General information and introduction. The following paragraph shall be included:

"General information. This section contains complete requirements for special inspections, overhaul and retirement schedule, and standards of serviceability applicable to the aircraft. The inspections prescribed in this section shall be accomplished at specified periods by AVUM activities, with the assistance of AVIM activities when required. Complete Daily, Intermediate, Periodic, or Phased inspections are contained in the (insert applicable aircraft inspection checklist TM)."

5.9.2.4.2 Standards of serviceability. The following paragraph shall be included:

"Standards of serviceability to be utilized in the day-to-day inspection and maintenance of the aircraft can be found as fits, tolerances, wear limits, and specifications in the aircraft maintenance manuals. Standards of serviceability for transfer to aircraft are contained in TM 1-1500-328-25."

5.9.2.4.3 Special inspections.

a. Definition and general information. The following shall be included: "This section supplements scheduled inspections as outlined in the applicable aircraft inspection checklists. This section also includes inspection of items which are required to be inspected at intervals not compatible with airframe operating time or airframe inspection intervals. Refer to DA PAM 738-751 for applicable forms, records, and worksheets required for these inspection intervals. Typical examples of this type of inspection are:

(1) Inspections which are solely contingent upon specific conditions or incidents that occur (e.g., hard landings, overspeed, or sudden stoppage), wherein immediate inspection is required to ensure safe flight.

(2) Inspection of components or airframe on a calendar basis: e.g., first aid kits, weight and balance check, aircraft inventory."

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b. Requirements. The requirements of this section shall include items which qualify under the criteria for special inspections, e.g., hard landings, sudden stoppage, overspeed. These requirements shall be grouped under area headings only and shall be inserted in a columnar listing on the inspection checksheet format, in such a manner as to permit local reproduction of the entire section.

5.9.2.5 Section VIII - overhaul and retirement schedule.

5.9.2.5.1 Introduction. The introduction shall contain the following statement:

"Overhaul and retirement schedule. This section lists units of operating equipment that are to be overhauled or retired at the period specified. Unless otherwise specified in TM 1-1500-328-25, removal of equipment for overhaul may be accomplished at the inspection nearest the time when overhaul is due."

5.9.3 Maintenance instructions chapter(s). The chapter(s) shall describe in detail the maintenance allocated by the MAC. Chapters shall be arranged to align with the MAC.

5.9.3.1 Section ( ) - troubleshooting. Unless requirement is specifically excluded by the contracting activity, this section shall be included along with adequate theory of operations information to support the troubleshooting procedures. When data are extensive, and when specified by the contracting activity, troubleshooting may be published as a separate (-T) manual. Troubleshooting information published as a separate manual shall comply with all requirements of this paragraph.

5.9.3.2 Maintenance procedures. Refer to 5.5.7.

5.9.4 Appendixes. In addition to the requirements of 5.18, the following shall be included.

5.9.4.1 Aircraft inventory master guide appendix. This appendix shall include sufficient information on standard inventory procedures to allow determination of inventoriable items of installed and loose equipment authorized and required by the specific aircraft in performance of its mission. A sample presentation of inventoriable items is shown in figure 11. The general arrangement of the appendix shall be as follows:

5.9.4.1.1 Introduction. This section shall include a short explanation of scope and purpose of the appendix. Information pertaining to necessary steps to ensure the list is accurate, exact, and complete (e.g., research of authorized changes (MWOs, additions/deletions for special mission requirements)) shall be included and the introduction shall refer to DA PAM 738-751 for applicable forms and records.

5.9.4.1.2 Security. This paragraph shall state that aircraft inventory records should be unclassified but that any

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classification of the contents, if necessary, should be in accordance with the existing security regulation.

5.9.4.1.3 Inventoriable items. The selection of inventoriable items is without regard to the agency, governmental or contractual, furnishing the items.

a. Items to be listed are:

(1) Items essential to the execution of the designated mission of the aircraft, such as electronic, photographic, armament, special mission instruments, and safety and comfort equipment.

(2) Loose equipment delivered with the aircraft and items subject to pilferage or readily converted to personal use.

(3) Modification kits which are issued or distributed to using organizations for installation and which are not immediately placed in work. These shall be recorded on the affected aircraft's DA Form 2408-17, Aircraft Inventory Record, and identified as loose equipment until modification is completed.

(4) Equipment required for operation in a specific environment.

b. Items to be excluded are:

(1) Nonaccountable items coded as expendable in the applicable stock lists.

(2) Personal issue or items furnished on unit allowance or other authority.

(3) Items or components considered as basic or integral parts of the airframe or basic aircraft, such as engines, propellers, wheels, and standard instruments.

(4) Equipment publications, checklists, and aircraft forms.

5.9.4.1.4 Periods of inventory. This portion shall include coverage of the following information and shall include a statement similar to the following:

Inventoriable items shall be checked against the Aircraft Inventory Record, DA Form 2408-17, at the following periods:

a. Upon receipt.

b. Prior to transfer of the aircraft to another organization.

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c. Upon placing aircraft in storage and upon removal from storage. Aircraft need not be inventoried while in storage.

d. Twelve months after last inventory.

5.9.4.2 Storage of aircraft appendix.

5.9.4.2.1 Section I - general information. This portion shall include a statement similar to the following at the beginning of the text:

Components involved in an accident. Any component removed for reason of accident shall not be preserved, but shall be shipped in the same condition it was in after the accident.

5.9.4.2.1.1 Categories of storage.

a. The categories of storage and the timeframes involved shall be listed as follows:

(1) Flyable storage - no time limit.

(2) Short term (administrative storage) - 1 to 45 days.

(3) Intermediate storage - 46 to 180 days.

b. This paragraph shall include a general discussion of each category of aircraft storage, to include considerations for selection of the appropriate category (e.g., ground operation, motoring of engines, and other required maintenance for which personnel and materials are needed) and steps to be taken for care of the aircraft during exceptionally wet weather.

5.9.4.2.1.2 Sections II, III, and IV - storage procedures. Each of these sections shall include all essential information and procedures for preparing the complete aircraft for storage and removal from storage, excluding any information on when or why the aircraft are stored. Each section shall make reference to inspection documents and inspection procedures to be conducted prior to, during, and after storage. Sections shall be numbered as follows:

Section II - Flyable storage.

Section III - Short term storage.

Section IV - Intermediate storage.

5.9.4.3 Wiring diagrams appendix. This appendix shall present a description of wiring provisions contained in the aircraft, including all systems or equipment which can be installed or removed later (e.g., mission-related systems/equipment). A statement shall be included that explains that wiring diagrams

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and essential wiring information are provided for all electrical and electronic systems and circuits.

5.9.4.3.1 Wiring data. Subparagraphs shall be used as necessary to include any critical wire and cable data. Applicability of diagrams shall be explained in relation to aircraft configuration. The following information shall be provided:

a. Wire identification. Identification of wires by number shall be explained. A list of circuit designators and a wire identification diagram shall be included.

b. Abbreviations. A statement shall be included that abbreviations are in accordance with MIL-STD-12 and AR 310-50, except when the abbreviation stands for a marking actually found in the aircraft.

c. Wiring diagrams. Wiring diagrams for all electrical and electronic systems and circuits shall be included.

NOTE

Additional appendixes, including the following, shall be required when specified by the contracting activity. When an appendix does not apply to an item of equipment, the remaining appendixes shall be relettered so that no gaps appear in the lettering sequence. Letters I and O shall not be used.

5.9.4.4 Weight and balance (AVIM) appendix. This appendix shall refer to TM 55-1500-342-23 for general information and shall include any additional information peculiar to the aircraft which is not otherwise available to the user. The weight and balance appendix shall include three sections.

5.9.4.4.1 Section I - general information. This section shall include a statement such as the following:

This appendix contains brief instructive information for use with the forms and charts which provide for continuous control of weight and balance of the aircraft. The data to be inserted on the charts and forms are applicable only to the individual aircraft, the serial number of which appears on the various forms and charts. The weight and balance records are to remain with the aircraft in accordance with existing directives. The charts and forms referred to herein may differ in nomenclature and arrangement from those shown in previously published copies, since these charts are revised from time to time; however, the general principles of use will not change.

5.9.4.4.2 Section II - instructions for use of the forms and charts. This portion shall refer to TM 55-1500-342-23 and to the aircraft files for preparation of Chart E.

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5.9.4.4.3 Section III - weighing instructions. This section shall include a reference to AR 95-16 for periodic aircraft weighing requirements.

a. Preliminary weighing instructions. This portion shall include any general and/or specific preliminary weighing instructions not covered in TM 55-1500-342-23 and appropriate for the particular aircraft covered in the TM.

b. Weighing equipment. Additional instruction for use of the weighing equipment in the individual aircraft, if not contained in TM 55-1500-342-23, shall be covered in this portion.

5.10 AVUM and AVIM TMs for troubleshooting (-T). The -T manual shall be used in lieu of including the troubleshooting instructions in the -23 level maintenance TM when the troubleshooting coverage exceeds 1000 pages. The -T manual shall describe in detail the AVUM and AVIM troubleshooting tasks as allowed by the MAC. In addition to complying with paragraphs 5.5.1b(4) and 5.5.6 of this standard, the -T manual shall comply with the provisions of MIL-STD-361-9.

5.11 Unit, direct support, and general support maintenance TM (-24). This TM shall describe in detail the unit, direct, and general support maintenance prescribed by the LSA/MAC and SMR coded items. The actual number of chapters in a TM depends on the maintenance concept and complexity of the equipment. (Refer to MIL-M-63008.) Required theory of operation for direct and general support maintenance only shall be included in the applicable maintenance chapter. Levels of maintenance shall be separated by chapter.

5.11.1 Front matter. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.11.2 Chapter 1 - introduction. Refer to MIL-STD-361-6.

5.11.2.1 Section I - general information. Refer to 5.6.2.1.

5.11.2.2 Section II - equipment description and data. This section shall include a paragraph referencing the equipment data in the operator's manual. Any data required by the unit, direct support, or general support levels, but not included in the operator's TM, shall be included. Reference may be made to material in other publications, provided they are available at the maintenance levels covered. Information will not be duplicated except for emphasis or clarity.

5.11.3 Inspection and test of conventional and chemical ammunition or components containing radioactive materials chapter. See 5.6.3.

5.11.4 Maintenance instructions chapter(s). See 5.5 through 5.5.8 for Sections I through VII and adjust to include direct support and general support maintenance.

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a. Provide instructions for missile system equipment units under test in accordance with MIL-M-63044.

b. For ammunition, procedures shall prescribe the responsibilities for performing such operations as care and preservation of items and stocking authorized replacement components; packing material and consumable supplies; and recovering items such as cardboard inner boxes and wooden boxes needed for maintenance of stocks. Reference shall be made to the RPSTL for replacement parts and any special tools required, to the expendable and durable items list appendix for the list of expendable materials, and to the ammunition marking appendix for appropriate marking information, as applicable.

c. See 5.6.4c and adjust as necessary.

5.11.4.1 Section VIII (or as applicable) - pre-embarkation inspection of materiel in units alerted for overseas movement. When applicable, the information for this section shall be included as provided by the contracting activity.

5.11.5 Appendixes. Refer to 5.18.

5.12 Direct support and general support maintenance TM (-34). This TM shall describe in detail the direct support and general support maintenance prescribed by the LSA/MAC and SMR coded items. The actual number of maintenance chapters in a TM depends on the maintenance concept and complexity of the equipment. (Refer to MIL-M-63008.) Required theory of operation shall be included in the applicable maintenance chapter. Levels of maintenance shall be separated by chapter.

5.12.1 Front matter. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.12.2 Chapter 1 - introduction. Refer to MIL-STD-361-6.

5.12.2.1 Section I - general information. Refer to 5.6.2.1.

5.12.2.2 Section II - equipment description and data. This section shall include a paragraph referencing the equipment data in the operator's TM. Any data required by the direct or general support levels, but not included in the operator's or unit TM shall be included. References may be made to material in other publications, provided they are available at the maintenance levels covered. Information will not be duplicated except for emphasis and clarity.

5.12.3 Inspection and test of conventional and chemical ammunition or components containing radioactive materials chapter. See 5.6.3.

5.12.4 Maintenance instructions chapter(s). See 5.5 through 5.5.8 for Sections I through VII and adjust for direct support and general support maintenance.

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a. Provide instructions for missile system equipment units under test in accordance with MIL-M-63044.

b. For ammunition, procedures shall prescribe the responsibilities for performing such operations as care and preservation of items and stocking authorized replacement components; packing material and consumable supplies; and recovering items such as cardboard inner boxes and wooden boxes needed for maintenance stocks. Reference shall be made to the RPSTL for replacement parts and any special tools required, to the expendable and durable items list appendix for the list of expendable materials, and to the ammunition marking appendix for appropriate marking information, as applicable.

c. See 5.6.4c and adjust as necessary.

5.12.4.1 Section VIII (or as applicable) - pre-embarkation inspection of materiel in units alerted for overseas movement. When applicable, the information for this section shall be included as provided by the contracting activity.

5.12.5 Appendixes. Refer to 5.18.

5.13 Combined operator's and maintenance TMs. These TMs shall describe in detail the scope of operation and maintenance prescribed by the LSA/MAC and SMR coded items. The actual number of chapters in a TM depends on the maintenance concept and complexity of the equipment. Required theory of operation shall be included in the applicable operation or maintenance chapter(s). Levels of maintenance shall be separated by chapter.

5.13.1 Operator's and unit, or operator's and AVUM TM (-12) (other than aircraft).

a. Chapter data requirements for AVUM shall be as stated in 5.13.1.2 through 5.13.1.7. When listed chapter data requirements are not applicable, the remaining chapters shall be renumbered so that no gaps appear in the chapter number sequence.

b. Appendix data requirements for AVUM shall be as stated in 5.13.1.8 through 5.13.1.8.9. When appendix data requirements are deleted or not applicable, appendixes shall be relettered as appropriate so that no gaps appear in the appendix letter sequence.

5.13.1.1 Front matter. Refer to MIL-STD-361-1 and MIL-STD-361-6.

5.13.1.2 Chapter 1 - introduction. Refer to MIL-STD-361-6.

5.13.1.3 Chapter 2 - operating instructions. Refer to MIL-STD-361-7, except the PMCS titles shall be preceded by the word "Operator."

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5.13.1.4 Chapter 3 - maintenance instructions. Refer to MIL-STD-361-7, except the title shall be changed to read "Operator Maintenance."

5.13.1.5 Chapter 4 - maintenance of auxiliary equipment (when applicable). Refer to MIL-STD-361-7.

5.13.1.6 Chapter 5 - ammunition/class V material (when applicable). Refer to MIL-STD-361-7 and 5.5.7.2.

5.13.1.7 Chapter 6 - unit maintenance or aviation unit maintenance (AVUM). For unit and aviation unit maintenance (other than aircraft), refer to 5.5.1. For aviation unit maintenance (aircraft), refer to 5.9. Section number sequence shall be by the top-down generation sequence listed in the LSA/MAC for the applicable functional group codes. For lubrication, refer to 5.5.7.17.

5.13.1.8 Appendixes. Include as applicable.

5.13.1.8.1 References appendix. This appendix shall be prepared in accordance with 5.18.1.

5.13.1.8.2 Maintenance allocation chart (MAC) appendix. This appendix shall be prepared in accordance with 5.18.2.

5.13.1.8.3 Repair parts and special tools list (RPSTL) appendix. If this appendix is required by the contracting activity, it shall be prepared in accordance with MIL-STD-361-11.

5.13.1.8.4 Components of end item (COEI) and basic issue items (BII) appendix. For other than aircraft, refer to MIL-STD-361-7. This appendix is not a requirement for aircraft maintenance TMs.

5.13.1.8.5 Additional authorization list (AAL) items appendix. For other than aircraft, refer to MIL-STD-361-7. This appendix is not a requirement for aircraft TMs.

5.13.1.8.6 Expendable and durable items appendix. This list shall be prepared in accordance with MIL-STD-361-7.

5.13.1.8.7 Illustrated list of manufactured items appendix. This list shall be prepared in accordance with 5.18.6.

5.13.1.8.8 Torque limits appendix. If this appendix is required by the contracting activity, it shall be prepared in accordance with 5.18.7.

5.13.1.8.9 Mandatory replacement parts appendix. If this appendix is required by the contracting activity, it shall be prepared in accordance with 5.18.9.

5.13.1.9 Glossary. A glossary shall be included only when the terms are uncommon and are not adequately defined in the text. Refer to MIL-STD-361-1.

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5.13.1.10 Alphabetical index. An alphabetical index shall be prepared when the number of titled paragraphs in the publication exceeds 100, or as specified by the contracting activity, regardless of the number of paragraphs. Refer to MIL-STD-361-1.

5.13.2 Operator's, unit, and direct support maintenance TM (-13). All requirements in 5.13.1 are applicable. In addition, this manual shall include the following: "Chapter 7 - direct support maintenance or aviation intermediate maintenance (AVIM)." (Refer to 5.6.)

5.13.3 Operator's, unit, direct support, and general support maintenance TM (-14). All requirements in 5.13.1 and 5.13.2 apply. In addition, this TM shall include the following: "Chapter 8 - general support maintenance." (Refer to 5.7.)

5.14 Depot maintenance work requirement (DMWR) (overhaul/depot level maintenance). The manual shall contain those depot maintenance tasks identified by the applicable LSA/MAC and programmed to be performed at depot level. These tasks shall be developed and presented to include RCM as prescribed in AR 750-2 and produce the intended/expected reliability, availability, and maintainability-durability (RAM-D). The manual shall consist of the following divisions:

Front Matter
Chapter 1 - Introduction
Chapter 2 - Technical Support Requirements
Chapter 3 - Preshop Analysis
Chapter 4 - Overhaul Procedures
Chapter 5 - Quality Assurance Requirements
Chapter 6 - Preservation, Packaging, and Marking
APPENDIXES
- References
- Repair Parts and Special Tools List
- Expendable and Durable Items
- Depot Mobilization Requirements

Glossary
Alphabetical Index

5.14.1 Front matter. Front matter shall be prepared in accordance with MIL-STD-361-1 and MIL-STD-361-6.

5.14.2 Chapter 1 - introduction. Chapter 1 shall consist of the following sections:

Section I - General Information
Section II - Equipment Description
Section III - Mobilization Requirements

5.14.2.1 Section I - general information. In addition to the requirements of MIL-STD-361-6, the following shall be included.

a. Engineering change proposals. This paragraph shall contain the following statement:

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"Engineering change proposals (ECP) will be submitted using DD Form 1693 (Engineering Change Proposal [Short Form]). (See MIL-STD-481 for instructions.) Completed forms should be mailed direct to (enter the name and address of the responsible command or activity). A reply will be furnished to you."

b. Deviations and exceptions. This paragraph shall contain the following statement:

"Requests for deviations or exceptions to this DMWR will be processed in accordance with DESCOM-R 702-1."

5.14.2.2 Section II - equipment description. In addition to the requirements of MIL-STD-361-6, an end item DMWR shall have a paragraph(s) that identifies components of the end item/system that are not covered by the DMWR, but are covered in a lower level maintenance manual(s). The components and their manuals shall be identified or referenced in narrative or tabular format, as applicable.

5.14.2.3 Section III - mobilization requirements. This section shall have the following statement:

"All requirements of this DMWR will be exempted or revised in the event of mobilization. Only those procedures necessary to return the (insert equipment name) to a serviceable condition will be performed. The exemptions and revisions are explained in Appendix __."

5.14.3 Chapter 2 - technical support requirements. This chapter shall consist of the following sections:

Section I - Support Items
Section II - Mandatory Replacement Parts, Critical Safety Items, Repair Parts, and Modifications
Section III - Standards

5.14.3.1 Section I - support items. This section shall consist of the following paragraphs:

a. Facilities. This paragraph shall describe all facilities, such as test stands, test tracks, clean rooms, shielded rooms, or other facilities, that are required to do the work specified in the DMWR. The paragraph shall reference any specifications or standards that these facilities must meet.

b. Special tools and equipment. This paragraph shall contain a list of all special tools and TMDE not contained in lower TMs, or in the applicable RPSTL, required to perform the procedures in the DMWR. This list shall include any special inspection equipment used only for the item that the DMWR covers. The list shall contain the following data: part number and CAGEC, paragraph or figure number where the tool or equipment is referenced, nomenclature,

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use or application of the tool/equipment, and a reference to any applicable calibration documents. See figure 12 for an example.

c. Fabricated tools and equipment. This paragraph shall briefly describe any fabricated tools, equipment, or fixtures that are needed to perform the procedures in the DMWR. If applicable, it shall refer to the appendix prepared to provide specific information on fabricated tools and equipment. See 5.18.6.

5.14.3.2 Section II - mandatory replacement parts, critical safety items, repair parts, and modifications. This section shall consist of the following paragraphs:

a. Mandatory replacement parts list. This paragraph shall contain a table listing all items that must be replaced during the repair and overhaul of the equipment whether they have failed or not. All items that must be replaced during overhaul or repair procedures (based on usage intervals such as miles, time, or rounds fired, or replaced on a time between overhaul (TBO) interval) shall be included in the table. The table shall contain the following data elements: item description, part number, CAGEC, NSN, location and replacement interval, and total quantity per equipment. The table shall be arranged in alphabetical sequence by item name. See figure 13 for an example.

b. Critical safety items list. As applicable, this paragraph shall include a tabular listing of all critical safety items required by AMC-R 702-32. The location of the critical safety procedures or processes within the DMWR shall be referenced.

c. Repair parts and special tools. This paragraph shall refer to the lower level manual or RPSTL appendix of the DMWR where the repair parts and special tools list can be found for the items covered by the DMWR.

d. Modifications. This paragraph shall identify all modifications which have been incorporated into the work required by the DMWR. Reference shall be made to DA PAM 25-30 and DA PAM 750-10 which list all published modification work orders (MWO).

5.14.3.3 Section III - standards. This section shall consist of the following paragraphs:

a. Quality of material. This paragraph shall include a statement(s) similar to the following:

Material used for replacement, repair, or modification must meet the requirements of this DMWR. If quality of material requirements are not in this DMWR, the material must meet the requirements of the drawings, standards,

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specifications, or approved engineering change proposals applicable to the subject equipment.

b. Electromagnetic compatibility standards. When applicable, reference shall be made to the electromagnetic compatibility standards (such as MIL-STD-461 or MIL-STD-462) that apply to the equipment covered in the DMWR.

c. Electrostatic discharge (ESD). When applicable, this paragraph shall identify the electronic parts, assemblies, and equipment, and their ESD classes that require ESD protection. The paragraph shall reference the appropriate documents, such as MIL-HDBK-263 and DOD-STD-1686, that contain ESD control procedures and material necessary to protect these items.

d. Nuclear hardness. As applicable, this paragraph shall contain nuclear hardness information and requirements in accordance with MIL-STD-361-1.

e. Reliability, availability, maintainability (RAM) requirements. When specified by maintenance engineering of the contracting activity, the DMWR shall have a RAM table listing the pertinent measurable RAM ranges for the major overhauled components covered under the DMWR for overhaul of the item/system. The RAM requirements shall be prescribed by maintenance engineering of the contracting activity and when established by maintenance engineering shall include critical measurement factors such as meantime between failures (MTBF), meantime between corrective maintenance (MTBCM), maximum time to repair (MTR), availability, and maintenance ratio. The reliability and availability portion of the table shall give the minimum acceptable values while the maintainability portion shall provide the maximum allowable rates. Availability may be expressed as a probability versus a qualified number.

5.14.4 Chapter 3 - preshop analysis. A preshop analysis chapter shall apply when data indicates that an inspection or test is more effective in determining useful life of a system, subsystem, or component than a mandatory disassembly. When the chapter is not required, the words "NOT APPLICABLE" shall follow the chapter heading. When applicable, this chapter shall consist of the following sections:

5.14.4.1 Section I - general. This section shall consist of the following paragraphs:

a. Scope. This paragraph shall state the purpose and coverage of the preshop analysis.

b. Checking attached documents. Instructions shall be provided for checking all tags, forms, and documents attached to the item to determine the reason for its return and to identify any other obvious faults or damage.

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c. Unpacking and special handling. This paragraph shall contain the procedures for removing the item, assemblies, subassemblies, or components from the shipping containers and packaging material. This paragraph shall contain needed handling requirements for hazardous material, electrostatic sensitive devices, precious metal content, classified material, or critical material and any special condemnation procedures for the item and its assemblies and subassemblies.

d. External inspection. This paragraph shall contain the procedures for external inspection of the item to determine if it is complete and if there are any obvious external damages.

e. Component checklist. When specified by the contracting activity, a component checklist shall be prepared as an appendix. The appendix shall be referenced in this paragraph and located in the back of the DMWR, so it may be removed or copied for completion. The checklist shall be preceded by a brief explanation of its use. The checklist shall contain the following data, as applicable:

Name/nomenclature of the equipment/item_____.
Serial number_____.
Date received_____.
Received from (identify unit)_____.
Component name_____.
NSN_____.
Part number_____.
Quantity required_____.
Quantity received_____.
Visual damage found_____.

f. Cleaning and preservation. Instructions shall be included for cleaning the item to prepare it for the preshop analysis testing. They shall include the procedures for any temporary preservation or corrosion protection measures needed to protect the item until the work required by the DMWR is started.

5.14.4.2 Section II - preshop analysis procedures. This section shall provide the detail procedures to do a preshop analysis. These procedures shall be prepared in either narrative format or tabulated checklist format, as specified by the contracting activity.

5.14.4.2.1 Narrative format. When specified by the contracting activity, procedures shall be presented in narrative format. Test and analysis procedures shall be presented in a logical sequence, not to cause any unnecessary disassembly, and in the order in which they should be done, usually in top-down sequence. Each procedure shall be identified by a step number. When applicable, procedures shall be arranged in groups by major components, assemblies, and subassemblies. Each group shall be

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headed by an applicable title. References to information contained elsewhere shall be in accordance with MIL-STD-361-1.

5.14.4.2.2 Preshop analysis checklist. When specified by the contracting activity, a preshop analysis checklist shall be prepared and include the following:

a. Cover sheet. The cover sheet shall contain spaces to record the following item information: part number, serial number, NSN, modifications required, reason for overhaul or repair, unpacking of secondary items required, review of tags or forms with the item, name and signature of person doing the analysis, and date. Refer to figure 14.

b. Table of tests and inspections. This table shall have an entry for each test and inspection procedure. Each entry shall have, as a minimum, the following information: name of item to inspect or test, test or inspection procedure, results of the test or inspection procedure, and recommended repair or overhaul action. Refer to figure 15.

5.14.5 Chapter 4 - overhaul procedures. This chapter shall consist of section I - Scope, General Safety, and General Inspection and one or more overhaul procedure sections. (The exact number of sections will be determined by the complexity of the equipment.) A section shall be included to cover each major assembly, each subassembly or component of the equipment, and a section to cover final assembly and testing. The overhaul procedure sections shall appear in a top-down breakdown sequence, beginning with the end item if practical. If not practical, the first section shall start at the highest possible level of assembly. The last section of the chapter shall cover the final assembly and testing of the highest assembly of the item after the overhaul procedures are completed.

5.14.5.1 Section I - scope, general safety, and general inspection. Section I shall consist of these paragraphs:

a. Scope. This paragraph shall contain a brief summary of the procedures in the chapter. It shall explain the sequence of the procedures, the identification of quality assurance procedures, and any unusual or critical procedures. It shall also describe the measurement system used and provide references to any conversion tables provided in the text or appendixes.

b. General safety precautions. When specified by the contracting activity, this paragraph shall contain general shop related safety precautions. These precautions shall not be repeated in the text of the procedures. For example, use safety glasses while soldering, use care when lifting heavy objects, or a fire extinguisher must be on hand. The precautions should be those that concern common tasks encountered while working on the equipment. If

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there is a doubt about a precaution, it shall become a warning in the text.

c. General inspection procedures. When specified by the contracting activity, this paragraph shall contain general inspection procedures that apply to the equipment and are expected to be practiced by the technician during overhaul. These procedures shall include inspecting solder joints on an electronic item, welds on an armored vehicle, fluid leakage on vehicles, connectors on electronic devices, and other items to identify defects that must be corrected.

d. Overhaul operation flowchart. When specified by the contracting activity, a flowchart shall be provided to show typical work flow to do the prescribed tasks.

5.14.5.2 Section II - overhaul procedures for (enter item/assembly). This section shall contain the overhaul and repair procedures for the highest assembly, subassembly, or component of the equipment. Reference to the appropriate manual for instructions on removal, disassembly, etc., shall be made when duplication of this information would exceed two pages in length. (See MIL-STD-361-1.) Illustrations shall be used to augment the text when necessary. The procedures shall, as applicable to the item, include the following paragraphs:

a. Removal. Procedures for the removal of the assembly or subassembly from the next higher assembly shall be provided. Instructions for recording the condition of the item/assembly, marking, handling, and storing the item shall be provided. The procedures shall identify items which must be matched or precision mated when installed at a later time.

b. Disassembly. Instructions for the required disassembly of the equipment for inspection and maintenance purposes shall be provided. These instructions shall ensure that matched parts are identified. Routine tools and procedures shall not be covered. Illustrations shall be used to augment the text and show the use of special tools or jigs that are necessary for disassembly. Instructions shall not include separation of bonded, welded, soldered, press-fitted, riveted, or other permanently attached parts unless unavoidable for cleaning, inspecting, testing, or accessing lower assemblies.

c. Cleaning. General and specific cleaning instructions for the item shall be provided. All cleaning materials required and instructions for their use, if necessary, shall be provided. Temporary corrosion prevention instructions for metal parts which have been cleaned shall be included.

d. Inspection. All inspection procedures required for the item shall be provided. All procedures that have a

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quality impact shall be identified with the boldface letters "QA" preceding them. The paragraph shall also contain instructions for tagging all rejected parts, other than mandatory replacement parts, and listing the reasons for the rejection. The instructions shall be sufficient to determine whether or not items conform to the standards in the DMWR and referenced standards. Visual inspection procedures to detect defects such as burrs, cracks, bends, or dents shall be included in the paragraph. The paragraph shall not contain vague terms such as "excessive wear" or "operates smoothly," but shall contain requirements which can be accurately determined or measured. The inspection procedures in the paragraph shall apply only to the item covered by the section. Accurate and measurable accept or reject requirements and standards shall be provided which will allow the user of the DMWR to determine if the item under inspection conforms to the tolerances, wear limits, fit, or other standards and requirements presented. Support items used to make these inspections shall be those listed in chapter 2.

e. Overhaul inspection procedure (OIP) tables. Unless requirement is specifically excluded by the contracting activity, OIP tables shall be prepared for items with critical characteristics or tolerances. As applicable, illustrations shall be integrated within the text of the OIP table. The tables shall be assigned table numbers in accordance with MIL-STD-361-1 and shall be formatted as shown on figure 16.

f. Mandatory replacement parts. Instructions for the replacement of mandatory replacement parts listed in Chapter 2 that are applicable to the equipment, assembly, or subassembly covered in this section shall be provided. Guidance for the disposition of the replaced parts shall be included. If figures with item numbers are used in the text of the section, the figure and item numbers shall also be included in a table. The table shall be arranged in alphabetical sequence by item name. Items in the table that must be replaced each time the equipment is disassembled, disturbed, or exposed during overhaul shall be identified with an asterisk next to the item name. A table note shall state the purpose of the asterisk. The text shall clearly identify and provide complete instructions for replacing these items. See figure 17 for an example of the table.

g. Critical safety items. If critical safety items are used in the equipment, instructions to replace and dispose of the item replaced shall be included.

h. Repair, replacement, and adjustment. Repair, replacement, and adjustment instructions shall be included. Instructions shall be presented in a logical sequence. When it is necessary, illustrations shall be used to

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support the text. The instructions shall consist of step-by-step procedures and list all jigs, fixtures, materials, test equipment, and other items required to perform the tasks. Instructions for automatic test or diagnostic equipment shall be included. Schematics, wiring diagrams, parts location drawings, and other visual aids shall be used where necessary. Tolerances, torque values, clearances, adjustment information, and other parameters shall be provided. All procedures or steps in this paragraph which have a major quality assurance effect shall be prefixed with the boldface letters "QA" to identify them.

i. Assembly. Step-by-step instructions for assembling the item shall be included. Illustrations shall be used to support the text where necessary. Drawings may be referenced to support the text. Tolerances, clearances, torque values, and all other parameters required for completing the assembly of the item shall be provided. For example see figure 18. All assembly procedures which have a major quality assurance effect shall be prefixed by the boldface letters "QA" to identify them. Instructions for lubrication that must be performed during the assembly procedure shall be included. The lubrication instructions shall provide a list of the lubricants required and the lubrication methods to be used.

j. Lubrication. Lubrication instructions for the item after it is assembled shall be provided. A list of the lubricants and methods required to do the tasks shall be furnished. The instructions in the assembly instructions above shall not be duplicated, except to check that the previous tasks were completed. All lubrication procedures which have a major quality assurance effect shall be prefixed by the boldface letters "QA" to identify them.

k. Testing. The testing procedures required to be done before the installation of the item into the next higher assembly shall be provided. Instructions for recording the results of the testing shall be provided. The tools and test equipment required for the procedures shall be listed. Any automatic or diagnostic test equipment required to complete the procedures shall be included or referenced. All testing procedures that have a major quality assurance effect shall be prefixed by the boldface letters "QA" to identify them.

l. Painting, refinishing, and marking. Instructions for painting, refinishing, and marking the item prior to its installation in the next higher assembly of the equipment shall be provided. The materials and facilities required for the tasks shall be listed.

m. Final assembly or installation. The instructions for installing or assembling the item with the next higher assembly of the equipment shall be provided. Any

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materials, such as sealing compounds, lubricants, or corrosion prevention compounds used during the final assembly shall be listed.

n. Data plates required. For data plates which require replacement, the type of material shall be indicated. Detailed preparation and attachment instructions shall be included. The instructions for stamping data plates shall include the initials of the facility performing the overhaul or modification, contract number (if applicable), date of overhaul or modification, part number, and total operating time since new (if applicable). The instructions shall specify the letter and figure sizes and indicate their placement (adjustment to manufacturer's data). The following statement shall be included in this paragraph:

"When sufficient space is not available on the existing data plate to add information, the plate shall be replaced and all pertinent data transferred to the new plate. Data shall not be stamped directly on any part, assembly, or item of equipment except when approved by the Government."

5.14.5.3 Section (last section of Chapter 4) - final testing. This section shall appear after the overhaul procedure section(s) and shall be numbered accordingly. The section shall cover the final testing of the highest assembly or equipment/end item that the DMWR covers to assure that the parameters of reliability, availability, maintainability, and durability are met. The section shall contain the following paragraphs:

a. Scope. A general explanation of the arrangement of the paragraph and the procedures in it shall be provided.

b. Inspection. The inspection procedures that are required prior to final testing to assure that the item is complete and ready for final testing shall be provided. When applicable, instructions for any minor preparation tasks for final testing shall be included.

c. Lubrication. Any final lubrication procedures that need to be done prior to final testing shall be provided. All necessary lubrication materials and instructions for the lubrication methods shall be listed.

d. Final testing. Test procedures, performance standards, and tolerances to be met before the equipment is adequately overhauled and ready for issue without qualifications shall be provided. The procedures shall list all tools, TMDE, jigs, fixtures, and other support items required for the test. Operating instructions for special test equipment shall be included where necessary. Illustrations, charts, and checklists shall be used to augment the text where necessary. Procedures for minor adjustments that can be done without disassembling equipment shall be included in the paragraph. Complete

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procedures for burn in or run in tests shall also be included.

e. Final painting, refinishing, and marking. Procedures for any final painting, refinishing, and marking that could not be done during the overhaul procedures shall be provided. The materials and tools required to do the job shall be identified.

5.14.6 Chapter 5 - quality assurance requirements. This chapter shall consist of these general information paragraphs:

a. Responsibility statement. This paragraph shall consist of the following statement:

"The depot/contractor is responsible for complying with the quality assurance requirements contained in this chapter in accordance with MIL-Q-9858 (Quality Program Requirements), and DESCOM-R 702-1 (DESCOM Product Assurance Program). The commodity manager reserves the right to perform inspections or make changes that ensure the depot work being done meets the quality standards of the DMWR and preserves the inherent reliability of the item."

b. Definitions. This paragraph shall define all quality assurance terms extensively used in the DMWR. These definitions shall be in accordance with MIL-STD-109 and MIL-W-63150.

c. Special requirements for inspection tools and equipment. When specified by the contracting activity, this paragraph shall list any special requirements for the maintenance and calibration of tools and test equipment used for quality assurance inspections. These requirements shall be met in accordance with the applicable standards, specifications, or regulations such as MIL-STD-120, MIL-I-45607, MIL-STD-45662, or DESCOM-R 702-1.

d. Certification requirements. When specified by the contracting activity, this paragraph shall list any certification or licensing requirements for processes, procedures, materials, equipment, or personnel skills. The list shall include appropriate standards, specifications, regulations, or laws that apply. The list shall reference the paragraph in the DMWR where there is a requirement for a soldering, welding, or magnetic particle inspecting certification, radioactive substance, or test driver licenses.

e. In-process inspections statement. This paragraph shall consist of the following statement:

"In-process quality assurance inspections are contained throughout Chapter 4 of this DMWR. These inspections are prefixed with the boldfaced letters "QA" to identify them,

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and they are the minimum inspections required. Additional quality assurance inspections may be established by the depot or the commodity manager."

f. Acceptance inspections statement. This paragraph shall consist of the following statement:

"Items overhauled in accordance with this DMWR will be accepted based on the following criteria:

(1) Conformance to quality of material requirements.

(2) Conformance to all in-process quality assurance inspections.

(3) Conformance to all final assembly testing requirements.

(4) Conformance to the preservation, packaging, and marking requirements."

g. First article inspection. When specified by the contracting activity, first article inspection/test shall be included in the DMWR in accordance with DESCOM-R 702-1.

5.14.7 Chapter 6 - preservation, packaging, and marking. This chapter shall consist of the following two paragraphs:

a. Packaging information. The first paragraph shall consist of the following:

"Packaging information. Refer to the packaging segment of the Army Reader Microform System (ARMS) file for detailed packaging information. If ARMS does not contain the information that you need, you must contact the AMC Packaging, Storage, and Containerization Center, Tobyhanna Army Depot, for packaging information."

b. Special instructions. The second paragraph shall contain any special or unique preservation, packaging, or marking instructions that apply to the equipment covered by the DMWR. These instructions shall include cautions or references concerning electrostatic discharge, nuclear materiel, hazardous substances, special marking instructions, or any other instructions required that are not covered in the standard packaging and preservation information.

5.14.8 Appendixes. Include as applicable.

5.14.8.1 References appendix. Refer to 5.18.1.

5.14.8.2 Repair parts and special tools list (RPSTL) appendix. When specified by the contracting activity, this appendix shall be included in the DMWR. The RPSTL shall be prepared in accordance with MIL-STD-361-11.

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5.14.8.3 Expendable and durable items appendix. This appendix shall be prepared in accordance with MIL-STD-361-7.

5.14.8.4 Depot mobilization requirements appendix. When specified and provided by the contracting activity, this appendix shall include the modifications, deletions, or additions to the procedures in Chapters 3 and 4 required during mobilization. The appendix shall begin with the following statement: "The purpose of this appendix is to streamline and accelerate the overhaul process during the mobilization of the depot." The appendix shall include a list of instructions for modifying the procedures, and the list shall be presented in a format similar to the following:

PAGE	PARAGRAPH	ACTION
1-3	1.5	Add: The extent of disassembly will be based on preshop analysis only.
4-12	4.14c	Delete all procedures in the paragraph.
4-27	4.31	Change to read: Painted metal will be touched up only.

5.14.8.5 Other appendixes. When specified by the contracting activity, other applicable appendixes shall be added to the DMWR.

5.14.9 Glossary. A glossary shall be included only when the terms are uncommon and are not adequately defined in the text. Refer to MIL-STD-361-1.

5.14.10 Alphabetical index. An alphabetical index shall be prepared when the number of titled paragraphs in the publication exceeds 100, or as specified by the contracting activity, regardless of the number of paragraphs. Refer to MIL-STD-361-1.

5.14.11 Authentication page. The authentication page, provided by the contracting activity, shall be the last page of the DMWR. See figure 19.

5.15 Auxiliary equipment maintenance instructions. When auxiliary equipment (e.g., MTOE items, etc.) maintenance TMs or maintenance requirements cards are not procured for peculiar support equipment furnished by the contractor, maintenance instructions for such equipment shall be included. Instructions for required fabrication of special tools, when such fabrication is approved by the contracting activity, shall be included in the proper appendix. Concise step-by-step procedures shall be included for proper care of auxiliary equipment while in and out of service. These procedures shall include instructions for storage, preventive maintenance, lubrication, operating checks, and adjustments.

5.16 Ammunition maintenance instructions. As applicable, maintenance instructions shall reference or contain all procedures required for care and handling of ammunition,

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including hazard distances, storage, special requirements, prevention of deterioration due to rough handling, exposure to adverse weather conditions or other hazards. Visual inspection criteria shall be provided to determine item serviceability. Instructions shall also contain procedures for disposition of defective ammunition. These procedures shall include use of cleaning materials and paint authorized for use in the specified maintenance operations. Reference shall be made to the RPSTL for replacement parts and any special tools required, to the expendable and durable items list appendix for the list of expendable and durable materials, and to the ammunition marking appendix for appropriate marking information, as applicable. (See 5.18.8.)

5.17 Administrative storage. Information pertaining to administrative storage of equipment shall be included as provided by the contracting activity.

5.18 Appendixes. Appendixes shall be identified alphabetically throughout the document in the order of their reference in the text (e.g., Appendix A, Appendix B, etc.). The following appendixes shall be prepared, as applicable.

5.18.1 References appendix. This appendix shall list all publications referenced in the TM and required by the user to maintain the equipment. The appendix shall have an introduction or scope paragraph to provide a brief statement(s) concerning its use and content. The publications shall be listed in groups by publication types. If the publication is nongovernment, the source shall be provided. The complete name and number of each publication shall be used. See figure 20. If a list of applicable publications (LOAP) exists, it may be referenced.

5.18.1.1 Related TMs. The introduction shall list, in tabular form, all related publications, with the exception of those publications that are currently unpublished. This list 75 shall identify the TMs, by number and title, that affect the operation, maintenance, and inspection procedures of the equipment.

5.18.2 Maintenance allocation chart (MAC) appendix. This appendix shall be included as specified by the contracting activity. The MAC shall be prepared from the equipment top-down breakdown to consolidate and identify those groups on the list which involve maintenance functions identified below. LSA Output Report 04, Maintenance Allocation Summary, and LSA 20 Report will be used, when available, as source data for the final approved MAC.

5.18.2.1 Maintenance functions. The MAC shall list the applicable maintenance functions assigned to each maintenance level and shall be published only in equipment TMs containing unit or aviation unit maintenance instructions. Maintenance functions shall be assigned based on guidance contained in AR 750-1.

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5.18.2.2 MAC entries.

a. The basic entries in the MAC shall be a list of functional groups applicable to the end item which require maintenance. The term functional group applies to reparable assemblies and subassemblies (spares[1]), but not to repair parts[2]) The end item group shall be numbered "00," or its equivalent "AA."

b. Entries shall be item names[3] and, where applicable, type designators, without stock or part numbers if possible, in order to minimize need for subsequent change; however, entries shall contain positive identification. Lamps, electron tubes, fixed resistors, rotor caps, ignition coils, "throwaway" modules, and repair parts of a like category are not considered to be functional groups; are not subject to maintenance; and shall not be listed in the MAC.

c. All item names of functional groups shall be the same in the MAC, RPSTL, PMCS, and narrative maintenance instructions. Official reverse word order shall be used in the MAC and RPSTL. Normal word order shall be used in maintenance instructions.

d. The maintenance code entered in the third position of the SMR code in the RPSTL (or parts listing) shall identify the lowest category of maintenance that is authorized to remove, replace, and use the spare or repair parts. SMR codes are further defined in MIL-STD-361-11.

e. If the maintenance function is a replace function only for a repair part, the repair part shall not be listed in the MAC, unless not listing the repair part would result in omission of the next higher assembly (NHA) group number; in this case, the part shall be listed in order to list the NHA functional group number. Such parts shall be included in the appropriate RPSTL, which is automatic authority to replace at the lowest maintenance level to which the part is authorized (indicated by the code in the third position of the SMR code).

f. All items in the MAC shall specify the maintenance level(s) to which a function is authorized.

g. Exception is authorized for ammunition MACs to permit use of maintenance function headings that better describe or identify ammunition peculiar maintenance functions. The headings used and their definitions shall be included in the appropriate ammunition TM(s).

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[1] A spare part is any reparable component (assembly or subassembly) required for the maintenance or repair of an end item.

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[2] A repair part is any consumable (i.e., nonreparable) component (assembly or subassembly) required for the maintenance or repair of an end item.

[3] An item name consists of a basic name (e.g., transformer), and a noun word or phrase modifier (e.g., pulse, low power).

5.18.2.3 MAC sequence of entries. The sequence of entries in the MAC shall prescribe the sequence of group numbers and functional groups followed in the RPSTL(s) for the end item(s).

NOTE

The MAC, RPSTL, and maintenance instructions in equipment TMs shall be complete and consistent with regard to level of maintenance. In equipment TMs containing replace or repair maintenance instructions, the replace or repair procedures will be arranged to coincide with the functional group number sequence followed in the MAC and RPSTL.

5.18.2.4 MAC introduction (standard format). The introduction to the MAC shall be prepared as follows:

"APPENDIX __

MAINTENANCE ALLOCATION CHART (MAC)

SECTION I. INTRODUCTION

__-1 THE ARMY MAINTENANCE SYSTEM MAC.

a. This introduction (Section I) provides a general explanation of all maintenance and repair functions authorized at various maintenance levels under the standard Army Maintenance System concept.

b. The Maintenance Allocation Chart (MAC) in Section II designates overall authority and responsibility for the performance of maintenance functions on the identified end item or component. The application of the maintenance functions to the end item or component will be consistent with the capacities and capabilities of the designated maintenance levels, which are shown on the MAC in Column (4) as:

Unit - includes two subcolumns, C (operator/crew) and O (unit) maintenance.

Direct Support - includes an F subcolumn.

General Support - includes an H subcolumn.

Depot - includes a D subcolumn.

c. Section III lists the tools and test equipment (both special tools and common tool sets) required for each maintenance function as referenced from Section II.

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d. Section IV contains supplemental instructions and explanatory notes for a particular maintenance function.

__-2 **MAINTENANCE FUNCTIONS.**

Maintenance functions are limited to and defined as follows:

a. **Inspect.** To determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel). This includes scheduled inspection and gagings and evaluation of cannon tubes.

b. **Test.** To verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards on a scheduled basis, i.e., load testing of lift devices and hydrostatic testing of pressure hoses.

c. **Service.** Operations required periodically to keep an item in proper operating condition; e.g., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases. This includes scheduled exercising and purging of recoil mechanisms.

d. **Adjust.** To maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.

e. **Align.** To adjust specified variable elements of an item to bring about optimum or desired performance.

f. **Calibrate.** To determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. Consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.

g. **Remove/Install.** To remove and install the same item when required to perform service or other maintenance functions. Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module (component or assembly) in a manner to allow the proper functioning of an equipment or system.

h. **Replace.** To remove an unserviceable item and install a serviceable counterpart in its place. "Replace" is authorized by the MAC and assigned maintenance level is shown as the 3rd position code of the SMR code.

i. **Repair.** The application of maintenance services[1] including fault location/troubleshooting,[2] removal/installation, disassembly/assembly[3] procedures, and maintenance actions[4] to identify troubles and restore serviceability to an item by

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correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system.

j. Overhaul. That maintenance effort (service/action) prescribed to restore an item to a completely serviceable/operational condition as required by maintenance standards in appropriate technical publications. Overhaul is normally the highest degree of maintenance performed by the Army. Overhaul does not normally return an item to like new condition.

---

[1]Services - Inspect, test service, adjust, align, calibrate, and/or replace.

[2]Fault location/troubleshooting - The process of investigating and detecting the cause of equipment malfunctioning; the act of isolating a fault within a system or unit under test (UUT).

[3]Disassembly/assembly - The step-by-step breakdown (taking apart) of a spare/functional group coded item to the level of its least component, that is assigned an SMR code for the level of maintenance under consideration (i.e., identified as maintenance significant).

[4]Actions - Welding, grinding, riveting, straightening, facing, machining, and/or resurfacing.

k. Rebuild. Consists of those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. Rebuild is the highest degree of materiel maintenance applied to Army equipment. The rebuild operation includes the act of returning to zero those age measurements (e.g., hours/miles) considered in classifying Army equipment/components.

__-3 EXPLANATION OF COLUMNS IN THE MAC, SECTION II.

a. Column 1, Group Number. Column 1 lists functional group code numbers, the purpose of which is to identify maintenance significant components, assemblies, subassemblies, and modules with the next higher assembly.

b. Column 2, Component/Assembly. Column 2 contains the item names of components, assemblies, subassemblies, and modules for which maintenance is authorized.

c. Column 3, Maintenance Function. Column 3 lists the functions to be performed on the item listed in Column 2. (For detailed explanation of these functions see paragraph __-2.)

d. Column 4, Maintenance Level. Column 4 specifies each level of maintenance authorized to perform each function listed in Column 3, by indicating work time required (expressed as manhours in whole hours or decimals) in the appropriate subcolumn. This work-time figure represents the active time required to perform that maintenance function at the indicated level of maintenance. If the number or complexity of the tasks within the listed maintenance function varies at different

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maintenance levels, appropriate work-time figures are to be shown for each level. The work-time figure represents the average time required to restore an item (assembly, subassembly, component, module, end item, or system) to a serviceable condition under typical field operating conditions. This time includes preparation time (including any necessary disassembly/assembly time), troubleshooting/fault location time, and quality assurance time in addition to the time required to perform the specific tasks identified for the maintenance functions authorized in the maintenance allocation chart. The symbol designations for the various maintenance levels are as follows:

C.....Operator or crew maintenance

O.....Unit maintenance

F.....Direct support maintenance

L.....Specialized repair activity (SRA)[5]

H.....General support maintenance

D.....Depot maintenance

---

[5]This maintenance level is not included in Section II, Column (4) of the Maintenance Allocation Chart. Functions to this level of maintenance are identified by a work-time figure in the "H" Column of Section II, Column (4), and an associated reference code is used in the Remarks Column (6). This code is keyed to Section IV, Remarks, and the SRA complete repair application is explained there.

e. Column 5, Tools and Equipment Reference Code. Column 5 specifies, by code, those common tools sets (not individual tools), common TMDE, and special tools, special TMDE and special support equipment required to perform the designated function. Codes are keyed to tools and test equipment in Section III.

f. Column 6, Remarks Code. When applicable, this column contains a letter code, in alphabetical order, which is keyed to the remarks contained in Section IV.

-4 EXPLANATION OF COLUMNS IN TOOLS AND TEST EQUIPMENT REQUIREMENTS, SECTION III.

a. Column 1, Tool or Test Equipment Reference Code. The tool or test equipment reference code correlates with a code used in the MAC, Section II, Column 5.

b. Column 2, Maintenance Level. The lowest level of maintenance authorized to use the tool or test equipment.

c. Column 3, Nomenclature. Name or identification of the tool or test equipment.

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d. Column 4, National Stock Number. The National Stock Number of the tool or test equipment.

e. Column 5, Tool Number. The manufacturer's part number, model number, or type number.

__-5 EXPLANATION OF COLUMNS IN REMARKS, SECTION IV.

a. Column 1, Remarks Code. The code recorded in Column 6, Section II.

b. Column 2, Remarks. This column lists information pertinent to the maintenance function being performed as indicated in the MAC, Section II."

5.18.2.5 MAC introduction (three-level for Army aviation). The introduction for a three-level Army aviation MAC shall be prepared as follows:

"APPENDIX __

MAINTENANCE ALLOCATION CHART

SECTION I. INTRODUCTION

__-1 AVIATION MAINTENANCE ALLOCATION CHART.

a. This Maintenance Allocation Chart (MAC) assigns maintenance functions in accordance with the Aviation Maintenance concept for Army aviation. These maintenance levels - Aviation Unit Maintenance (AVUM), Aviation Intermediate Maintenance (AVIM), and Depot Maintenance - are depicted on the MAC as:

AVUM - corresponds to an O code in the Repair Parts and Special Tools List (RPSTL).

AVIM - corresponds to an F code in the Repair Parts and Special Tools List (RPSTL).

DEPOT - corresponds to a D code in the Repair Parts and Special Tools List (RPSTL).

b. The maintenance to be performed below depot and in the field is described as follows:

(1) Aviation Unit Maintenance (AVUM). AVUM activities will be staffed and equipped to perform high frequency "On-Aircraft" maintenance tasks required to retain or return aircraft systems to a serviceable condition. The maintenance capability of the AVUM will be governed by the Maintenance Allocation Chart (MAC) and limited by the amount and complexity of ground support equipment (GSE), facilities required, authorized manning strength, and critical skills available. The range and quantity of authorized spare modules/components will be consistent with the mobility requirements dictated by the air mobility concept. (Assignments of maintenance tasks to divisional company size

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aviation units will consider the overall maintenance capability of the division, the requirement to conserve personnel and equipment resources, and air mobility requirements.)

(a) Company Size Aviation Units: Perform those tasks which consist primarily of preventive maintenance and maintenance repair and replacement functions associated with sustaining a high level of aircraft operational readiness. Perform maintenance inspections and servicing to include preflight, daily, intermediate, periodic (or phased), and special inspections, as authorized by the MAC or higher headquarters. Identify the cause of equipment/system malfunctions using applicable technical manual troubleshooting instructions, built-in test equipment (BITE), installed aircraft instruments, or test, measurement, and diagnostic equipment (TMDE). Replace worn or damaged modules/components that do not require complex adjustments or system alignment and which can be removed/installed with available skills, tools, and ground support equipment. Perform operational and continuity checks and make minor repairs to the electrical system. Inspect, service, and make operational, capacity, and pressure checks to hydraulic systems. Perform servicing, functional adjustments, and minor repair/replacement to the flight control, propulsion, power train, and fuel systems. Accomplish airframe repair that does not require extensive disassembly, jigging, or alignment. The manufacture of airframe parts will be limited to those items which can be fabricated with tools and equipment found in current air mobile tool and shop sets. Evacuate unserviceable modules/components and end items beyond the repair capability of AVUM to the support AVIM.

(b) Less than Company Size Aviation Units: Aviation elements organic to brigade, group, battalion headquarters, and detachment size units are normally small and have less than 10 aircraft assigned. Maintenance tasks performed by these units will be those which can be accomplished by the aircraft crew chief or assigned aircraft repairman and will normally be limited to preventive maintenance, inspections, servicing, spot painting, module/component fault diagnosis, and replacement of selected modules/components. Repair functions will normally be accomplished by the support AVIM unit.

(2) Aviation Intermediate Maintenance (AVIM).

(a) Provides mobile, responsive "one-stop" maintenance support. (Maintenance functions which are not conducive to sustaining air mobility will be assigned to depot maintenance.)

(b) May perform all maintenance functions authorized to be done at AVUM. Repair of equipment for return to user will emphasize support of operational readiness requirements. Authorized maintenance includes replacement and repair of modules/components and end items which can be accomplished efficiently with available skills, tools, and equipment.

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(c) Establishes the Direct Exchange (DX) program for AVUM units by repairing selected items for return to stock when such repairs cannot be accomplished at the AVUM level.

(d) Inspects, troubleshoots, performs diagnostic tests, repairs, adjusts, calibrates, and aligns aircraft system modules/components. AVIM units will have capability to determine the serviceability of specified modules/components removed prior to the expiration of the Time Between Overhaul (TBO) or finite life. Module/component disassembly and repair will support the DX program and will normally be limited to tasks requiring cleaning and the replacement of seals, fittings, and items of common hardware. Airframe repair and fabrication of parts will be limited to those maintenance tasks which can be performed with available tools and test equipment. Unserviceable reparable modules/components and end items which are beyond the capability of AVIM to repair will be evacuated to Depot Maintenance.

(e) Performs aircraft weight and balance inspections and other special inspections which exceed AVUM capability.

(f) Provides quick response maintenance support, including aircraft recovery and air evacuation, on-the-job training, and technical assistance through the use of mobile maintenance contact teams.

(g) Maintains authorized operational readiness float aircraft.

(h) Provides collection and classification services for serviceable/unserviceable materiel.

(i) Operates a cannibalization activity in accordance with AR 750-50. (The aircraft maintenance company within the maintenance battalion of a division will perform AVIM functions consistent with air mobility requirements and conservation of personnel and equipment resources. Additional intermediate maintenance support will be provided by the supporting nondivisional AVIM unit.)

__-2 USE OF THE MAINTENANCE ALLOCATION CHART (SECTION II).

NOTE

Approved item names are used throughout this MAC. Generic terms/nomenclature (if any) are expressed in parentheses and are not to be considered as official terminology.

a. This Maintenance Allocation Chart assigns maintenance functions to the lowest level of maintenance, based on past experience and the following considerations:

(1) Skills available.

(2) Work time required.

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(3) Tools and test equipment required and/or available.

b. Only the lowest level of maintenance authorized to perform a maintenance function is indicated. If the lowest maintenance level cannot perform all tasks of any single maintenance function (e.g., test, repair), then the higher maintenance level(s) that can accomplish additional tasks will also be indicated.

c. A maintenance function assigned to a maintenance level will automatically be authorized to be performed at any higher maintenance level.

d. A maintenance function that cannot be performed at the assigned level of maintenance for any reason may be evacuated to the next higher maintenance level. Higher maintenance levels will perform the maintenance functions of lower maintenance levels when required by the commander who has the authority to direct such tasking.

e. The assignment of a maintenance function will not be construed as authorization to carry the related repair parts or spares in stock. Information to requisition or otherwise secure the necessary repair parts will be as specified in the associated Repair Parts and Special Tools List (RPSTL).

f. Normally there will be no deviation from the assigned level of maintenance. In cases of operational necessity, at the request of a lower maintenance level and on a one-time basis, transfer of maintenance functions to the lower level may be accomplished by specific authorization of the maintenance officer of the higher level of maintenance to which the function is assigned. The special tools, equipment, etc., required by the lower level of maintenance to perform this function will be furnished by the maintenance level to which the function is assigned. This transfer of a maintenance function to a lower maintenance level does not relieve the higher maintenance level of the responsibility for the function. The higher level of maintenance will provide technical supervision and inspection of the function being performed at the lower level.

-3 **MAINTENANCE FUNCTIONS**. Maintenance functions will be limited to and defined as follows:

a. **Inspect**. To determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel).

b. **Test**. To verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards.

c. **Service**. Operations required periodically to keep an item in proper operating condition, i.e., to clean (includes

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decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases.

d. **Adjust**. To maintain or regulate, within prescribed limits, by bringing into proper or exact position, or by setting the operating characteristics to specified parameters.

e. **Align**. To adjust specified variable elements of an item to bring about optimum or desired performance.

f. **Calibrate**. To determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipments used in precision measurement. Consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.

g. **Remove/Install**. To remove and install the same item when required to perform service or other maintenance functions. Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module (component or assembly) in a manner to allow the proper functioning of an equipment or system.

h. **Replace**. To remove an unserviceable item and install a serviceable counterpart in its place. "Replace" is authorized by the MAC and assigned maintenance level is shown as the 3rd position code of the SMR code.

i. **Repair**. The application of maintenance services,[1] including fault location/troubleshooting,[2] removal/installation and disassembly/assembly procedures,[3] and maintenance actions[4] to identify troubles and restore serviceability to an item by correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system.

---

[1]Services - Inspect, test, service, adjust, align, calibrate, and/or replace.

[2]Fault locate/troubleshoot - The process of investigating and detecting the cause of equipment malfunctioning; the act of isolating a fault within a system or unit under test (UUT).

[3]Disassemble/assemble - The step-by-step taking apart (or breakdown) of a spare/functional group coded item to the level of its least componency identified as maintenance significant (i.e., assigned an SMR code) for the level of maintenance under consideration.

[4]Actions - Welding, grinding, riveting, straightening, facing, machining, and/or resurfacing.

j. **Overhaul**. That maintenance effort (service/action) prescribed to restore an item to a completely serviceable/operational condition as required by maintenance standards in appropriate technical publications. Overhaul is

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normally the highest degree of maintenance performed by the Army. Overhaul does not normally return an item to like new condition.

k. **Rebuild**. Those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. Rebuild is the highest degree of materiel maintenance applied to Army equipment. The rebuild operation includes the act of returning to zero those age measurements (e.g., hours/miles) considered in classifying Army equipment/components.

__-4 **EXPLANATION OF COLUMNS IN THE MAC, SECTION II**.

a. **Functional Groups (Columns 1 and 2)**. The functional groupings shown in the sample below identify maintenance significant components, assemblies, subassemblies, and modules with the next higher assembly.

GROUP NUMBER	DESCRIPTION
04	POWER PLANT
0401	ENGINE GENERAL Servicing, handling inspection requirements, overhaul and retirement schedules. External lines & hoses. (As applicable.)
0402	COMPRESSOR SECTION (COLD SECTION MODULE) Rotor, blades, vanes, impeller, stators, inlet guide vanes, main frame, particle separator, bleed valve, bearings, seals, external lines and hoses.
0403	COMBUSTION SECTION (HOT SECTION MODULE) Liners, nozzles, stators, rotor, seals, couplings, blades.
0404	POWER-TURBINE (POWER TURBINE MODULE) Nozzles, rotors, blades, exit guide vanes, exhaust frame, drive shaft, bearings, seals, external lines and hoses.
0405	ACCESSORY GEAR BOX (ACCESSORY SECTION MODULE) Input and output gears, seals, chip detector, housings, drive shaft, bearings.
0406	FUEL SYSTEM Fuel control, fuel boost pump, governors, fuel filter assembly, sequence valve, fuel manifold, fuel nozzle, external lines and hoses.

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0407 ELECTRICAL SYSTEM	0408 OIL SYSTEM
Electrical control units, exciters, thermocouples, ignition harness, electrical cables, history record, torque overspeed sensor, Np sensor, external lines and hoses.	Tanks, oil filter, oil cooler, lube and scavenger pumps, oil filter bypass sensor, external lines and hoses.

b. **Maintenance Function (Column 3)**. Column 3 lists the functions to be performed on the items listed in Column 2.

c. **Maintenance Level (Column 4)**. The maintenance levels AVUM, AVIM, and DEPOT are listed on the Maintenance Allocation Chart with individual columns that include the work times for maintenance functions at each maintenance level. Work time presentations such as "0.1" indicate the average time (expressed in manhours in whole hours or decimals) it requires a maintenance level to perform a specified maintenance function. If a work time has not been established, the columnar presentation will indicate "--." Maintenance levels higher than the level of maintenance indicated are authorized to perform the indicated function.

d. **Tools and Equipment Reference Code (Column 5)**. Column 5 specifies, by code, those common tool sets (not individual tools), common TMDE, and special tools, special TMDE, and special support equipment required to perform the designated function.

e. **Remarks Code (Column 6)**. When applicable, this column contains a letter code, in alphabetical order, which is keyed to the remarks contained in Section IV.

__-5 EXPLANATION OF COLUMNS IN TOOLS AND TEST EQUIPMENT REQUIREMENTS, SECTION III.

a. **Column 1, Tool or Test Equipment Reference Code**. The tool or test equipment reference code correlates with a code used in the MAC, Section II, Column 5.

b. **Column 2, Maintenance Level**. The lowest level of maintenance authorized to use the tool or test equipment.

c. **Column 3, Nomenclature**. Name or identification of the tool or test equipment.

d. **Column 4, National Stock Number**. The National Stock Number of the tool or test equipment.

e. **Column 5, Tool Number**. The manufacturer's part number.

__-6 EXPLANATION OF COLUMNS IN REMARKS, SECTION IV

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a. Column 1, Remarks Code. The code recorded in Column 6, Section II.

b. Column 2, Remarks. This column lists information pertinent to the maintenance function being performed as indicated in the MAC, Section II."

5.18.2.6 MAC format. The maintenance allocation chart shall appear in the format shown in figure 21 (standard) or figure 22 (aviation), as applicable.

a. For an explanation of data to be listed in columns of this chart, refer to boilerplate information presented in 5.18.2.4 or 5.18.2.5, as applicable.

b. The group number shall be entered in Column 1, the nomenclature of the spare (component/assembly) shall be entered in Column 2, and the maintenance function shall be listed in Column 3 of the MAC.

c. Column 4 of the standard maintenance allocation chart shall be divided into four main headings, one for each level of maintenance (i.e., Unit, Direct Support, General Support, and Depot). Column 4 of the aviation maintenance allocation chart shall be divided into three main headings (i.e., AVUM or Unit, AVIM or Intermediate, and Depot).

d. A work time figure must appear in the subcolumn for the maintenance level authorized to perform the maintenance listed in Column 3.

(1) The active repair time specified for each level of maintenance shall be the average aggregate time to restore an item (subassembly, assembly, component, module, end item, or system) to a serviceable condition under typical field operating conditions and shall include troubleshooting time, time to accomplish preliminary (equipment condition) and follow-on tasks, preparation time, and quality assurance/quality control time in addition to the time required to perform the specific maintenance function authorized in the MAC.

(2) This time shall be the established time standard developed through LSA, or it shall be derived from the calculation of a statistically weighted time estimate, incorporating the optimistic (a), most likely (m), and pessimistic (b) estimates for the work to be accomplished using the formula:

$$t = \frac{a + 4m + b}{6}$$

(3) This work time shall be carried to one decimal place (tenths of hours).

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e. Reference numbers for all required tools and test equipment shall be listed in Column 5 of the MAC. A list of all tools and test equipment, both special and common, required to maintain the equipment shall be prepared in accordance with the format shown in figure 23 or figure 24, as applicable. Common tools shall not be included on this list when they are part of an existing set, kit, or outfit authorized to the intended user; however, the authorized set, kit, or outfit which contains the prescribed common tools shall be listed.

f. Reference letters for applicable remarks shall be listed in Column 6 of the MAC. Remarks pertinent to maintenance functions shall be prepared in accordance with the format shown on figure 23 or figure 24, as applicable.

5.18.3 Tool identification list appendix. This appendix shall have two sections: Section I, Introduction, and Section II, Tool Identification List (tabular listing). This appendix shall not include illustrations. This list shall include the tools authorized to the levels of maintenance covered in the narrative portion of the TM and as referenced by the initial setups.

5.18.3.1 Section I - introduction. The introduction shall appear as follows:

"APPENDIX _

UNIT MAINTENANCE
TOOL IDENTIFICATION LIST

SECTION I. INTRODUCTION

_-1 SCOPE.

This appendix lists all common tools and supplements and special tools/fixtures needed to maintain the (insert equipment name).

_-2 EXPLANATION OF COLUMNS.

a. Column 1 - Item Number. This number is assigned to the entry in the listing and is referenced in the initial setup to identify the item (e.g., "Crowbar (item 32, App C)").

b. Column 2 - Item Name. This column lists the item by noun nomenclature and other descriptive features (e.g., "Gage, belt tension").

c. Column 3 - National Stock Number. This is the national stock number (NSN) assigned to the item; use it to requisition the item.

d. Column 4 - Part Number. Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity) which controls the design and characteristics of the item by means of its engineering drawings,

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specifications, standards, and inspection requirements to identify an item or range of items.

e. Column 5 - Reference. This column identifies the authorizing supply catalog or RPSTL for items listed in this appendix."

5.18.3.2 Section II - tool identification list. The list shall be formatted and shall include the applicable information under the headings shown on figure 25.

5.18.4 Repair parts and special tools list appendix. This appendix shall be prepared in accordance with MIL-STD-361-11.

5.18.5 Expendable and durable items appendix. This list shall be prepared in accordance with MIL-STD-361-7.

5.18.6 Illustrated list of manufactured items appendix. This appendix shall include simplified line drawing illustrations for each item authorized to be manufactured or fabricated by unit, direct support, or general support maintenance personnel (e.g., all MO, MF, and MH source coded items authorized in the applicable RPSTL). Supporting text, consisting of all instructional criteria needed to manufacture or fabricate the item shall be provided. A list of bulk materials to be used in manufacture or fabrication of the item shall be included on each illustration. In addition, this appendix shall list (as applicable) the part numbers of the items to be manufactured. At a minimum, this appendix shall provide the following:

5.18.6.1 Section I - introduction. The following statements shall be included:

"a. This appendix includes complete instructions for making items authorized to be manufactured or fabricated at the (enter applicable maintenance level).

b. A part number index in alphanumeric order is provided for cross-referencing the part number of the item to be manufactured to the figure which covers fabrication criteria.

c. All bulk materials needed for manufacture of an item are listed by part number or specification number in a tabular list on the illustration."

5.18.6.2 Section II - index of manufactured items. A part number index shall be furnished which lists part numbers, in alphanumeric order, of all items illustrated in this appendix. The part number of each manufactured item shall be cross-referenced to applicable figure.

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5.18.6.3 Section III - illustrations of manufactured items.

a. Line drawings in clear, simple, complete format (figure 26) shall be used as illustrations and shall contain sufficient views to portray all features of the item.

b. More than one illustration may appear on one page to save space.

c. Each illustration on a page shall be assigned a separate figure number.

d. Consecutive figure numbers shall be assigned to the illustrations.

e. All instructions (explanatory text and list of bulk materials) needed by maintenance personnel to manufacture the item (See figure 26.) shall be included on the illustrations as follows:

(1) All dimensional, locational, and processing instructions (e.g., 30-in. long, top surface, primer coating) needed to manufacture the item shall be included.

(2) A description of the item to be manufactured, including the part number and name, shall be furnished.

(3) A list of bulk materials needed to manufacture the item shall be furnished. The list of bulk materials shall consist of the part number, national stock number, or specification number of the raw bulk material to be used in manufacture of the item and shall cite the technical characteristics (e.g., standards, specifications, conditions, dimensions, and any other pertinent data).

(4) When applicable, reference shall be made to the associated RPSTL TM, or RPSTL appendix (for combined TMs).

5.18.7 Torque limits appendix. Unless requirement is specifically excluded by the contracting activity, this appendix shall be included to provide applicable torque values (expressed in lb-ft or lb-in. terms), data as to bolt grade markings and their proper identification, and specific torque sequencing requirements, as applicable. For an example of the type of information presented in a torque limits appendix, see figure 27.

5.18.8 Ammunition marking information appendix. Unless requirement is specifically excluded by the contracting activity, a separate ammunition appendix shall be prepared. This appendix shall contain applicable information on ammunition marking, classification, care and handling, preservation, transportation, packing, and other data pertinent to ammunition usage and

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maintenance. Where appropriate, such data shall be presented in tabular format. See figures 28, 29, 30, and 31 for sample ammunition appendix data presentation.

5.18.9 Mandatory replacement parts appendix. Unless requirement is specifically excluded by the contracting activity, this appendix shall be included and shall list all mandatory replacement parts referenced in the task setups and procedures. Items in the list shall be identified by an item number, nomenclature, and applicable part number.

5.19 Glossary. A glossary shall be included only when the terms are uncommon and are not adequately defined in the text. Refer to MIL-STD-361-1.

5.20 Alphabetical index. An alphabetical index shall be prepared when the number of titled paragraphs in the publication exceeds 100, or as specified by the contracting activity, regardless of the number of paragraphs. Refer to MIL-STD-361-1.

5.21 Quality assurance provisions. Quality assurance and quality control for TMs, revisions, supplements, and changes shall be in accordance with MIL-STD-361-4.

5.22 Packaging and preparation for delivery. TMs, revisions, supplements, and changes shall be packaged in accordance with MIL-STD-361-5.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document prescribes the maintenance instructions applicable to various types of technical manuals, revisions, supplements, and changes prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.1.1.)

6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with MIL-M-63008.

6.4 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced. (See 2.1.1.)

c. TM maintenance level (5.1.2, 5.2.4).

d. Approval of combined operator's and maintenance TMs (5.2.3).

e. Approval of referencing within PMCS table to exceed 120 words or 10 lines (5.5.5.1).

f. Required abbreviations for lubrication instructions not covered in appropriate handbooks (5.5.7.17, 5.9.2.2.2).

g. Preparation for storage or shipment: security procedures, administrative storage instructions, and pre-embarkation inspection of materiel in units alerted for overseas movement (5.5.8, 5.6.4.1, 5.7.4.1, 5.8.4.1, 5.11.4.1, 5.12.4.1, 5.17).

h. Disposition of lots (5.6.3.7).

i. Mobilization requirements, if applicable (5.14.8.4).

j. Authentication page for DMWR (5.14.11).

k. Approval of fabrication of special tools for auxiliary equipment (5.15).

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6.5 Supersession data. This document consolidates, standardizes, and supersedes the maintenance instructions content preparation requirements contained in the following documents:

MIL-M-63036D(TM)	-	Manuals, Technical: Operator's, Preparation of.
MIL-M-63038C(TM)	-	Manuals Technical: Organizational or Aviation Unit, Direct Support or Aviation Intermediate, and General Support Maintenance.
MIL-M-63041E(TM)	-	Manuals, Technical: Preparation of Depot Maintenance Work Requirements.

6.6 Subject term (key word) listing. The following terms are to be used to identify this document during retrieval searches:

Alphabetical index
Ammunition maintenance procedures
Arrangement of combined manuals
Arrangement of procedures
Assembly of equipment
Cable fabrication
Common tools and supplements and special tools/fixtures list
Depot maintenance work requirement (DMWR)
Direct support maintenance instructions
Functional group code (FGC)
General support maintenance instructions
Glossary
Ground handling
Inspection and test of conventional and chemical ammunition
Inspection of installed items
Installation instructions
Lubrication instructions
MAC functional group entries
Maintenance procedures
Manufactured and fabricated items
Measurements and tolerances
Packaging and preparation for delivery
Pre-embarkation inspection of materiel in units alerted for overseas movement
Preliminary servicing and adjustment of equipment
Preparation for storage or shipment
Preparation of illustrations
Preventive maintenance checks and services (PMCS)
Processing unpacked equipment
Radio interference suppression
Service upon receipt of materiel
Site and shelter requirements
Special tools, TMDE, and support equipment
Storage of equipment and systems
Technical manual maintenance levels
Text content
Torque limits
Troubleshooting and fault isolation

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Unit maintenance instructions
Unpacking
Use of maintenance allocation chart (MAC)
Weight and balance

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7.10 MANUAL RAMP-LOCK RELEASE LEVER ASSEMBLY (EN) REPAIR.

INITIAL SETUP

Test Equipment:
Multimeter (Item 4, App C)

Tools and Special Tools:
Fixed open end wrench set (Item 47, App G)
Torque wrench, 3/8 inch drive, 0-600 lb-in. (Item 143, App G)
Torque wrench, 1/2 inch drive, 0-175 lb-ft (Item 144, App G)

Personnel Required: 2
ITV/IFV/CFV Sys Mech 63T10

References:
TM 9-2350-284-10-2

Equipment Conditions:
Engine stopped (TM 9-2350-284-10-1).

SECTION II. TOOL IDENTIFICATION LIST

(1) ITEM NUMBER	(2) ITEM NAME	(3) NATIONAL STOCK NUMBER	(4) PART NUMBER	(5) REFERENCE
1	Adapter, socket wrench, 1/2 inch-3/4 inch	5120-00-114-5207	11655788-3	TM 9-2350-252-20P-1
2	Adapter, socket wrench 3/4 inch-1 inch	5120-00-227-8104		SC 4910-95-CL-A72
3	Adapter, socket wrench, 3/8 inch-1/2 inch	5120-00-240-8703	EX503B	TM 9-2350-252-20P-1
4	Adapter, test	4910-01-138-9334	11629693-1	TM 9-1250-252-20P-1
46	Fire suppression test equipment	4910-01-165-0500	12323388	TM 9-2350-252-20P-1
47	Fixed open end wrench set	5120-00-148-7918		SC 4910-95-CL-A74
48	Fixture, track connecting	5120-00-708-3799	7083799	TM 9-2350-252-20P-1
49	Frame, hacksaw	5110-00-289-9657		SC 4910-95-CL-A74
50	Funnel	7240-00-559-7364		SC 4910-95-CL-A74
51	Welder's tool kit	5180-00-754-0661		SC 5180-90-CL-N39
141	Wrench, torque, 3/8 inch drive, 0-150 lb-in.	5120-00-230-6380	TQ12B	TM 9-2350-284-20P-1
142	Wrench, torque, 3/8 inch drive, 0-200 lb-in.	5120-00-853-4538		SC 4910-95-CL-A72
143	Wrench, torque, 3/8 inch drive, 0-600 lb-in.	5120-00-542-5681	B58	TM 9-2350-284-20P-1
144	Wrench, torque, 1/2 inch drive, 0-175 lb-ft	5120-00-640-6364		SC 4910-95-CL-A74
145	Wrench, torque, 3/4 inch drive, 0-600 lb-ft	5120-00-221-7983		SC 4910-95-CL-A74

FIGURE 1. Example of an initial setup and cross-reference to tool identification list appendix (reference 5.5.2.1b, 5.5.7.1).

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Table X-X. Inspection Criteria for Packaging.

COMPONENT	ACCEPTABLE	REPARABLE	IRREPARABLE
Wooden Boxes and Crates			
Hardware	Operative and tight. Nails, screws, and fasteners present and in good condition.	Inoperative or loose. Nails, screws, and fasteners that can be replaced or properly sealed.	None None
Ends	Free from damage.	Broken or missing cleats and handles.	Damage that requires disassembly of box.
Wood	Splits less than 3 inches long, no closer than 1 inch to edge of board or adjoining split. The board must be secured by at least one nail on each side of the split when it extends to the end of the board.	Splits more than 3 inches but no closer than 1 inch to edge of board or adjoining split, or 1/2-inch wide, that can be repaired by use of corrugated fasteners.	Splits closer than 1 inch to edge of board or adjoining split or over 1/2-inch wide.
Fiber Containers			
Metal Ends	Minor rust, cracks, indentations, or splits that would not impair water proofing or serviceability of container.	None	Perforations, excessive rust, or ends which are crushed or not securely crimped to body.
Body and Cap	No tears, cuts or gouges.	Cuts, tears, or gouges not closer than 1 inch to closure, less than 1/2 square inch in area, and unpenetrated layers that can be spot painted.	Cuts, tears, or gouges closer than 1 inch to closure, more than 1/2 square inch in area, or through all impregnated layers.

FIGURE 2. Example of inspection/service upon receipt (sheet 1 of 2) (reference 5.5.3b, 5.5.3.2).

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Table X-X. M29 and M30 Control Surfaces and Containers.

LOCATION	ITEM	ACTION	REMARKS
1. Container	Components	a. Inspect for rust, fungus, paint damage, and deformation.	Para 4.2
		b. Reject container if damage prevents it from functioning properly.	–
2. M29	Control Surfaces	a. Inspect for dents and scratches on post, trailing edge phenolic, skin, and closure plate.	Para 4.3
		b. Reject control surface:	–
		(1) If post dents or scratches exceed 0.002 in.	
		(2) If trailing edge phenolic dents exceed 0.040 in.	
		(3) If skin dents exceed 0.030 in. within 2 in. of post.	
		(4) If closure plate dents exceed 0.030 in. within 2 in. of post.	
3. M30	Control Surfaces	a. Inspect for dents and scratches on post and skin.	Para 4.18
		b. Skin dents or scratches up to 0.050 in. are allowable, but should be blended.	–
		c. Reject control surface if post dents or scratches exceed 0.002 in.	–

FIGURE 2. Example of inspection/service upon receipt (sheet 2 of 2) (reference 5.5.3b, 5.5.3.2).

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Table 2-1. Preventive Maintenance Checks and Services for the MLRS Launcher.

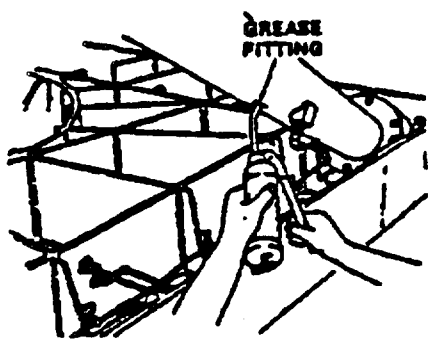
ITEM NO.	INTERVAL	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	NOT FULLY MISSION CAPABLE IF:
1	Quarterly	Elevation Actuator Ballscrew Nut	Perform lubrication as follows: a. Preparation. NOTE The following steps must be done on each elevation actuator. (1) Using boom controller, elevate LLM approximately 400 mils (22.5 degrees) until grease fitting appears in hole of actuator (paragraph 2.18). (2) Using cleaning cloth (item 4, App D), wipe grease fittings. (3) Before using grease gun, pump grease gun handle until a flow of clean grease comes out of grease gun nozzle. Use automotive and artillery grease (item 6, App D). b. Lubrication. CAUTION Do not overlubricate ballscrew nut. You can damage the inside seals. One pump of the grease gun handle is enough.  	

FIGURE 3. Example of PMCS tabular format with lubrication instructions (sheet 1 of 4) (reference 5.5.5, 5.5.5.1b).

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Table 2-1. Preventive Maintenance Checks and Services for the MLRS Launcher - Continued.

ITEM NO.	INTERVAL	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	NOT FULLY MISSION CAPABLE IF:
1	Quarterly	Elevation Actuator Ballscrew Nut (Coat)	(1) Push grease gun nozzle onto grease fitting. Pump grease gun handle one time. Remove grease gun nozzle from grease fitting. (2) Wipe fittings with cleaning cloth (item 4, App D). c. Stow LLM and shut down boom controller (paragraph 2.18).	
2	Quarterly	Boom Rack and Pinion Gears	Perform lubrication as follows: a. Preparation. (1) Turn on FCS and select LLM RIGHT REAR (TM 9-1425-646-10 or TM 9-1425-646-10-1). (2) Using boom controller, extend both booms all the way. (3) On FCP, set SYS PWR switch to OFF. b. Cleaning. CAUTION Do not use solvents to clean these gears. Solvents may deteriorate the plastic slides below each boom or the plastic buttons on the side of each boom. (1) Use cleaning cloth (item 4, App D) to clean teeth of boom rack gear and pinion gear. (2) Using speeder wrench, retract boom slightly to expose remaining gear teeth. c. Lubrication. (1) Using paint brush or grease gun with adapter, apply grease (item 6, App D) to exposed teeth of gears on both left and right booms and carriages. Use automotive and artillery grease. (2) Using speeder wrench, extend boom to original position. (3) Apply approximately 2 ounces of grease (item 6, App D) to pinion gears and using speeder wrench, retract boom while continuing greasing teeth until boom is completely retracted.	

FIGURE 3. Example of PMCS tabular format with lubrication instructions (sheet 2 of 4) (reference 5.5.5, 5.5.5.1b).

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Table 2-1. Preventive Maintenance Checks and Services for the MLRS Launcher - Continued

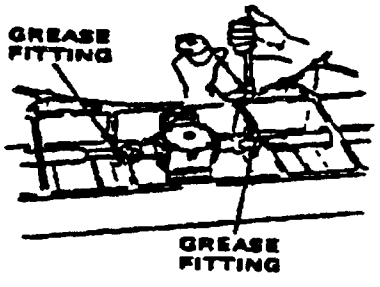
ITEM NO.	INTERVAL	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	NOT FULLY MISSION CAPABLE IF:
9	Annually	Elevation Drive Shaft (Cont)	b. Lubrication. (1) Push grease gun nozzle onto grease fitting. Pump grease gun handle until grease can be seen around universal joint. Disconnect grease gun. Use automotive and artillery grease (item 6, App D). (2) Wipe fittings with cleaning cloth (item 4, App D).  (3) If no further lubrication is required, proceed to paragraphs c and d below. c. Cycle LLM two times in azimuth and check for oil leaks if azimuth gear box was lubricated. d. Stow the LLM and shut down boom controller (paragraph 2.18).	
10	Annually	Boom Ball Nut Drive Assembly	Perform lubrication as follows: a. Preparation. (1) Position LLM at 3200 mils (180 degrees) (TM 9-1425-646-10). (2) Using boom controller, extend booms about 3 feet (paragraph 2.18). (3) Using cleaning cloth, (item 4, App D), wipe grease fittings clean. (4) Using grease gun with grease (item 6, App D), pump grease gun handle until a flow of clean grease comes out of grease gun nozzle.	

FIGURE 3. Example of PMCS tabular format with lubrication instructions (sheet 3 of 4) (reference 5.5.5, 5.5.5.1b).

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Table 2-1. Preventive Maintenance Checks and Services for the MLRS Launcher - Continued

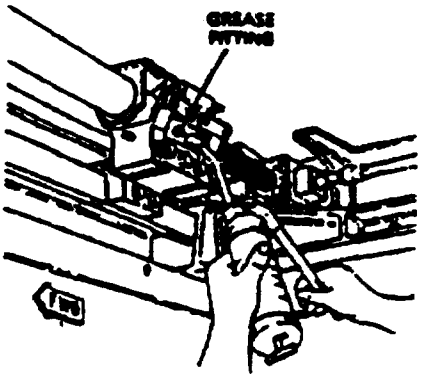
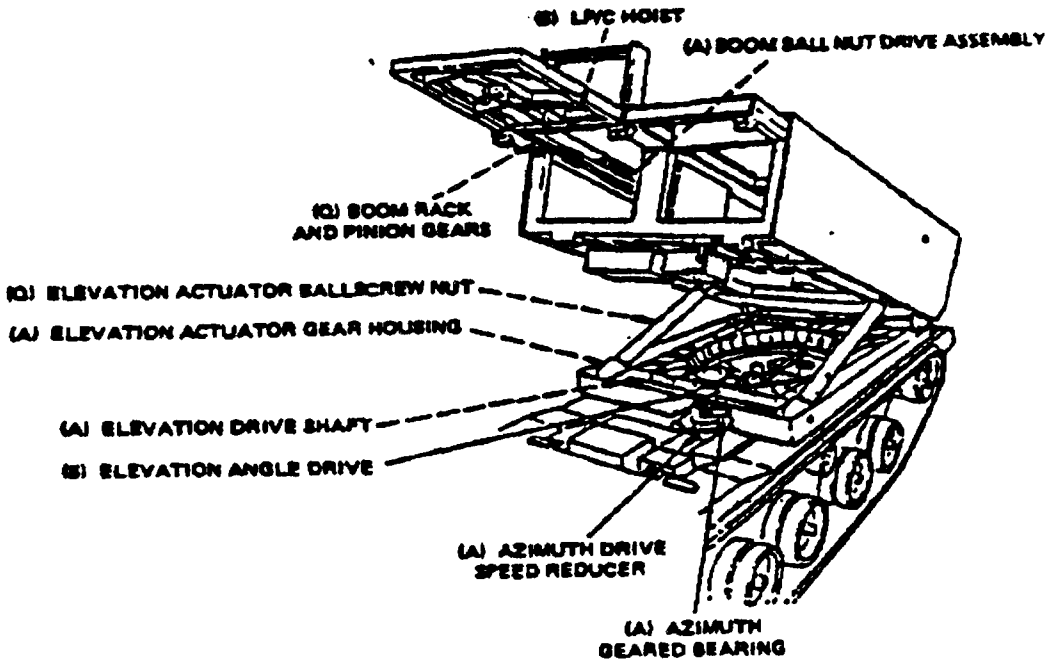
ITEM NO.	INTERVAL	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	NOT FULLY MISSION CAPABLE IF:
10	Annually	Boom Ball Nut Drive Assembly (Cont)	b. Lubrication. (1) Push grease gun nozzle onto grease fitting. Pump grease gun handle two times to lubricate the ball nut drive assembly. Remove grease gun nozzle from grease fitting. (2) Using cleaning cloth (item 4, App D), wipe excess grease from grease fitting.  (3) Using boom controller, retract booms (paragraph 2.18). (4) Stow LLM (TM 9-1425-646-10). c. Cycle LLM two times in azimuth and check for oil leaks if azimuth gear box was lubricated. d. Stow the LLM and shut down boom controller (paragraph 2.18). The following items must be selected for direct support PMCS: SEMIANNUAL: a. Elevating gearbox test and adjustment. b. Recuperator body cap servicing. c. Recuperator air intake filter servicing. d. Fire control instruments purging and servicing. e. Taboo gear adjustment. ANNUAL: Elevating gearbox lubrication.	

FIGURE 3. Example of PMCS tabular format with lubrication instructions (sheet 4 of 4) (reference 5.5.5, 5.5.5.1b).

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KEY
(Q) - QUARTERLY (S) - SEMIANNUALLY (A) - ANNUALLY – - INDICATES REQUIREMENT FOR LEFT AND RIGHT SIDE.

FIGURE 4. Example of lubrication instructions (sheet 1 of 2) (reference 5.5.5.1d).

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LUBRICATION TABLE

LUBRICANTS	COMPONENTS	REFILL CAPACITY (APPROX)	EXPECTED TEMPERATURES		INTERVALS	MAN-HOUR
			+5°F TO +125°F	+20°F TO -70°F		
GMD (MIL-G-21164) GREASE MOLYBDENUM DISULFIDE	Yoke Splines	As Req.	ALL TEMPERATURES		S-1500 Miles or Semi-annually	0.5
WTR (MIL-G-81322) WIDE TEMPERATURE RANGE	Propeller Shaft U-joints Speedometer Adapter Brake Torque Tube Rod Bearings Fan Drive Propeller Shaft	As Req.	ALL TEMPERATURES		S-1500 Miles or Semi-annually	0.5

FIGURE 4. Example of lubrication instructions (sheet 2 of 2) (reference 5.5.5.1d).

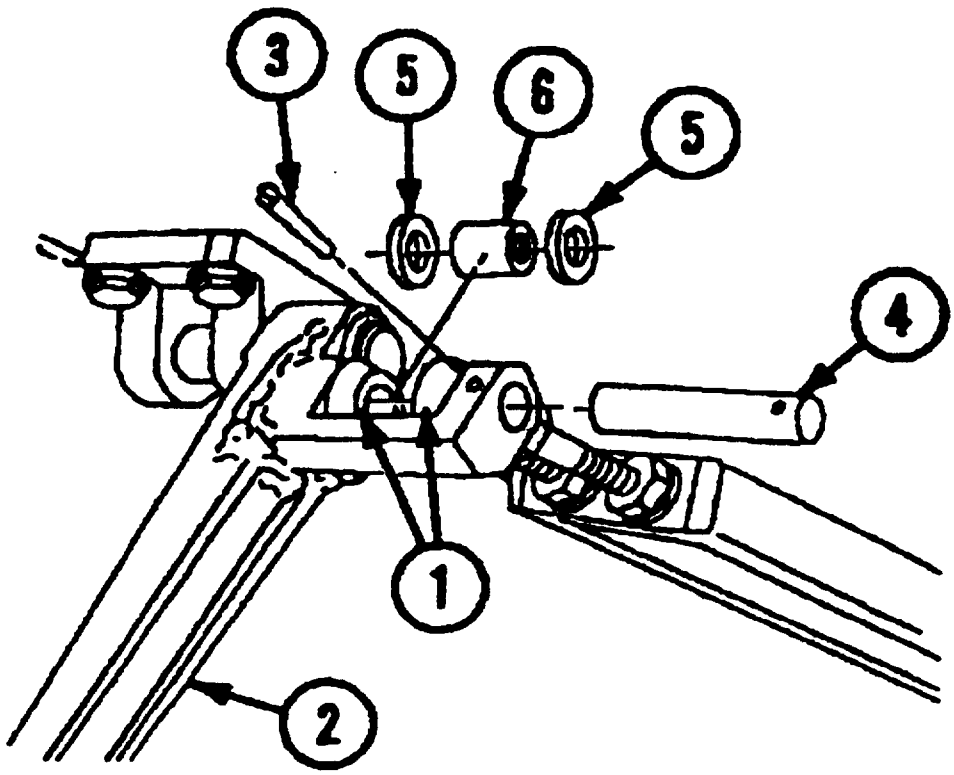
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Table X-X. PMCS Mandatory Replacement Parts List
Semiannual (1500 Mile)

ITEM NO.	PART NUMBER	NSN	NOMENCLATURE	QTY
1	HC7500Y144	4330-01-217-8184	FILTER ELEMENT, FLUID (M977/M985 ONLY)	1
2	HD223	2940-01-132-4842	FILTER ELEMENT, FLUID	1
3	MS35338-46	5310-00-637-9541	WASHER, LOCK (ALL EXCEPT M984A1)	2
4	MS35802-3	2940-00-580-6283	FILTER ELEMENT, FLUID	1
5	S-268-1	5306-01-084-5390	BOLT, MACHINE (ALL EXCEPT M984A1)	2
6	WA110	5310-01-061-5302	WASHER, LOCK	4
7	10232C	5330-01-168-8707	GASKET (M978 ONLY)	1
8	11007	5330-01-046-1990	GASKET	1
9	1112310	5330-01-225-4803	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
10	1124510	5330-01-143-0135	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
11	1128242	5330-01-145-1112	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
12	11350	5330-01-147-6003	PACKING, PREFORMED	1
13	1199478	5330-01-234-7625	GASKET	1
14	1199738	4330-01-192-7664	SCREEN, BY-PASS FILTER (M977/M985/M984A1 ONLY)	1
15	1300766	4330-01-232-8305	FILTER ELEMENT, FLUID (M984A1 ONLY)	1
16	1300767	4330-01-192-8832	FILTER ELEMENT, FLUID (M977/M985/M984A1 ONLY)	1
17	2020SM	4330-01-046-3399	FILTER ELEMENT, FLUID	1
18	2463HX	5310-01-054-5896	WASHER, FLAT (ALL EXCEPT M984A1)	2
19	25010778	2910-01-022-8183	FILTER ELEMENT, FLUID	1
20	4-1/2IN77/	5330-01-163-5849	GASKET (M978 ONLY)	2
21	45D020P6	5330-01-156-3764	PACKING, PREFORMED (M978 ONLY)	1
22	941107	4330-01-163-7326	FILTER ELEMENT, FLUID (M978/M983/M984/M985E1 ONLY)	1

FIGURE 5. Example format for mandatory replacement parts for PMCS (reference 5.5.5.2).

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WARNING

Soldiers can be killed or injured if ramp is lowered without warning. Make sure no one is in ramp zone before you lower ramp. If tactical situation permits, sound horn before dropping ramp.

a. Lower ramp. (See TM 9-2350-284-10-1.)

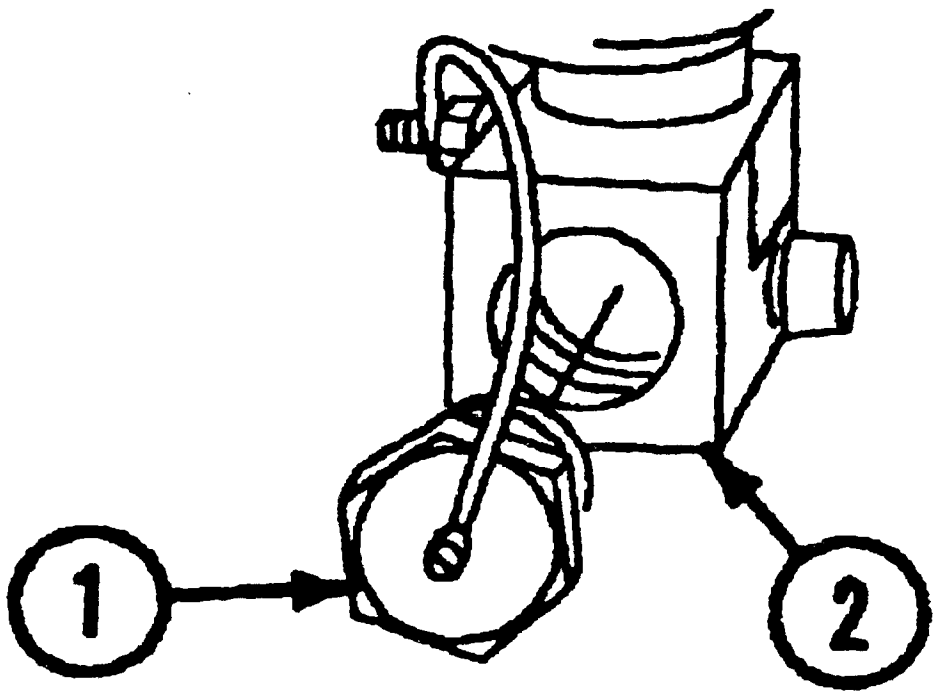
b. Remove two rod end bearings (1) from handle (2) as follows:

(1) Remove spring pin (3) from pin (4). Discard spring pin.

(2) Remove pin (4), two washers (5), spacer (6), and two rod end bearings (1) from handle (2).

FIGURE 6. Example of a typical illustrated, step-by-step maintenance procedure (reference 5.5.7.4).

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8.9.1 Removal.

a. Move fire suppression switch to manual. (See TM-9-2350-284-10-1.)

WARNING

Accidental discharge of fire bottles could injure you. Remove vehicle ground cable.

b. Remove vehicle ground cable (para 7.13).

WARNING

Fire extinguisher can discharge and injure you. Insert antirecoil plugs, lock pins, and cotter pins before you work near fire extinguishers.

c. Install plug (1) in extinguisher (2).

FIGURE 7. Example of a typical illustrated, step-by-step removal procedure (reference 5.5.7.9).

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d. Place gear clutch assembly (1) in assembly tool (5) with the clutch spur gear (2) on top as shown.

e. Rotate gear clutch assembly (1) until timing marks (6) are in position as shown.

f. Grasp small spur gear (7) and pull down while lifting the clutch spur gear (2) from spur gear shaft (8).

g. Push in on handle (9) of tool (5) and hold tightly while turning small spur gear (7) until lugs (10) are in line with shaft (11).

h. Slowly release pressure on handle (9) and remove roller (12) next to shaft (11).

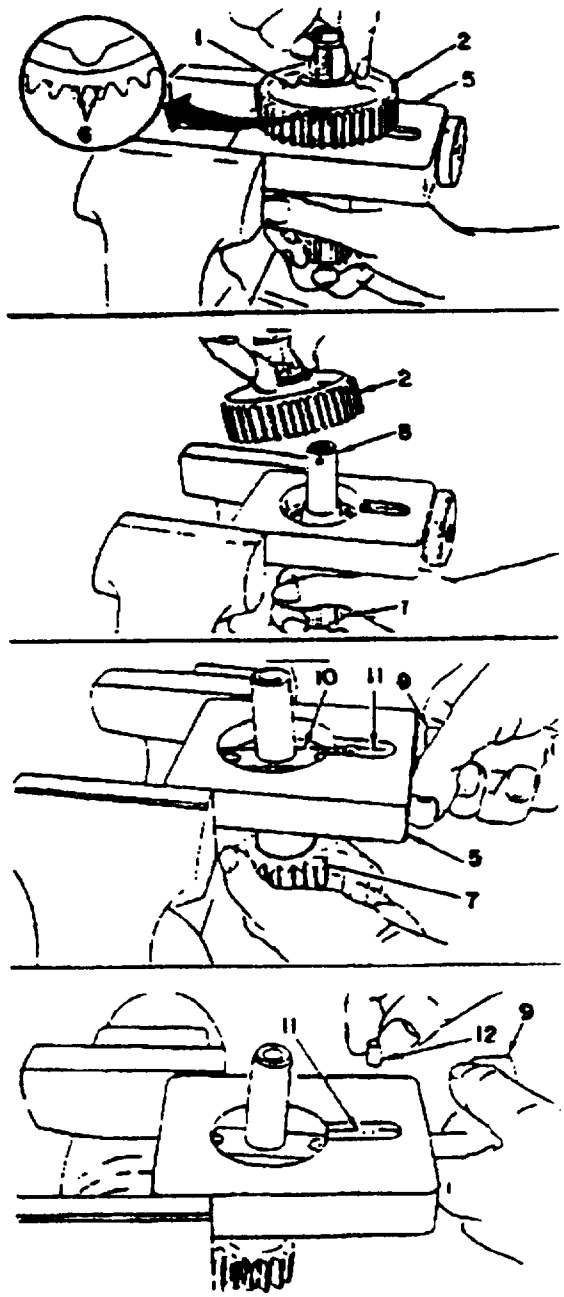
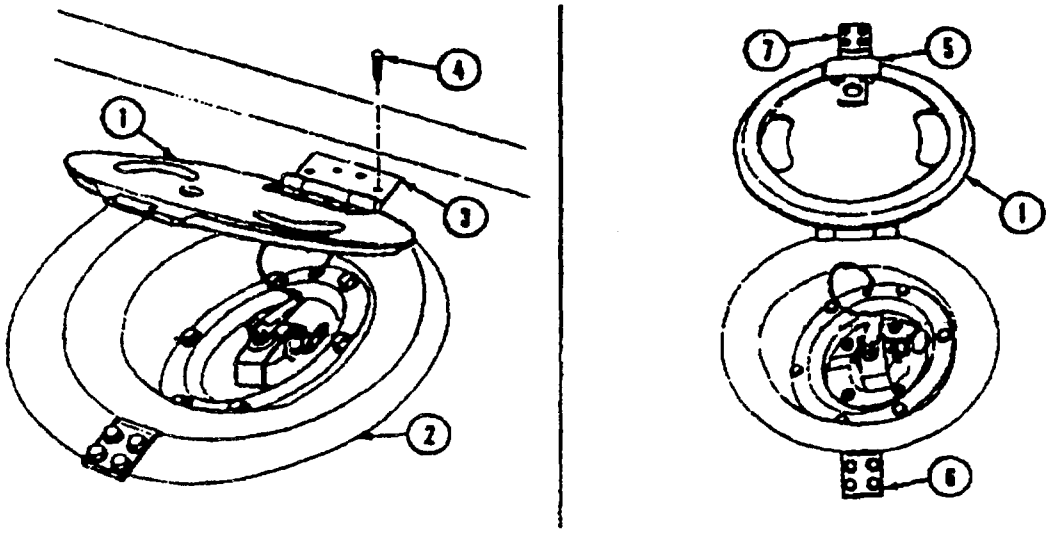


FIGURE 8. Example of a typical illustrated, step-by-step disassembly procedure (reference 5.5.7.10c).

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i. Install cover (1) on ramp (2) as follows:

(1) Position hinge (3) and cover (1) on ramp (2).

(2) Apply sealing compound (item 34, App D) to threads of four screws (4).

(3) Apply sealing compound (item 32, App D) to threads of four screws (4).

(4) Install cover (1), hinge (3), and four screws (4) on ramp (2).

j. Check operation of cover (1) and latch (5):

(1) Check that cover (1) latches securely to two strikes (6 and 7).

(2) If cover (1) does not latch securely, repeat e., f., and g. to adjust two strikes (6 and 7).

k. Raise ramp. (See TM 9-2350-252-10-1.)

FIGURE 9. Example of a typical illustrated, step-by-step assembly procedure (reference 5.5.7.18).

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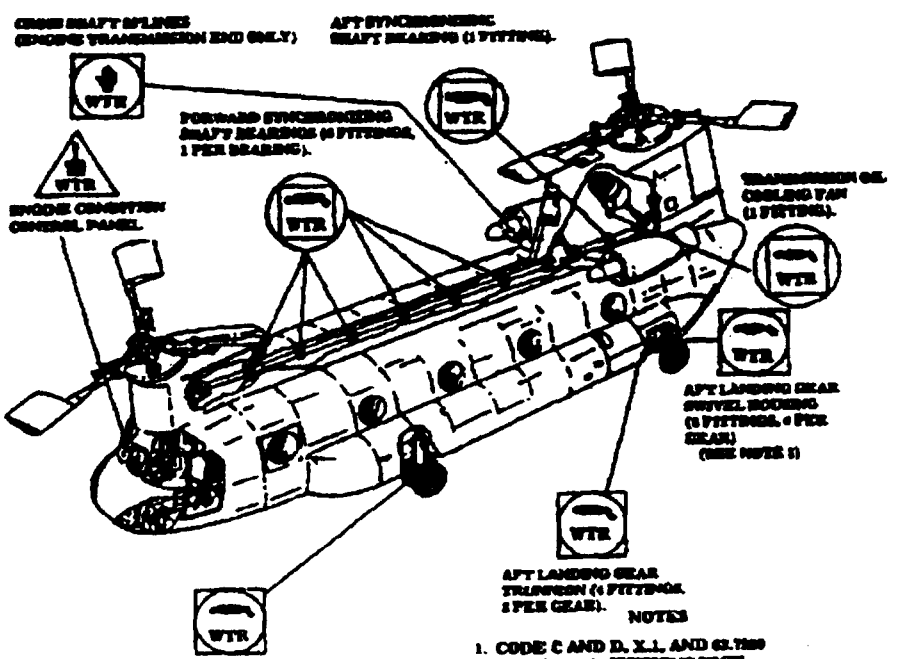
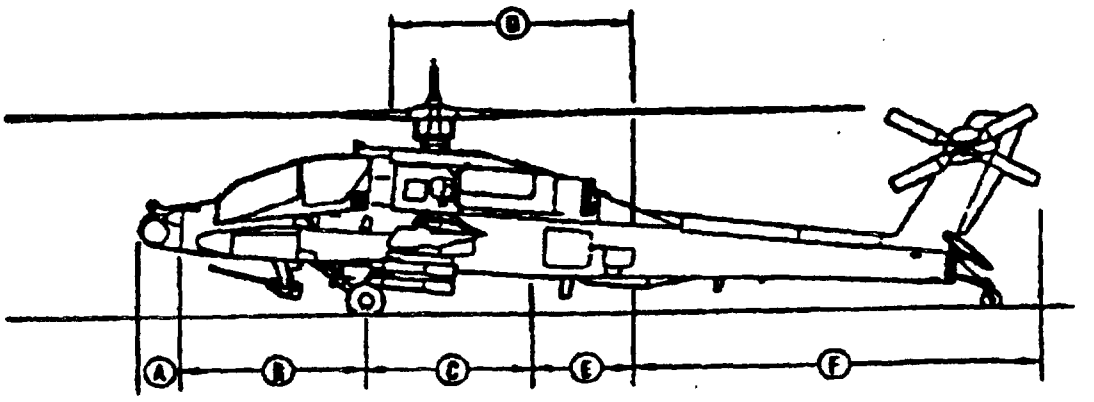


TABLE OF LUBRICANTS		
IDENTIFICATION LETTER	SPECIFICATION	TYPE OF LUBRICANT
WTR	MIL-G-81322B (Cl2)	GREASE, AIRCRAFT GENERAL PURPOSE, WIDE TEMPERATURE RANGE

SYMBOLS	FREQUENCY	METHOD OF APPLICATION
	25 HOURS	
	100 HOURS	
	500 HOURS	

FIGURE 10. Example of a lubrication chart for AVUM and AVIM maintenance (reference 5.9.2.2.2).

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NOMENCLATURE	PART NUMBER	NSN	QTY PER
SECTION A - NOSE ELECTRONICS			
TADS Turret	130760000	1270-01-188-4138	1
PNVS Turret and Fairing	130900000	1090-01-169-9415	1
Multiplex Terminal - Type 3	4032298-956	7025-01-210-7768	1
Signal Data Converter - Eng.	7-311C20019	6620-01-160-3518	1
Radar Jammer Transmit Antenna	SM-A-919685-2		1
Radar Warning Antenna	AS2891/APR-39V	1020-00-024-7608	2
Power Transformer	7-219510089	5950-01-186-8032	1
SECTION B - RIGHT FORWARD AVIONICS BAY			
Gun Control Box	7-317222500-603	1005-01-211-4165	1
Multiplex Terminal - Type 1	4032297-955	7025-01-211-0130	1
Turret Control Box	7-317222004	5930-01-239-2391	1
PNVS Electronic Box	13080274	4931-01-169-9369	1
Display Electronics - IHADSS	7-319430041	1270-01-183-0518	1
Sight Electronic Unit - IHADSS	7-319430031	1270-01-183-0519	1
Fire Control Computer	7-319200001	1430-01-211-0023	1
Remote Electronic Unit	7-317141001	1270-01-187-5778	1
Portable Fire Bottle	7-211180002-501	4210-00-555-8837	1
Electromagnet Relay	7-211812033	5945-01-160-5639	1

FIGURE 11. Example of inventoriable items presentation (reference 5.9.4.1.).

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Table 2-1. Special Tools and Test Equipment.

PART NO. (CAGEC)	PARA OR FIGURE NO.	NOMENCLATURE	USE AND APPLICATION (REF DOCUMENT)
T-608301 (06481)	5.32	Gyro and accelerometer torquer	Control bucking magnet flux density (TB XXX-XXX)
TE-113303 (06481)	2.15	Accelerometer driver magnet surge tester	To charge accelerometer drive magnets
TE-118525 (06481)	2.16	Torquer magnet stabilizer	To stabilize accelerometer torquer magnets (TB XXX-YY)
657932 (06481)	2.17	Instrument fill station	To fill accelerometers with floation fluid
661641 (06481)	2.18	Accelerometer precheck station	To null excitation yoke assemblies during assembly of accelerometer subassembly

FIGURE 12. Example of special tools and test equipment list format (reference 5.14.3.1b).

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Table 2-4. Mandatory Replacement Parts List for (enter equipment name).

ITEM DESCRIPTION	PART NUMBER (CAGEC)	NSN	LOCATION ITEM/FIG. NOS. (INTERVAL)	QTY
Gasket (Large prism front)	7688712 (19200)	5330-00-138-9791	16, fig. 4-3 3, fig. 4-7 (50,000 mi)	1
Packing, preformed (HV connector)	7688792 (19200)	5330-00-297-5353	15, fig. 4-3 (50,000 mi)	1
Packing, preformed (eccentric pin)	MS2877-112 (96906)	5330-00-599-2934	16, fig. 4-3 (50,000 mi)	2

FIGURE 13. Example of a mandatory replacement parts list (reference 5.14.3.2a).

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PRESHOP ANALYSIS FOR P/N _ _ _ _ _

Serial No. _ _ _ _ _ _ _

NSN _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

MWOs Required _

Reason(s) for Overhaul/Repair _____

_____

_____

_____

_____

Unpacking Secondary Items Required ? _ _ _ _

Reviewed Tags ? _ _ _ _

Reviewed Forms ? _ _ _

Name (please print) _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Signature _ _ _ _ _ _ _ _ _ _ _ _ Date _ _ _ _ _ _

FIGURE 14. Example of a cover for preshop analysis checklist (reference 5.14.4.2.2a).

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Table 3-1. Preshop Analysis Checklist.

INSPECTION POINT	CONDITION	ACTION	REMARKS	DONE
GENERAL				
Gear Box	Reason(s) for removal from service.	Record:		
	Modification only.	Paragraph 3.4a(5)		
	Burned.	Paragraph 3.4a(6)		
	Crash damage, accident damage, hard landing, sudden stoppage.	Paragraph 3.4a(7)		
	Metal on magnetic plug, contamination, internal failure.	Paragraph 3.4a(8),(9)		
	Dropped.	Paragraph 3.4a(10)		
	No records or historical data.	Paragraph 3.4a(11)		
EXTERIOR INSPECTION				
Case Assembly	Cracks, corrosion, nicks, dents, scratches, pits, broken studs.	Visual, OIP-16		
Output Sleeve	Cracks, corrosion, nicks, dents, burrs, scratches.	Visual, OIP-22		
Output Shaft	Cracks, nicks, dents, or pits on spline; wear over pin dimension of spline not within allowable limits; corrosion.	Visual, OIP-29		
Input Sleeve	Cracks.	Visual, OIP-1		

FIGURE 15. Example of a preshop analysis checklist (reference 5.14.4.2.2b).

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Table 3-21. OIP for Gear (Item 15, fig. 3-21).

	NO.	REF LTR	CHARACTERISTIC	INSP METHOD	REQUISITE
	1		Serviceability	Visual	No cracks, chipped grooves, or missing gear teeth
(QA)	2		Metal fatigue	Magnetic Particle Inspection	No cracks
(QA)	3	A	Tooth wear	Measure with micrometer (599-30-11) over two 0.1920 in. (4.877mm) gage pins	2.256 in. (57.30 mm) min.

FIGURE 16. Example of an overhaul inspection procedure (OIP) table (reference 5.14.5.2e).

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Table 4-2. Mandatory Replacement Parts for (enter name of item).

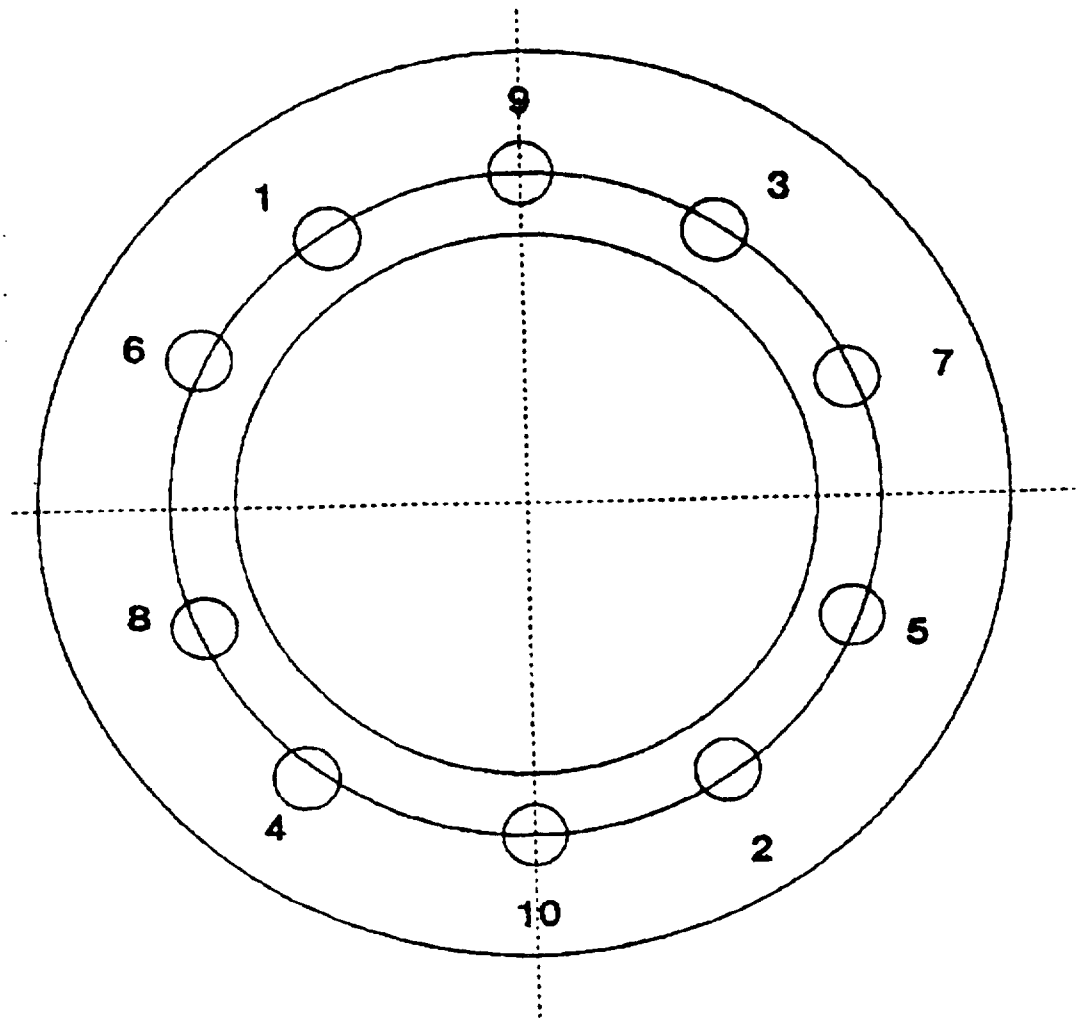
ITEM DESCRIPTION	PART NUMBER (CAGEC)	NSN	LOCATION ITEM/FIG. NO. (INTERVAL)	QTY
* Bearing kit-cam	324143 (4N530)	3120-01-266-6801	34, 4-17	1
Bearing-radial ball	32578 (4N530)	3120-01-224-8082	8, 4-17 (Overhaul)	1
* Cup-sealing drilled	323818 (4N530)	5340-01-266-5116	18, 4-17	1
* Gasket	382490 (4N530)	5330-01-267-1231	53, 4-17	1
Spacer-crank oil seal	324163 (4N530)	5365-01-224-9095	4, 4-17 (Overhaul)	1

NOTE: Items annotated with an asterisk (*) must be replaced when (enter component name) is disassembled.

FIGURE 17. Example of a table of mandatory replacement parts (when item is disassembled) (reference 5.14.5.2f).

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SCREW TORQUING SEQUENCE: WITH SCREWS NUMBERED 1 THROUGH 10 AS SHOWN (STARTING POINT OPTIONAL) TORQUE 1 & 2, 3 & 4, 5 & 6, 7 & 8, 9 & 10.



NOTE

FULL TORQUE ON SCREWS IS TO BE ACHIEVED IN INCREASING INCREMENTS OF 10 POUND-INCHES PER SCREW PER SEQUENCE SPECIFIED ABOVE. DO NOT APPLY FULL TORQUE IMMEDIATELY TO ANY SCREW.

FIGURE 18. Example of screw torquing sequence (reference 5.14.5.2i).

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DMWR 9-1005-318-4

FOR THE COMMANDER:

LARRY D. BACHELOR
Colonel, GS
Chief of Staff

OFFICIAL:

R.D. HUSSON
Director of Maintenance

DISTRIBUTION:

COMMANDER RED RIVER ARMY DEPOT ATTN: SDSRR-ME TEXARKANA, TX 75507-5000	5 CYS
COMMANDER U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND ATTN: AMSMC-MAE-WA AMSMC-MAW-AD AMSMC-QAW AMSMC-MAS-B AMSMC-MAF-AP ROCK ISLAND, IL 61299-6000	 2 CYS 2 CYS 2 CYS 15 CYS 2 CYS 2 CYS
COMMANDER AMC MATERIEL READINESS SUPPORT ACTIVITY ATTN: AMXMD-MP LEXINGTON, KY 40511-5101	2 CYS

FIGURE 19. Example of an authentication page (reference 5.14.11).

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ARMY TM 10-1670-274-23&P
AIR FORCE T.O. 14D1-2-446-2
MARINE CORPS TM 1670-23&P/1

APPENDIX A

REFERENCES

A-1 SCOPE.

This appendix lists all forms, TMs, and miscellaneous publications referenced in this manual.

A-2 PUBLICATION INDEXES.

The following publication indexes should be consulted frequently for the latest changes or revisions of references given in this appendix and for new publications relating to the material covered in this manual:

Consolidated Index of Army Publications and Blank Forms . . . . . DA PAM 25-30

The Army Maintenance Management System (TAMMS) . . . . . DA PAM 738-750

A-3 TECHNICAL MANUALS.

General Maintenance of Parachutes and Other Airdrop Equipment. . . . . TM 10-1670-201-23/ T.O. 13C-1-141/ NAVAIR 13-1-17

Miscellaneous Airdrop Canvas, Webbing, Metal, and Wood Items . . . . . TM 10-1670-240-20/ T.O. 13C7-49-11

Preservation, Packaging, Packing of Military Supplies and Equipment (Vols 1 and 2) . . . . . TM 38-230-1 and TM 38-230-2

Procedures for the Destruction of Air Delivery Equipment to Prevent Enemy Use. . . . . TM 43-0002-1

A-4 FIELD MANUALS.

First Aid for Soldiers . . . . . FM 21-11

A-5 ARMY REGULATIONS.

Dictionary of United States Army Terms . . . . . AR 310-25
Authorized Abbreviation and Brevity Codes . . . . . AR 310-50
Packaging of Material . . . . . AR 700-15
Army Materiel Maintenance Concepts and Policies . . . . . AR 750-1

A-6 TECHNICAL BULLETINS.

Maintenance Expenditure Limits for FSC Group 16 . . . . . TB 43-0002-4 (To be superseded in part by TB 43-0002-43)

Use of Material Condition Tags and Labels on Army Aeronautical and Air Delivery Equipment . . . . . TB 750-126

A-1/A-2 blank

FIGURE 20. Example of references appendix format (reference 5.18.1).

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SECTION II. MAINTENANCE ALLOCATION CHART FOR TSEC/ST-34

(1) GROUP NUMBER	(2) COMPONENT/ASSEMBLY	(3) MAINTENANCE FUNCTION	(4) MAINTENANCE LEVEL					(5) TOOLS AND EQUIPMENT REF CODE	(6) REMARKS CODE
			UNIT		DIRECT SUPPORT	GENERAL SUPPORT	DEPOT		
			C	O	F	H	D		
00	TSEC/ST-34	Inspect	0.1						A
		Service	0.2						B
		Replace	0.4					1	C,D
		Test	0.3						E
		Repair			1.5			1-2	F
		Repair				2.0		1-5	G,J
		Repair					2.0	1-9	H
		Overhaul					16.0	1-9	I
01	STP-34 POWER UNIT	Inspect	0.1						A
		Test	0.3						E
		Repair			1.8			1-2	F
		Repair				2.0		1-5	G,J
		Repair					2.0	1-9	H
0101	STP-34 PRINTED CIRCUIT BOARDS								
010101	E-EB0/1	Inspect			0.1				A
		Test			0.5			1,2	E
		Test					1.0	1-3,6-8	I
		Replace			0.5			1	
		Repair					2.0	1-4,6-8	G
010102	SWITCHING ASSEMBLY	Inspect			0.1			1	A
		Replace			0.5			1	
		Test					1.0	1-3,6-8	
		Repair					2.0	1-4,6-8	H
02	STB-34 LOGIC UNIT	Inspect	0.1						A
		Test	0.5						E
		Repair			1.0			1-2	F
		Repair				2.0		1-5	G,J
0201	STB-34 PRINTED CIRCUIT BOARDS	Inspect			1.0				A
		Test			0.2				E
		Test				1.0		1-3.5	J
		Replace			0.3			1	
		Repair				2.0		2-5	G

FIGURE 21. Example of a standard MAC appendix (reference 5.18.2.6).

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SECTION II. MAINTENANCE ALLOCATION CHART (AVIATION)
FOR
T-XXX TURBINE ENGINE

(1) GROUP NUMBER	(2) COMPONENT/ASSEMBLY	(3) MAINTENANCE FUNCTION	(4) MAINTENANCE LEVEL AVIM(O)	AVIM(F)	DEPOT(D)	(5) TOOLS & EQUIP REF CODE	(6) REMARKS CODE
04	POWER PLANT						
0401	TURBINE ENGINE	Inspect	–			1	1
		Test	–				2
		Test		–(3)			
		Test			–		
		Service	0.2				10
		Install	–				
		Replace	–				
		Repair	–				
		Repair		–(4)			A
		Overhaul			–		
040101	EXTERNAL LINES & HOSES	Inspect	–			3	11
		Test		–			
		Install	–				
		Replace	–				
		Repair		–			
0402	COMPRESSOR SECTION (COLD SECTION MODULE)	Inspect	0.1				11
		Inspect		0.2			
		Test		–			
		Service	0.2				
		Repair	0.4				
		Repair		0.6			
		Overhaul			–		

FIGURE 22. Example of aviation MAC appendix (reference 5.18.2.6).

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SECTION III. TOOLS AND TEST EQUIPMENT FOR TSEC/ST-34

TOOL OR TEST EQUIPMENT REF CODE	MAINTENANCE LEVEL	NOMENCLATURE	NATIONAL STOCK NUMBER	TOOL NUMBER
1	F	Tool, kit, electronic equipment	5180-00-610-8177	TK-105/6
2	F	Multimeter, digital	6625-01-139-2512	AN/PSM-45
3	H	Oscilliscope	6625-01-187-7847	AN/USM-488
4	H	Repair and soldering center (page)	4940-01-031-4541	PRC-350C/equip
5	H	Automatic test system ST-51	5810-00-089-4599	TSEC/ST-51
6	D	Power supply (0-35 VDC 2.4A)	6130-00-006-5224	HP 6434B86
7	D	Multimeter, digital	6625-01-145-2430	AN/USM 486
8	D	Power supply tester	N/A	ON502427

SECTION IV. REMARKS FOR TSEC/ST-34

REMARKS CODE	REMARKS
A	External.
B	Preventive maintenance checks and services (PMCS).
C	Replace rack installed unit, 0.4 hrs.
D	Bench top use only, 0.1 hrs.
E	Self-test.
F	Repair by PMA and authorized component replacement only.
G	Complete unit and subassembly repair (except STP-34 switching assembly and E-EBO/1).
H	Complete unit and subassembly repair.
I	In compliance with TSEC/ST-34 CIDOS.
J	Function performed by specialized repair activity (SRA). (Theater COMSEC Logistics Support Center-Europe or Lexington-Blue Grass Army Depot)

FIGURE 23. Example of standard MAC tools and test equipment and remarks sections (reference 5.18.2.6e).

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SECTION III. TOOLS AND TEST EQUIPMENT REQUIREMENTS FOR T-XXX TURBINE ENGINE

TOOL OR TEST EQUIPMENT REF CODE	MAINTENANCE LEVEL	NOMENCLATURE	NATIONAL STOCK NUMBER	TOOL NUMBER
1	AVUM/AVIM	Sling, aircraft maintenance	1730-00-903-5019	LTCT 773
2	AVUM/AVIM	Wrench, crowfoot	5120-00-034-6193	LTCT 4810
3	AVIM	Wrench, socket	5120-00-875-2588	LTCT 393
4	AVIM	Wrench, spanner	5120-00-886-1794	LTCT 9263

SECTION IV. REMARKS FOR T-XXX TURBINE ENGINE

REMARKS CODE	REMARKS
1	Diagnostic inspection using borescope.
2	Functional test at AVUM - engine in airframe.
3	Functional test at AVIM - engine in METS.
4	Repair at AVIM includes the engine assembly, individual line replacement units (LRU) (accessories) and modules.
5	Replace seal.
6	Repair limited to replacement of rotor assembly, stator, stage 1 nozzle, face type seal, and combustion liner.
7	Repair limited to replacement of external lines, hoses, and line replacement units (LRU) (accessories).
8	Replacment of carbon seal.
9	Reset button.
10	Water wash.
11	Visible inspection without detailed disassembly.
A	All repair and replacment of parts performed by AVUM is limited to authorized items listed in TM (cite specific TM-20P or -23P).

FIGURE 24. Example of aviation MAC tools and test equipment and remarks sections (reference 5.18.2.6e).

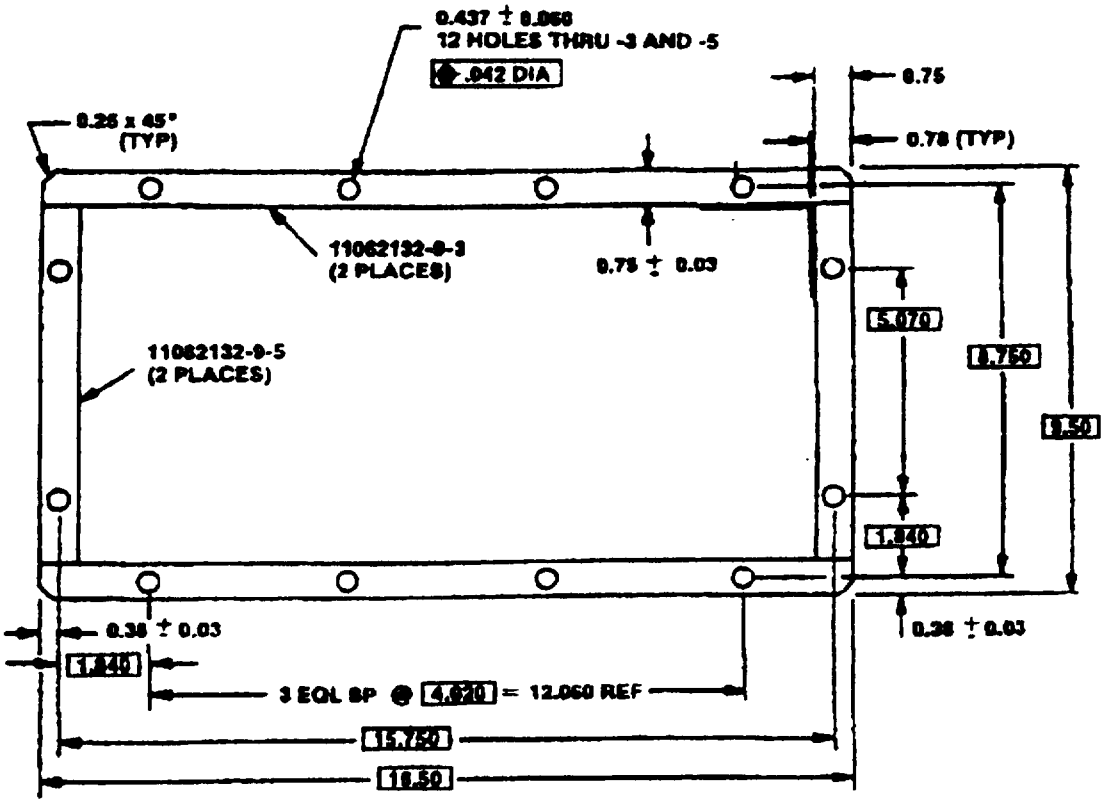
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SECTION II. TOOL IDENTIFICATION LIST

(1) ITEM NO.	(2) ITEM NAME	(3) NATIONAL STOCK NUMBER	(4) PART NUMBER	(5) REFERENCE
1	Adapter, socket wrench, 1/2 inch-3/4 inch	5120-00-114-5207	11655788-3	TM 9-2350-252-20P-1
2	Adapter, socket wrench, 3/4 inch-1 inch	5120-00-227-8104		SC 4910-95-CL-A72
3	Adapter, socket wrench, 3/8 inch-1/2 inch	5120-00-240-8703	EX503B	TM 9-2350-252-20P-1
4	Adapter, test	4910-01-138-9334	11629693-1	TM 9-1250-252-20P-1
5	Adapter, test	4910-01-138-9335	11629693-2	TM 9-1250-252-20P-1
6	Adapter, torque wrench, 1/2 inch drive, 1/2 inch	5120-00-399-1157	2588756	TM 9-2350-252-20P-1
7	Adapter, torque wrench, 1/2 inch drive, 3/4 inch	5120-00-399-1154	2588757	TM 9-2350-252-20P-1
8	Adapter, torque wrench, 1/2 inch drive, 5/16 inch	5120-01-115-1891	12298105-1	TM 9-2350-252-20P-1
9	Adapter, torque wrench, 1/2 inch drive, 15/16 inch	5120-00-215-8200	11663358-2	TM 9-2350-252-20P-1
10	Adjusting tool, belt	4910-01-128-2670	3375058	TM 9-2350-252-20P-1
11	Automotive electrical tool kit	5180-00-422-8594		SC 4910-95-CL-A74
12	Bit, screwdriver, 1/4 inch drive	5120-00-316-9228	TMC105A	TM 9-2350-252-20P-1
13	Bit, screwdriver, 3/8 inch drive, flat-tip	5120-00-243-7338		SC 4910-95-CL-A72
14	Blind hand riveter (kit)	5120-00-017-2849		SC 4910-95-CL-A74
15	Brush, wire, scratch	7920-00-291-5815		SC 4910-95-CL-A74
16	Bushing, pipe, 1/4 X 1/8 inch	4730-00-277-9386		SC 4910-95-CL-A74
17	C-clamp, 4 inch	5120-00-222-1612		SC 9410-95-CL-A72
18	C-clamp, 6 inch	5120-00-203-6431		SC 4910-95-CL-A72
19	Cap screw, hex head	5305-00-735-2317	MS90728-64	Improvised Tool
20	Chisel, hand cape	5110-00-271-9943		SC 4940-95-CL-A08
21	Compressor, power air	4310-00-752-9633		SC 4910-95-CL-A72
22	Crimping tool, hand	5120-00-132-6913	M22520/5-01	TM 9-2350-252-20P-1
23	Crimping tool, terminal	5120-01-165-0557	357-588	TM 9-2350-252-20P-1
24	Crimping tool, terminal	5120-00-432-7836	MY28	TM 9-2350-252-20P-1
25	Crowbar	5120-00-224-1390		SC 4910-95-CL-A72
26	Crowfoot attachment, 3/8 inch drive, 9/16 inch	5120-01-092-3276	FRH-180S	TM 9-2350-252-20P-1
27	Die set	5120-01-139-9674	Y554	TM 9-2350-252-20P-1
28	Die, crimping tool	5120-00-126-0860	M22520/5-100	TM 9-2350-252-20P-1
29	Drift pin, track	2530-01-075-8292	2590157	TM 9-2350-252-20P-1
30	Drill, portable electric, 1/2 inch	5130-00-889-9004		SC 4910-95-CL-A74
31	Electrical connector repair tool kit	5180-00-876-9336		SC 4910-95-CL-A74
32	Extractor, torsion bar	5120-01-115-4920	12298092	TM 9-2350-252-20P-1
33	Fire suppression test equipment	4910-01-165-0500	12323388	TM 9-2350-252-20P-1

FIGURE 25. Example of tool identification list (reference 5.18.3.2).

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MATERIALS	
DESCRIPTION	NSN
Cellular rubber, chem blwn, MIL-R-6130, Grade A, soft, Type 2	9320-00-034-7442

NOTES:
1. Dimensions shown are in inches.
2. Tolerances are 0.03 inch unless otherwise stated.
3. Angular tolerance is ± 2°.

PROCEDURE:
1. For fabricating cover gasket P/N XXXXXX, use cover 11062132-1 as template.
2. Cut rubber as shown, using template as guide.
3. Drill holes as shown, using template as guide.

Figure F-2. Fabrication of Cover Gasket.

FIGURE 26. Example of a manufactured items illustration (reference 5.18.6.3a).

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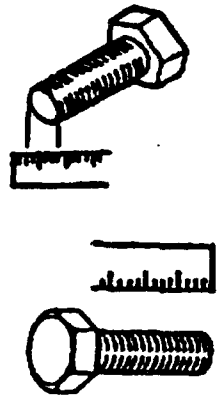
HOW TO USE TORQUE TABLE

a. Measure the diameter of the screw you are installing.

b. Count the number of threads per inch or use a pitch gage.

c. Under the heading SIZE, look down the left-hand column until you find the diameter of the screw you are installing. (There will usually be two lines beginning with the same size.)

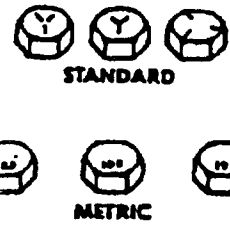
d. In the second column under SIZE, find the number of threads per inch that matches the number of threads you counted in step b. (Not required for metric screws.)



CAPSCREW HEAD MARKINGS

Manufacturer's marks may vary. These are all SAE Grade 5 (3-line).

Metric screws are of three grades: 8.8, 10.9, and 12.9. Grades and manufacturer's marks appear on the screw head.



e. To find the grade screw you are installing, match the markings on the head to the correct picture of CAPSCREW HEAD MARKINGS on the torque table.

f. Look down the column under the picture you found in step e. until you find the torque limit (LB-IN./FT or NM) for the diameter and threads per inch of the screw you are installing.

FIGURE 27. Example of a typical torque limits appendix (sheet 1 of 5) (reference 5.18.7).

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TORQUE LIMITS FOR DRY FASTENERS

CAPSCREW HEAD MARKINGS



Manufacturer's marks may vary.
These are all SAE Grade 5 (3-line).

SIZE			TORQUE							
			SAE GRADE NO. 2		SAE GRADE NO. 5		SAE GRADE NO. 6 OR 7		SAE GRADE NO. 8	
Dia. Inches	Threads Per Inch	Millimeters	Pound Feet	Newton Meters	Pound Feet	Newton Meters	Pound Feet	Newton Meters	Pound Feet	Newton Meters
1/4	20	6.35	5	7	8	11	10	14	12	16
1/4	28	6.35	6	9	10	14	12	16	14	19
5/16	18	7.94	11	15	17	23	21	28	25	34
5/16	24	7.94	12	16	19	26	24	32	25	34

TORQUE LIMITS FOR WET FASTENERS

CAPSCREW HEAD MARKINGS



Manufacturer's marks may vary.
These are all SAE Grade 5 (3-line).

SIZE			TORQUE							
			SAE GRADE NO. 2		SAE GRADE NO. 5		SAE GRADE NO. 6 OR 7		SAE GRADE NO. 8	
Dia. Inches	Threads Per Inch	Millimeters	Pound Feet	Newton Meters	Pound Feet	Newton Meters	Pound Feet	Newton Meters	Pound Feet	Newton Meters
1/4	20	6.35	4	6	6	8	8	11	9	12
1/4	28	6.35	5	7	7	9	9	12	10	14
5/16	18	7.94	8	11	3	18	16	22	18	24
5/16	24	7.94	9	12	14	19	18	24	20	27

FIGURE 27. Example of a typical torque limits appendix (sheet 2 of 5) (reference 5.18.7).

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TIGHTENING METAL FASTENERS

When torquing a fastener, select a wrench whose range fits the required torque value. A torque wrench is most accurate from 25% to 75% of its stated range. A wrench with a stated range of 0 to 100 will be most accurate from 25 to 75 pound feet. The accuracy of readings will decrease as you approach 0 pound feet or 100 pound feet. The following ranges are based on this principle.

TORQUE RANGES

STATED RANGE	MOST EFFECTIVE RANGE
0-200 lb-in.	4-13 lb-ft
0-600 lb-ft	50-450 lb-ft
0-170 lb-ft	44-131 lb-ft
15-75 lb-ft	30-60 lb-ft

FIGURE 27. Example of a typical torque limits appendix (sheet 3 of 5) (reference 5.18.7).

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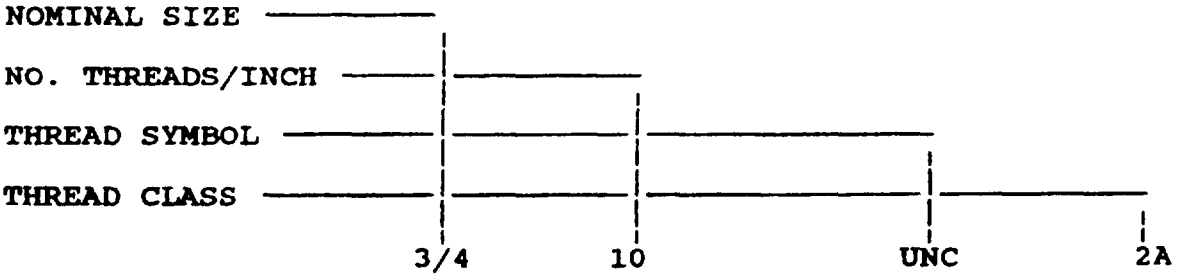
FASTENER SIZE AND THREAD PATTERN

Threaded fasteners are categorized according to diameter of the fastener shank. Thread styles are divided into broad groups, the two most common being coarse (Unified Coarse-UNC) and fine (Unified Fine-UNF). These groups are defined by the number of threads per inch on the bolt shanks. In addition, threads are categorized by thread class, which is a measure of the degree of fit between the threads of the bolt or screw (external threads) and the threads of the attaching nut or tapped hole (internal threads). The most common thread class for bolts and screws is Class 2.

THREAD CLASSES AND DESCRIPTION

EXTERNAL	INTERNAL	FIT
1A	1B	LOOSE FIT
2A	2B	MEDIUM FIT
3A	3B	CLOSE FIT

Thread patterns are designated as follows:



NOTE: Unless followed with -LH (e.g., 3/4-10UNC-2A-LH), threads are right-hand.

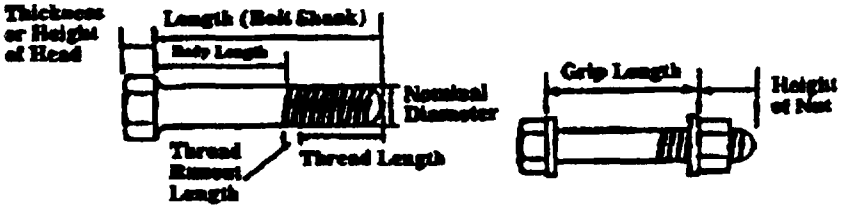


FIGURE 27. Example of a typical torque limits appendix (sheet 4 of 5) (reference 5.18.7).

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FASTENER GRADE

In addition to being classified by thread type, threaded fasteners are also classified by material. The most familiar fastener classification system is the SAE grading system.

SAE SCREW AND BOLT MARKINGS

SCREWS	BOLTS
SAE GRADE 2 NO MARKING	SAE GRADE 6 4 RADIAL DASHES 90° APART
SAE GRADE 3 2 RADIAL DASHES 180° APART	SAE GRADE 7 5 RADIAL DASHES 72° APART
SAE GRADE 5 3 RADIAL DASHES 120° APART	SAE GRADE 8 6 RADIAL DASHES 60° APART

NOTE
Torque values for grade 8.2 bolts are the same as for Grade B.

MARKINGS ON HEX LOCKNUTS

GRADE A - No Marks	GRADE A - No Mark
GRADE B - 3 Marks	GRADE B - Letter B
GRADE C - 6 Marks	GRADE C - Letter C
GRADE A - No Notches	
GRADE B - One Notch	
GRADE C - Two Notches	

FIGURE 27. Example of a typical torque limits appendix (sheet 5 of 5) (reference 5.18.7).

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Table X-X. Authorized Projectile Fuze Combinations for 8-inch Howitzer, SP, M110A2 Cannon M201A1.

TYPE AND MODEL NUMBER OF PROJECTILE	FUZE										
	PD		MT		MTSQ			PROX(VT)		ET	
	M739 SERIES	M557	M572	M565	M564	M567 SERIES	M582 SERIES	M728	M732	M762	M767
Agent GB, VX, M426	X	X	X					X 4	X 4		
HE, M106 (Shallow Cavity)	X	X	X		X 3		X		X		X
HE, M106 (Deep Cavity)	X	X	X		X 3		X	X 2	X		X
HE, M404 ICM				X		X				X	
HE, M509A1 ICM						X				X	
HERA, M650 (Rocket-On)	X	X	X								X
HERA M650[1] (Rocket-Off)	X	X	X		X 3		X		X		X

WARNING: 1-DO NOT FIRE THE M650 PROJECTILE IF THE OBTURATING BAND IS MISSING OR BROKEN. IF THE BAND IS DISPLACED AND CAN BE REPOSITIONED AND REMAIN IN THE GROOVE, THE PROJECTILE CAN BE FIRED.

NOTE: 2-AUTHORIZED, REQUIRES REMOVAL OF SUPPLEMENTARY CHARGE.

3-FUZE, MTSQ, M564 IS RESTRICTED FROM FIRING WITH ZONE 9 M188A1 PROPELLING CHARGE.

4-M728 AND M732 FIRED ONLY WITH "VX" PROJECTILE AND ONLY IN COMBAT EMERGENCY.

FIGURE 28. Example of an ammunition marking table (sheet 1 of 3) (reference 5.18.8).

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Table X-X. Authorized Propelling Charges Combinations for 8-inch Howitzer, SP, M110A2 with Common M201A1.

TYPE AND MODEL NUMBER OF PROJECTILE	M1 ZONE					M2 ZONE			M188 ZONE	M188A1 ZONE	
	1	2	3	4	5	5	6	7	8	8	9
Agent GB, VX, M426	X	X	X	X	X	X	X	X	X	X	X
HE, M106 (Shallow Cavity)	X	X	X	X	X	X	X	X	X	X	X 2
HE, M106 (Deep Cavity)	X	X	X	X	X	X	X	X	X	X	X 2
HE, M404 ICM	X	X	X	X	X	X	X	X			
HE, M509A1, ICM	X	X	X	X	X	X	X	X	X	X	X
HERA, M650 (Rocket-On)								X	X	X	X
HERA M650[1] (Rocket-Off)	X	X	X	X	X	X	X	X	X	X	X 2

WARNING: 1-DO NOT FIRE THE M650 PROJECTILE IF THE OBTURATING BAND IS MISSING OR BROKEN. IF THE BAND IS DISPLACED AND CAN BE REPOSITIONED AND REMAIN IN THE GROOVE, THE PROJECTILE CAN BE FIRED.

NOTE: 2-FUZE, MTSQ, M564 IS RESTRICTED FROM FIRING WITH ZONE 9 M188A1 PROPELLING CHARGE.

FIGURE 28. Example of an ammunition marking table (sheet 2 of 3) (reference 5.18.8).

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Table X-X. Model Number and Color coding of Projectiles for 8-inch Howitzer, M110A2.

TYPE AND MODEL NO. OF PROJECTILE	NEW MANUFACTURE			OLD MANUFACTURE		
	COLOR OF PROJECTILE	NO./COLOR OF BANDS	MARKING	COLOR OF PROJECTILE	NO./COLOR OF BANDS	MARKING
**Agent (GB) w/burster, M426	Gray	3/Green 1/Yellow	Green	Gray	1/Green	Green
**Agent (VX) w/burster, M426	Gray	3/Green 1/Yellow	Green	Gray	2/Green	Green
HE, (ICM) M404	Olive Drab	*Diamonds	Yellow	Olive Drab	None	Yellow

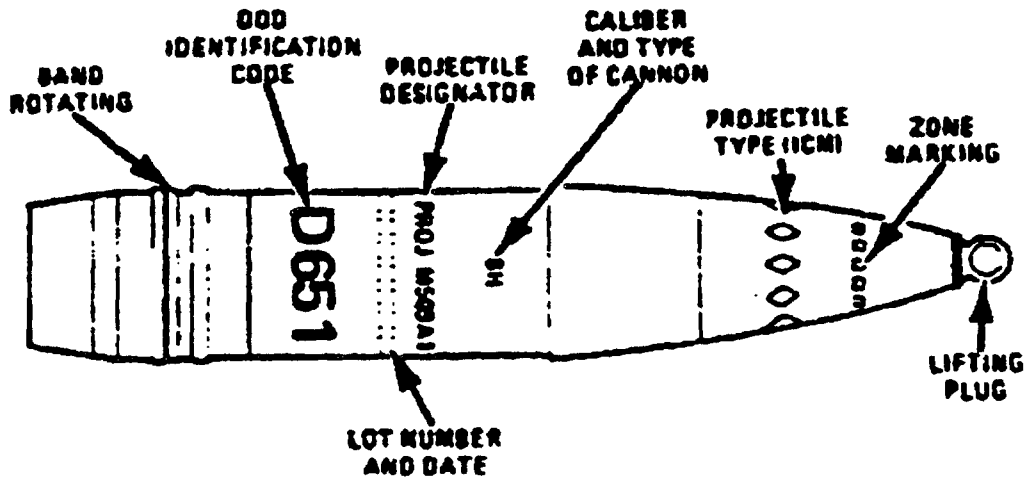
* Row of yellow diamonds between nose and bourrelet of projectile.

** Renovated or newly manufactured projectile (post 1976) will be marked with one green band and if burster, one yellow band.

FIGURE 28. Example of an ammunition marking table (sheet 3 of 3) (reference 5.18.8).

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X-10 MARKING OF 8-INCH DPICM, M509A1 PROJECTILE.

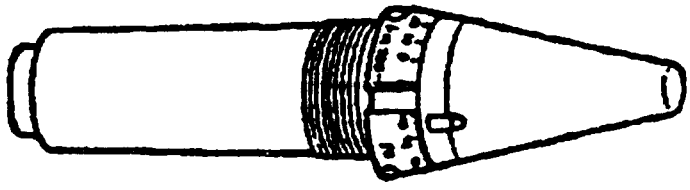


X-10.1 Projectile, 8-inch DPICM, M509A1. This projectile is used to deliver a concentration of grenades effective against personnel and materiel. The projectile is longer than the M404 projectile (requiring it to be reloaded over the rear of the vehicle) and requires separate registration. The projectile is provided with a fusible or universal lifting plug (yellow). The cargo consists of 180 M42 dual purpose grenades which are expelled from the projectile in flight when a preset fuze ignites an expulsion charge. Upon ejection from the carrier, the grenades are dispersed by the projectile spin and are oriented and armed by a stabilizing ribbon device. On impact, the grenade fuze initiates the high explosive charge producing an armor penetrating shaped-charge jet and a large number of small fragments. A spotting charge is available as a separate item of issue and, when substituted for the expulsion charge, permits observation of the projectile functioning in relation to the target. The M509A1 projectile weighs approximately 208 lb (94 kg). Refer to paragraph X-8 for authorized fuze combinations.

FIGURE 29. Example of an ammunition marking illustration (reference 5.18.8).

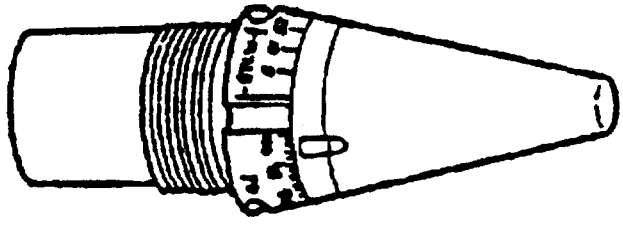
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X-20 FUZE, PROXIMITY, VARIABLE TIME (VT), M728.



The M728 proximity (VT) fuze is a long-intrusion fuze used with deep cavity projectiles and is essentially a self-powered radio and transmitting unit. The fuze can be set from 50 to 100 seconds. The time setting determines at what time along the trajectory the fuze will become activated. The M728 has an impact element that is armed 3 seconds after firing and will function either on proximity action or impact action, whichever occurs first. The M728 should be set for impact action by setting the time ring to 90.0 seconds or PD mark. A protective coating is on the fuze ogive to reduce the possibility of static electricity causing early downrange functioning.

X-21 FUZE, PROXIMITY, VARIABLE TIME (VT), M732.



This short intrusion proximity fuze has a plastic nose cone fitted to a movable steel ring which rotates on a steel sleeve. The movable ring has an index mark for setting time. The fuze is shipped with the index mark alined with the PD line on the sleeve. When set at any value between 5 and 150 seconds, the proximity arming occurs approximately 3 seconds prior to the set time. If the fuze fails to function in the proximity mode, it will function on ground impact.

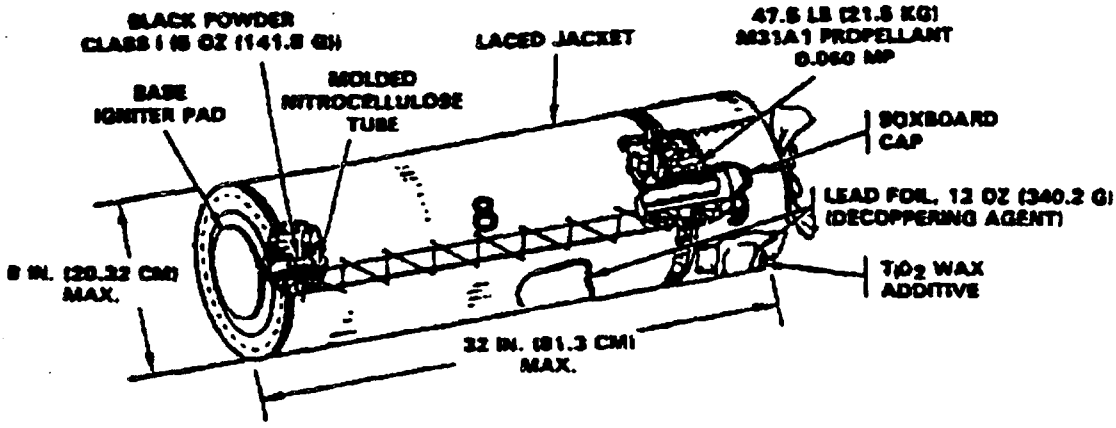
NOTE

The PD setting of the M732 VT fuze when fired into soft impact areas will produce less lethality than the superquick setting of the M739 series PD fuze.

FIGURE 30. Example of integrated ammunition illustration and text (reference 5.18.8).

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X-30 8-IN. PROPELLING CHARGE, M188A1.



NOTE

A yellow discoloration of the M188A1 bag may occur but does not affect performance and is safe to fire.

This is a two-increment "white bag" charge, 32 in. long by 8 in. (81.30 cm long by 20.32 cm) in diameter. A red cloth igniter charge is sewed to the rear (breech end) of increment 8. An igniter core extends through the center of the charge. An additive to reduce gun-tube wear is part of increment 9. A jacket made of acrylicviscose rayon is wrapped around increment 8 and laced up with cord. The jacket serves to maintain the proper propellant charge configuration.

FIGURE 31. Example of a propelling charge illustration and accompanying text (reference 5.18.8).

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